

13. An optical device which always produces images of $m = +1$ is :

- (A) Plane mirror
- (B) Convex mirror
- (C) Concave mirror
- (D) Convex lens

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(A)/ Plane mirror

- 15.** Concave mirror finds its application in which of the following cases ?
- (A) A solar furnace
 - (B) A rear-view mirror in vehicles
 - (C) To view full size erect image of distant tall buildings
 - (D) To produce divergent beam of light

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(A) / A solar furnace

- 25.** An object is placed 18 cm in front of a concave mirror of focal length 12 cm. Use mirror formula to determine the position of the image formed in this case.

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25. An object is placed 18 cm in front of a concave mirror of focal length 12 cm. Use mirror formula to determine the position of the image formed in this case.

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2

$$u = -18 \text{ cm}, f = -12 \text{ cm}, v = ?$$

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$\frac{1}{v} = \frac{1}{-12} - \frac{1}{-18}$$

$$\frac{1}{v} = \frac{-3 + 2}{36}$$

$$\frac{1}{v} = \frac{-1}{36}$$

$v = -36 \text{ cm}$ (image is formed at a distance 36 cm in front of the mirror)

$\frac{1}{2}$

$\frac{1}{2}$

1

- 25.** An object is placed 25 cm in front of a concave mirror of focal length 15 cm. Use mirror formula and new cartesian sign convention to determine the position of the image formed.

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2

$u = -25 \text{ cm}$ $f = -15 \text{ cm}$ $v = ?$	$\frac{1}{2}$
$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$	$\frac{1}{2}$
$\frac{1}{v} - \frac{1}{25} = \frac{-1}{15}$	
$\frac{1}{v} = \frac{-1}{15} + \frac{1}{25}$	
$= \frac{-5 + 3}{75} = \frac{-2}{75}$	
$v = \frac{-75}{2} = -37.5 \text{ cm}$ (Image is formed 37.5 cm in front of the mirror.)	1

11. A 5 cm long rod is placed vertically in front of a concave mirror. If its 5 cm long image is formed at 20 cm in front of the mirror, the focal length of the mirror is :

- (A) – 20 cm
- (B) – 10 cm
- (C) – 5 cm
- (D) 20 cm

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- (A) – 20 cm
- (B) – 10 cm
- (C) – 5 cm
- (D) 20 cm

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(B)/ –10 cm

20. *Assertion (A) :* The magnification of the virtual images produced by concave mirrors is always more than +1.

Reason (R) : The virtual images formed by concave mirrors are erect and magnified.

20. *Assertion (A) :* The magnification of the virtual images produced by concave mirrors is always more than +1.

Reason (R) : The virtual images formed by concave mirrors are erect and magnified.

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(A)/ Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

24. List four characteristics of the images formed by convex mirrors.

2

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24. List four characteristics of the images formed by convex mirrors.

2

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- Virtual
- Erect
- Diminished
- Behind the mirror.

$\frac{1}{2} \times$
4

39. A highly polished surface which reflects most of the light falling on it is called a mirror. Mirrors may be plane as well as spherical. The reflecting surface of a spherical mirror may be curved inwards or outwards. A spherical mirror where reflecting surface is curved inwards is called concave mirror, whereas spherical mirrors where reflecting surfaces are curved outwards are known as convex mirrors.

- (a) State the laws of reflection of light. 1
- (b) Define the term 'Principal axis of a concave mirror.' 1
- (c) (i) Write the path of the following light rays after reflection from a concave mirror :
 - (I) A ray of light moving parallel to the principal axis of a concave mirror.
 - (II) A ray incident obliquely to the principal axis towards the pole of the concave mirror making angle θ . 2

OR

- (c) (ii) An object is placed at 18 cm in front of a concave mirror of focal length 12 cm. Use mirror formula to determine the position of the image. 2

(a)

Laws of Reflection

- Incident ray, reflected ray and the normal to the surface at the point of incidence all lie in the same plane.

$\frac{1}{2}$

- Angle of incidence ($\angle i$) = Angle of reflection ($\angle r$)

$\frac{1}{2}$

(b) Principal Axis is an imaginary straight line passing through the pole and centre of curvature of a concave mirror.

1

(c) (i) (I) passes through the focus of the concave mirror
(II) is reflected obliquely making equal angle (θ) with the principal axis

1

1

OR

(c) (ii) Here $f = -12$ cm $u = -18$ cm $v = ?$

$\frac{1}{2}$

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u} \text{ or } \frac{1}{v} = \frac{1}{f} - \frac{1}{u}$$

$\frac{1}{2}$

$$\frac{1}{v} = \frac{1}{-12} - \frac{1}{-18} = \frac{-3+2}{36} = \frac{-1}{36}$$

$$v = -36 \text{ cm}$$

1

Position: Image is formed at 36 cm in front of the mirror

15. An object is placed at a distance of 30 cm from the pole of a concave mirror. If its real and inverted image is formed at 60 cm in front of the mirror, the focal length of the mirror is :

(A) -15 cm

(B) -20 cm

(C) $+20$ cm

(D) $+15$ cm

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15. An object is placed at a distance of 30 cm from the pole of a concave mirror. If its real and inverted image is formed at 60 cm in front of the mirror, the focal length of the mirror is :

(A) -15 cm

(B) -20 cm

(C) $+20$ cm

(D) $+15$ cm

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(B) / -20 cm

- 31.** A convex mirror used for rear view on an automobile has a focal length of 1.5 m. If a 3 m high bus is located at 6.0 m from the mirror, use mirror formula to determine the position and size of the image of the bus as seen in the mirror.

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- 31.** A convex mirror used for rear view on an automobile has a focal length of 1.5 m. If a 3 m high bus is located at 6.0 m from the mirror, use mirror formula to determine the position and size of the image of the bus as seen in the mirror.

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3

Here $f = +1.5$ m; $u = -6.0$ m; $h = 3$ m

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} + \frac{1}{-6} = \frac{1}{1.5}$$

$$\frac{1}{v} = \frac{1}{6} + \frac{1}{1.5}$$

$$v = \frac{6}{5} = 1.2 \text{ m}$$

$$\frac{h'}{h} = \frac{-v}{u}$$

$$\frac{h'}{3} = \frac{1.2}{6}$$

$$h' = 0.6 \text{ m}$$

$\frac{1}{2}$

$\frac{1}{2}$

1

$\frac{1}{2}$

$\frac{1}{2}$

13. Mirror 'X' is used to concentrate sunlight in solar furnace and Mirror 'Y' is fitted on the side of the vehicle to see the traffic behind the driver. Which of the following statements are true for the two mirrors ?

- (i) The image formed by mirror 'X' is real, diminished and at its focus.
- (ii) The image formed by mirror 'Y' is virtual, diminished and erect.
- (iii) The image formed by mirror 'X' is virtual, diminished and erect.
- (iv) The image formed by mirror 'Y' is real, diminished and at its focus.

(A) (i) and (ii)

(B) (ii) and (iii)

(C) (iii) and (iv)

(D) (i) and (iv)

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- (ii) The image formed by mirror 'Y' is virtual, diminished and erect.
- (iii) The image formed by mirror 'X' is virtual, diminished and erect.
- (iv) The image formed by mirror 'Y' is real, diminished and at its focus.

(A) (i) and (ii)

(B) (ii) and (iii)

(C) (iii) and (iv)

(D) (i) and (iv)

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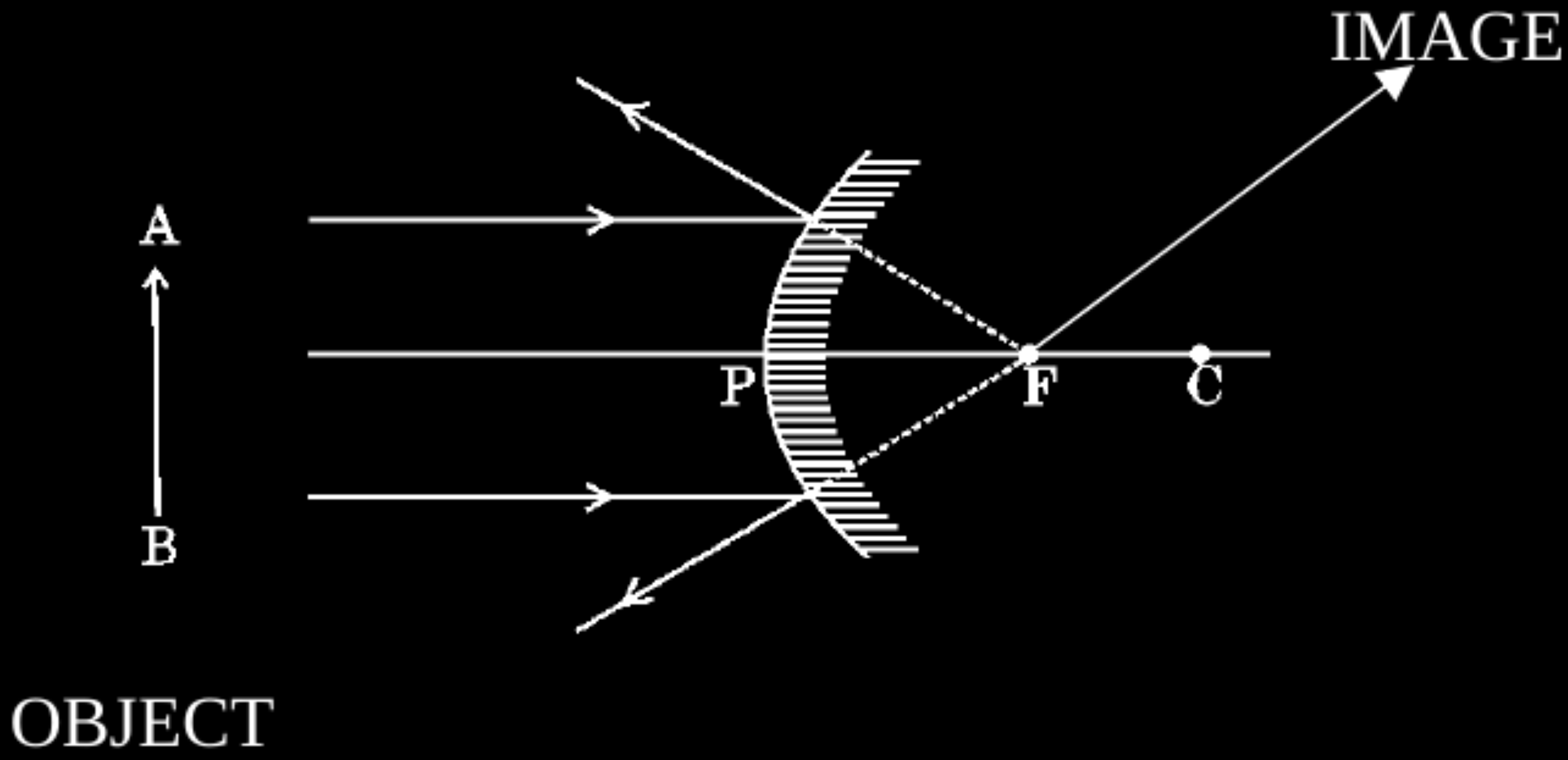
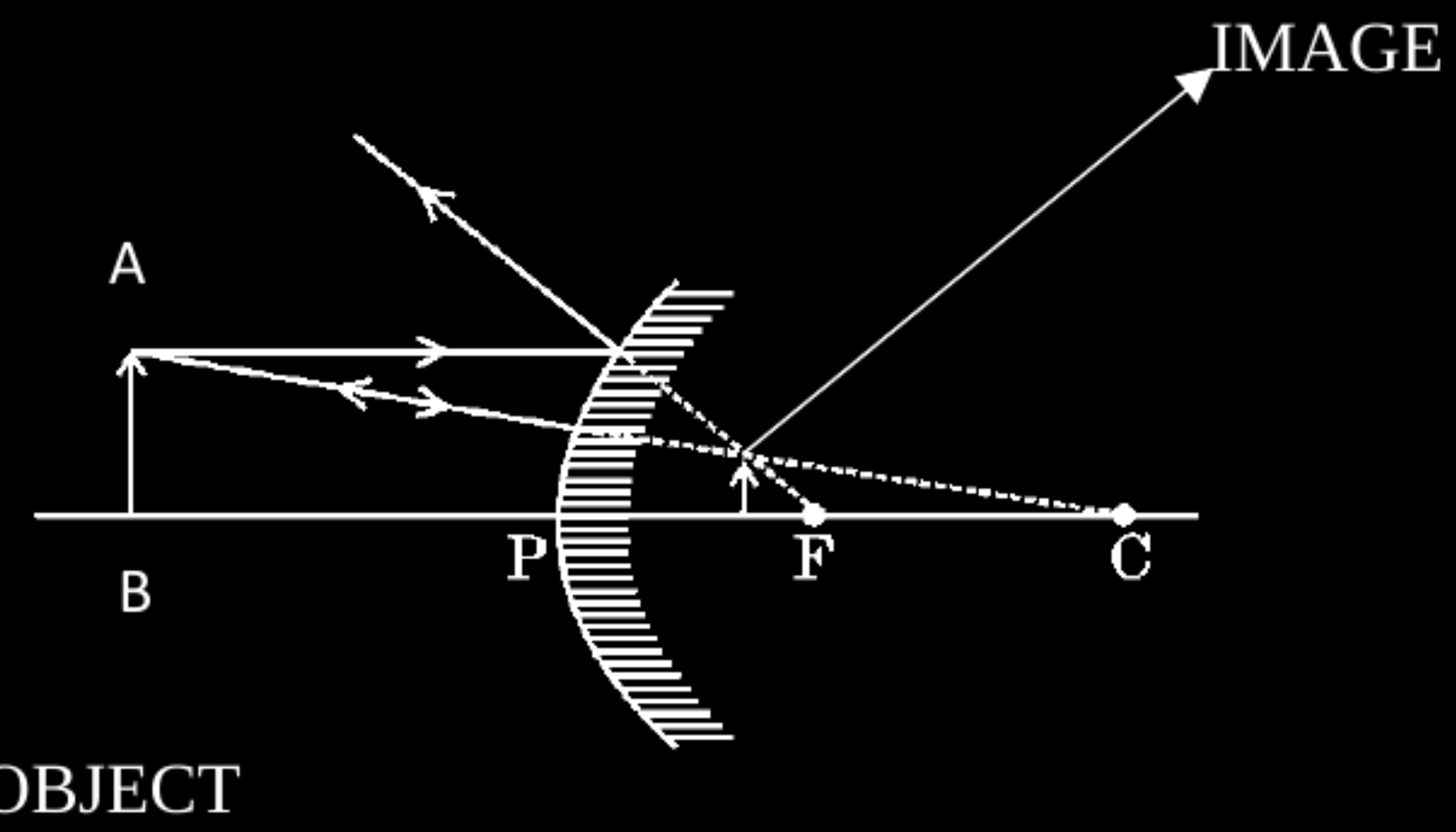
- 31.** Draw ray diagrams to show the nature, position and relative size of the image formed by a convex mirror when the object is placed (i) at infinity and (ii) between infinity and pole P of the mirror.

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3

(i)  OBJECT	1 ½
(ii)  OBJECT (Note: Deduct ½ mark if arrows are not drawn.)	1 ½

- 23.** An object is placed at a distance of 30 cm in front of a concave mirror of focal length 20 cm. Use mirror formula to determine the position of the image formed in this case.

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23. An object is placed at a distance of 30 cm in front of a concave mirror of focal length 20 cm. Use mirror formula to determine the position of the image formed in this case.

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2

$$f = -20 \text{ cm} ; \quad u = -30 \text{ cm}$$

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$\frac{1}{v} = \frac{1}{-20} - \frac{1}{-30}$$

$$\frac{1}{v} = \frac{1}{-20} + \frac{1}{30}$$

$$v = -60 \text{ cm}$$

Position of image is 60 cm in front of the mirror.

$\frac{1}{2}$

$\frac{1}{2}$

1

- 21.** An object is placed at a distance of 10 cm in front of a concave mirror of focal length 15 cm. Use mirror formula to determine the position of the image formed by this mirror.

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21. An object is placed at a distance of 10 cm in front of a concave mirror of focal length 15 cm. Use mirror formula to determine the position of the image formed by this mirror.

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2

Here $u = -10$ cm; $f = -15$ cm ;

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$\frac{1}{-15} = \frac{1}{v} + \frac{1}{-10}$$

$$\frac{1}{v} = \frac{1}{-15} + \frac{1}{10}$$

$$v = +30 \text{ cm}$$

Position of image is 30 cm from the mirror

$\frac{1}{2}$

$\frac{1}{2}$

1

- (b) (i) A student wishes to study the image formation by a concave mirror using candle flame as object. State the type of the image formed by the mirror and mention the change in the image formed, if any, that he observes when the candle flame is gradually moved away from the pole of the mirror. Draw a ray diagram to show the image formation when the object distance is nearly equal to the radius of curvature of the mirror.
- (ii) A convex mirror used for rear-view on an automobile has a focal length of 3.0 m. If a bus is located at 6.0 m from this mirror, use mirror formula to find the position of the image of the bus as seen in the mirror.

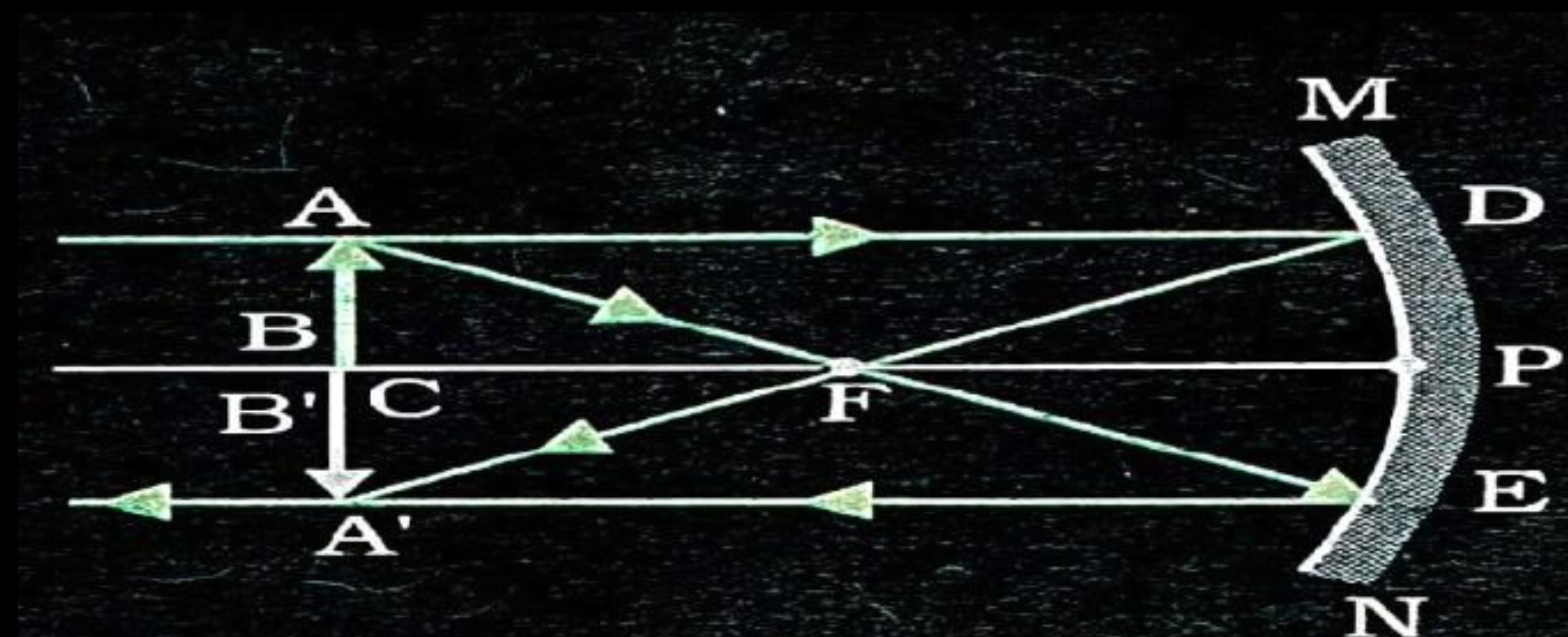
(b)

(i)

Type of Image

Real and Inverted (when candle is beyond F)/ Virtual and Erect (when candle is between P and F)

As the object is moved gradually away from the pole of the mirror, the image gets diminished.



(ii) Object distance, $u = -6.00 \text{ m}$

Image distance, $v = ?$

Focal length, $f = 3.00 \text{ m}$

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \quad \text{or} \quad \frac{1}{v} = \frac{1}{f} - \frac{1}{u}$$

$$\frac{1}{v} = \frac{1}{+3.00 \text{ m}} - \frac{1}{(-6.00 \text{ m})}$$

$$= \frac{1}{+3.00 \text{ m}} + \frac{1}{6.00 \text{ m}} = \frac{2+1}{6 \text{ m}}$$

$$\text{or } v = \frac{6}{3} = 2.0 \text{ m}$$

1

1

1

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

- 31.** If we want to obtain a virtual and magnified image of an object by using a concave mirror of focal length 18 cm, where should the object be placed? Use mirror formula to determine the object distance for an image of magnification +2 produced by this mirror to justify your answer. CBSE 2025 **3**

- 31.** If we want to obtain a virtual and magnified image of an object by using a concave mirror of focal length 18 cm, where should the object be placed? Use mirror formula to determine the object distance for an image of magnification +2 produced by this mirror to justify your answer. CBSE 2025 **3**

Object should be placed between F and P / At less than 18cm distance from the mirror.

$$\text{Mirror formula} = \frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

Magnification $m = +2$

$$f = -18 \text{ cm}$$

$$m = -\frac{v}{u} = +2$$

$$\therefore v = -2u$$

$$\frac{1}{-2u} + \frac{1}{u} = \frac{1}{-18 \text{ cm}}$$

$$\therefore \frac{1}{2u} = \frac{1}{-18 \text{ cm}}$$

$$u = -9 \text{ cm}$$

1

$\frac{1}{2}$

1

$\frac{1}{2}$

- 31.** If we want to obtain a real and magnified image of an object by using a concave mirror of focal length 18 cm. Where should the object be placed? Use mirror formula to determine the object distance for an image of magnification – 2 by this mirror to justify your answer.

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- 31.** If we want to obtain a real and magnified image of an object by using a concave mirror of focal length 18 cm. Where should the object be placed? Use mirror formula to determine the object distance for an image of magnification – 2 by this mirror to justify your answer.

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3

- Object should be placed between C and F / between 18 cm to 36 cm from the mirror.

- Mirror formula $= \frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

- Magnification $m = -2$

$$f = -18 \text{ cm}$$

$$m = -\frac{v}{u} = -2$$

$$\therefore v = 2u$$

- $$\frac{1}{2u} + \frac{1}{u} = \frac{1}{-18 \text{ cm}}$$

$$\therefore \frac{3}{2u} = \frac{1}{-18 \text{ cm}}$$

$$u = -27 \text{ cm}$$

1

$\frac{1}{2}$

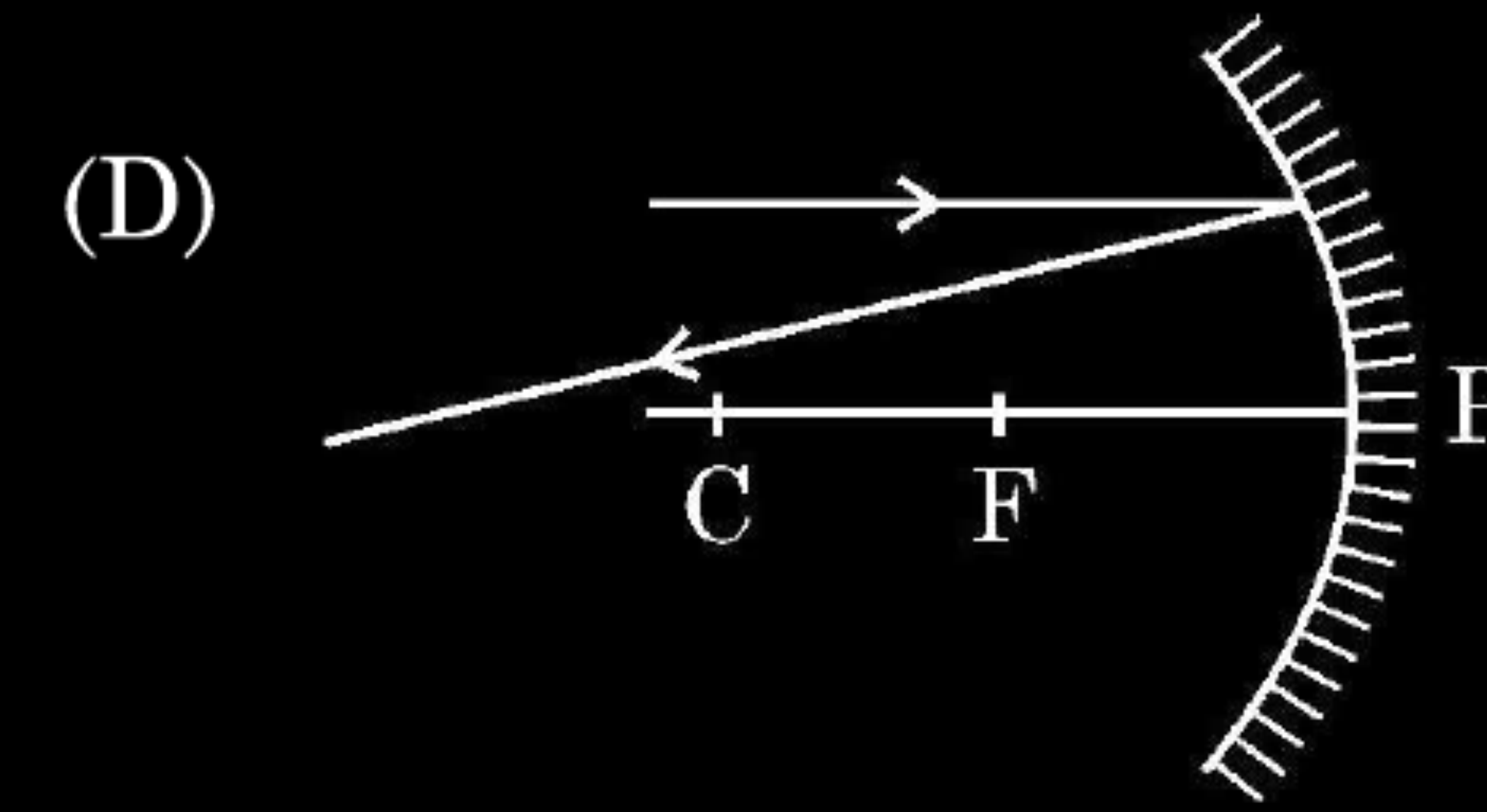
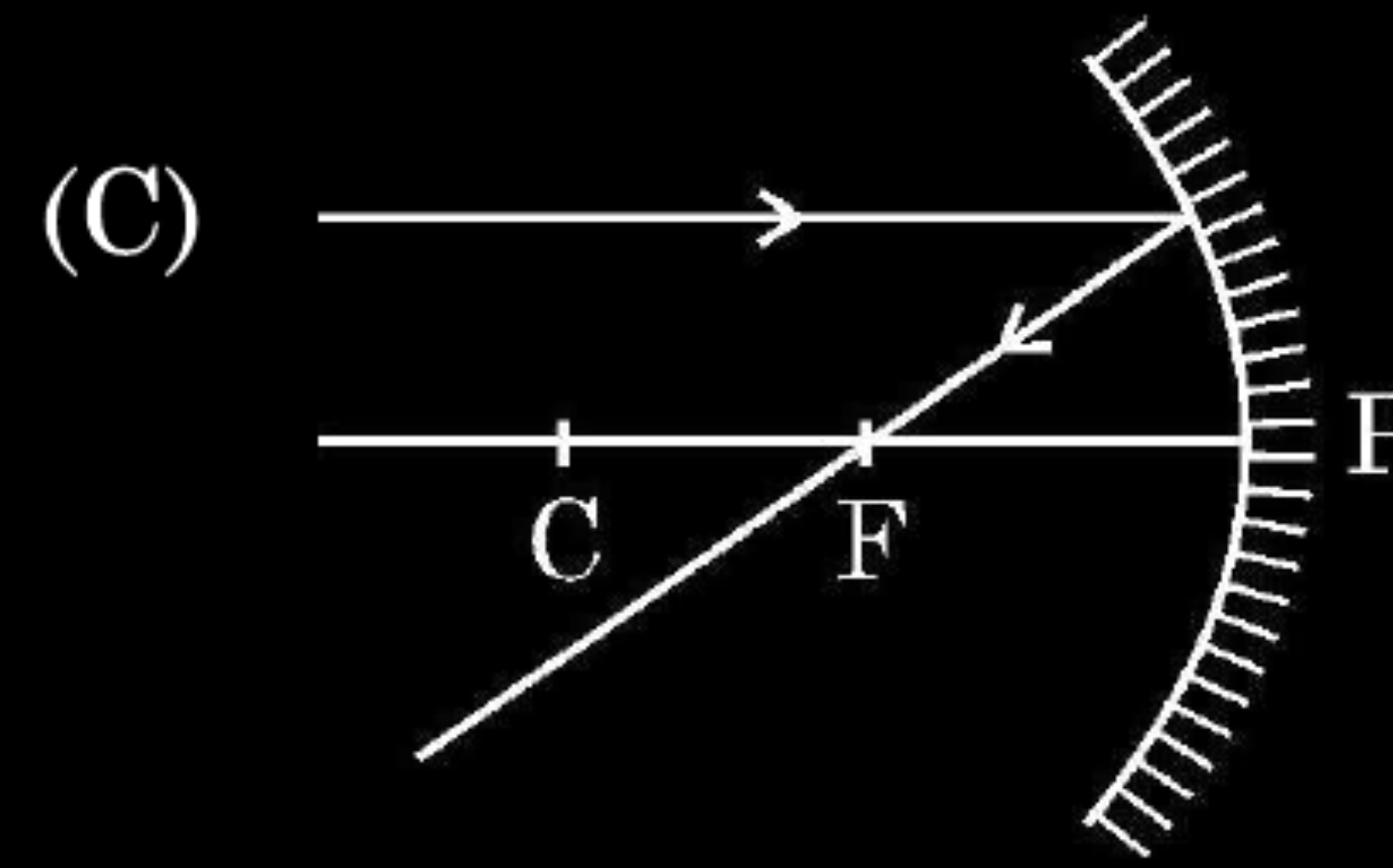
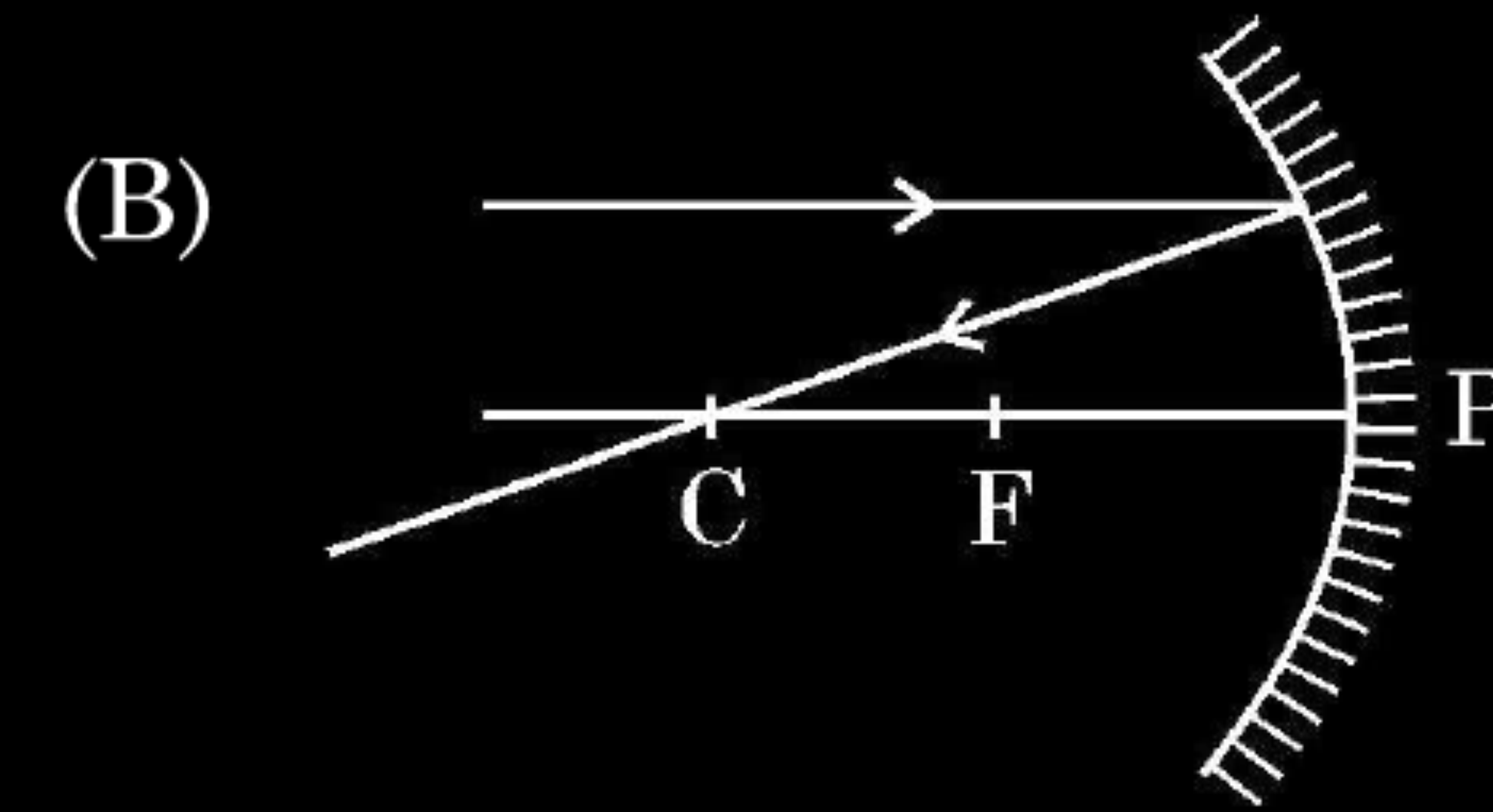
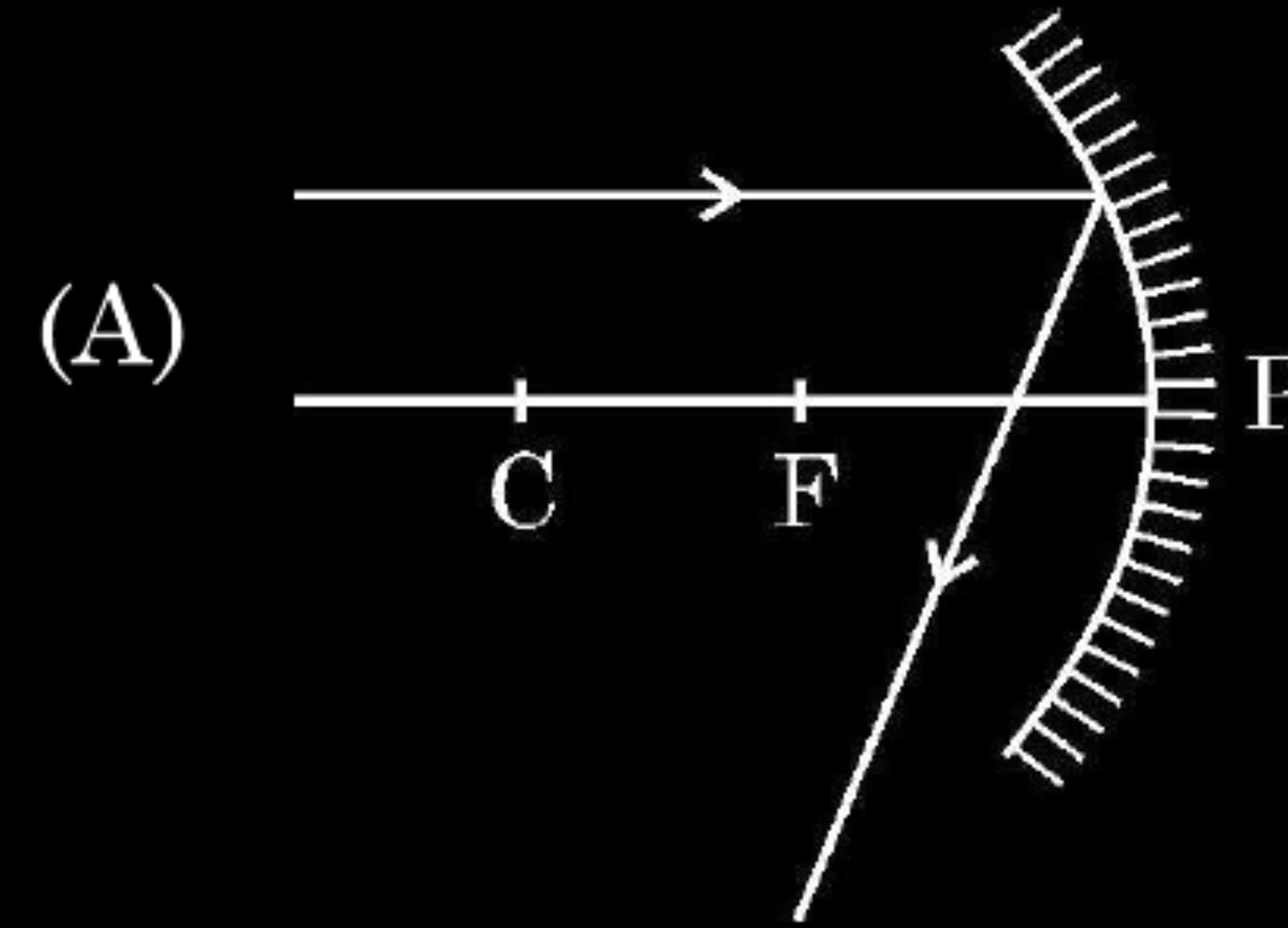
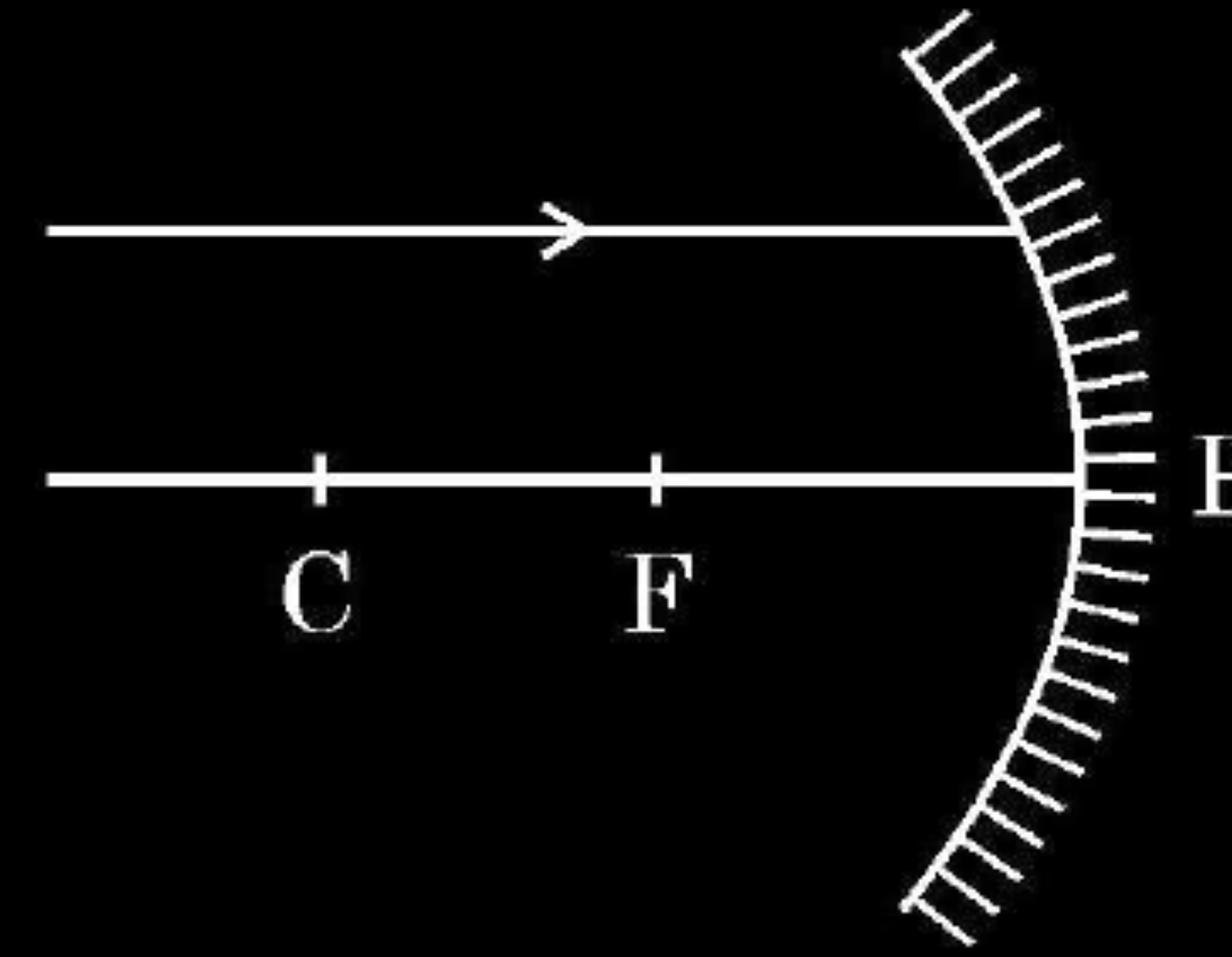
1

$\frac{1}{2}$

14. Identify from the following the ray diagram which shows the correct path of the reflected ray for the ray incident on a concave mirror as shown :

1

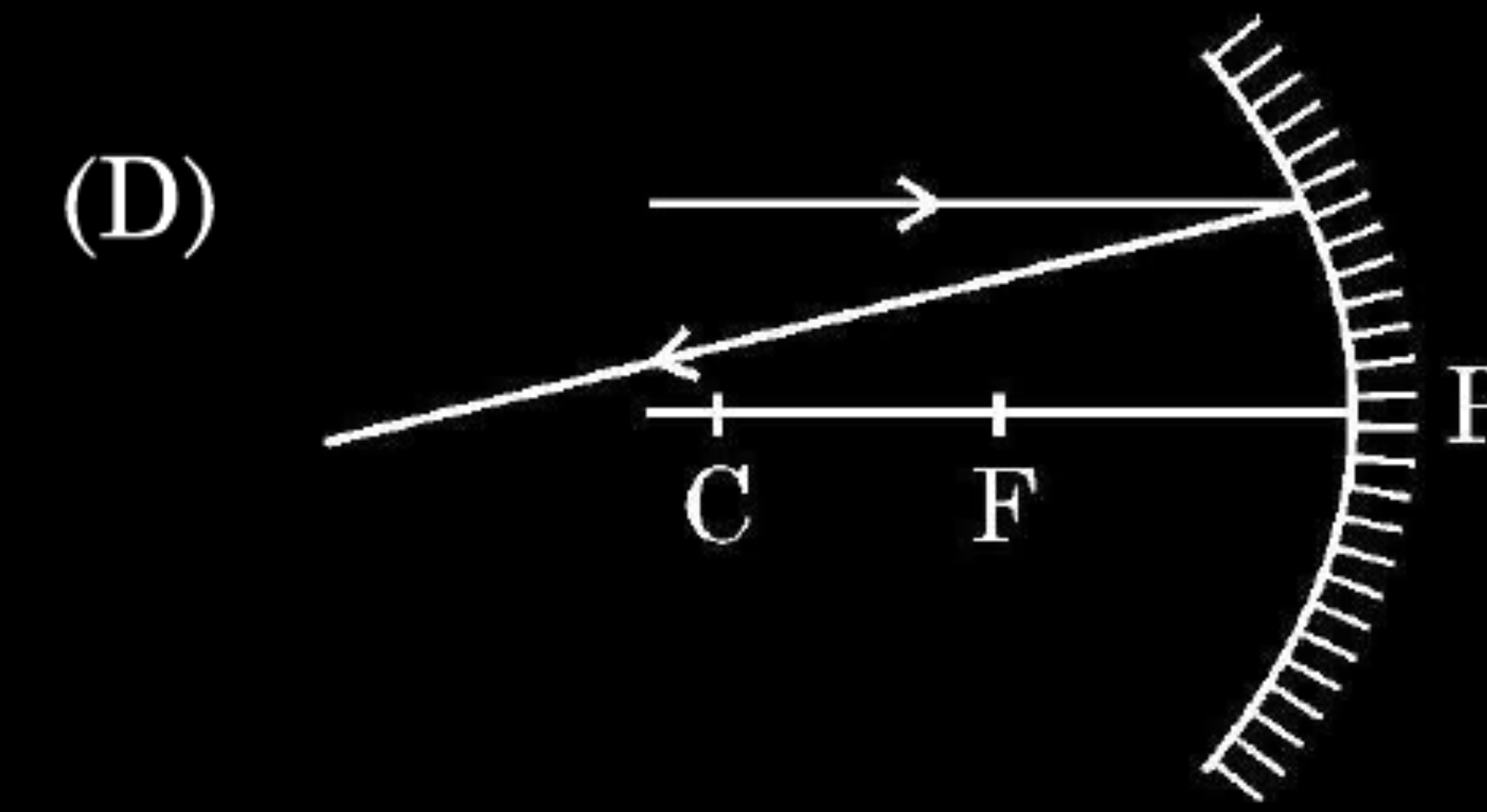
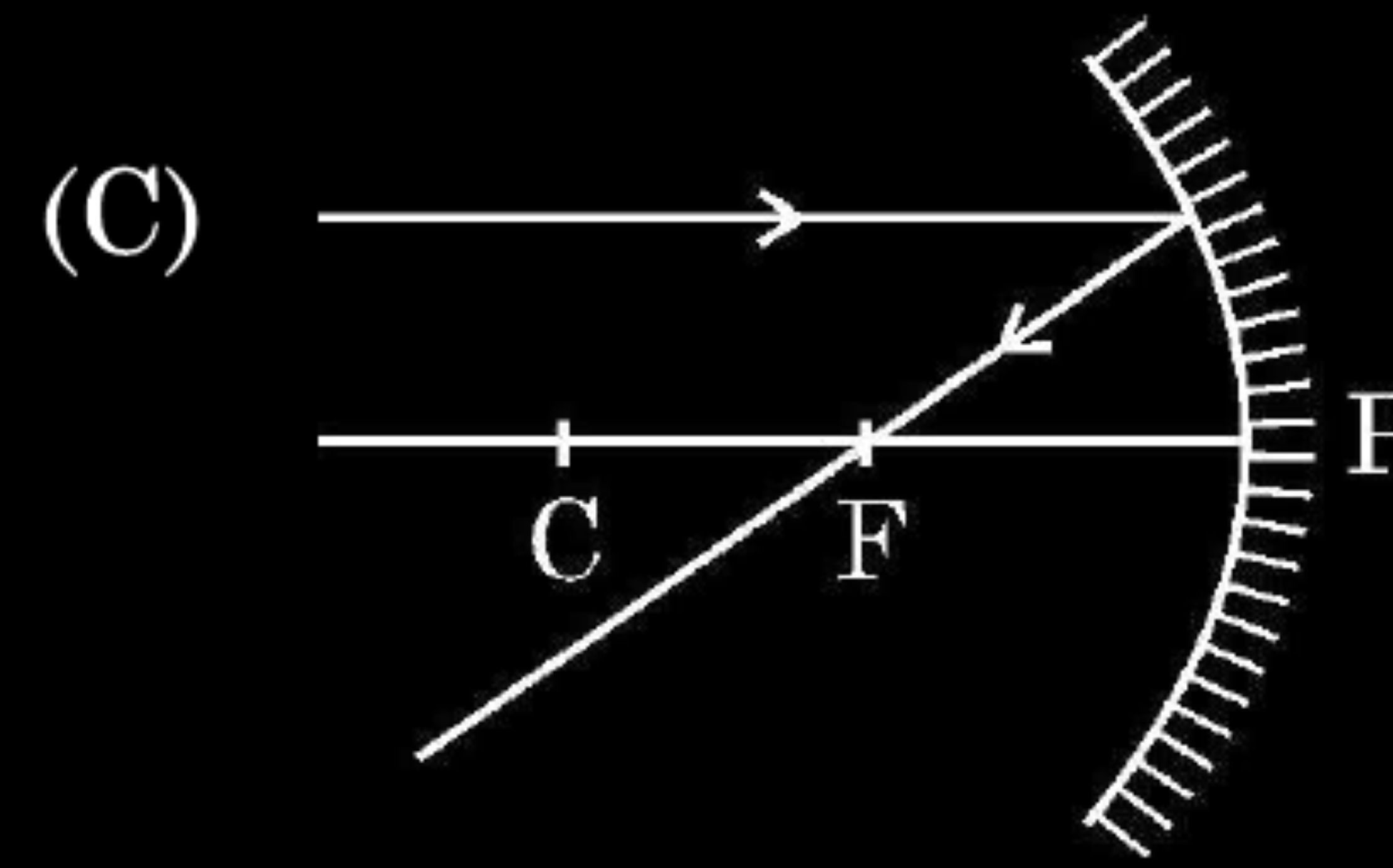
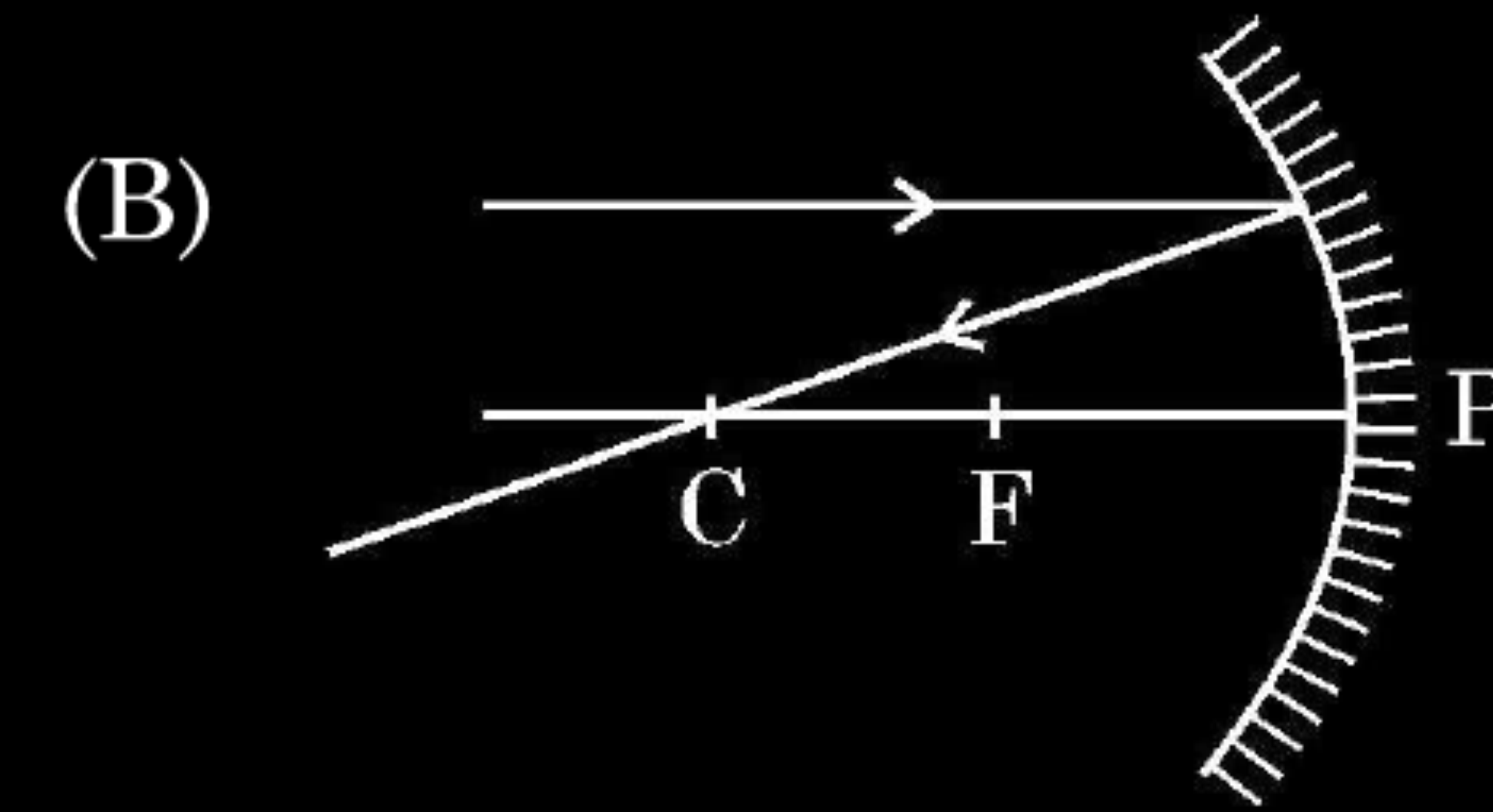
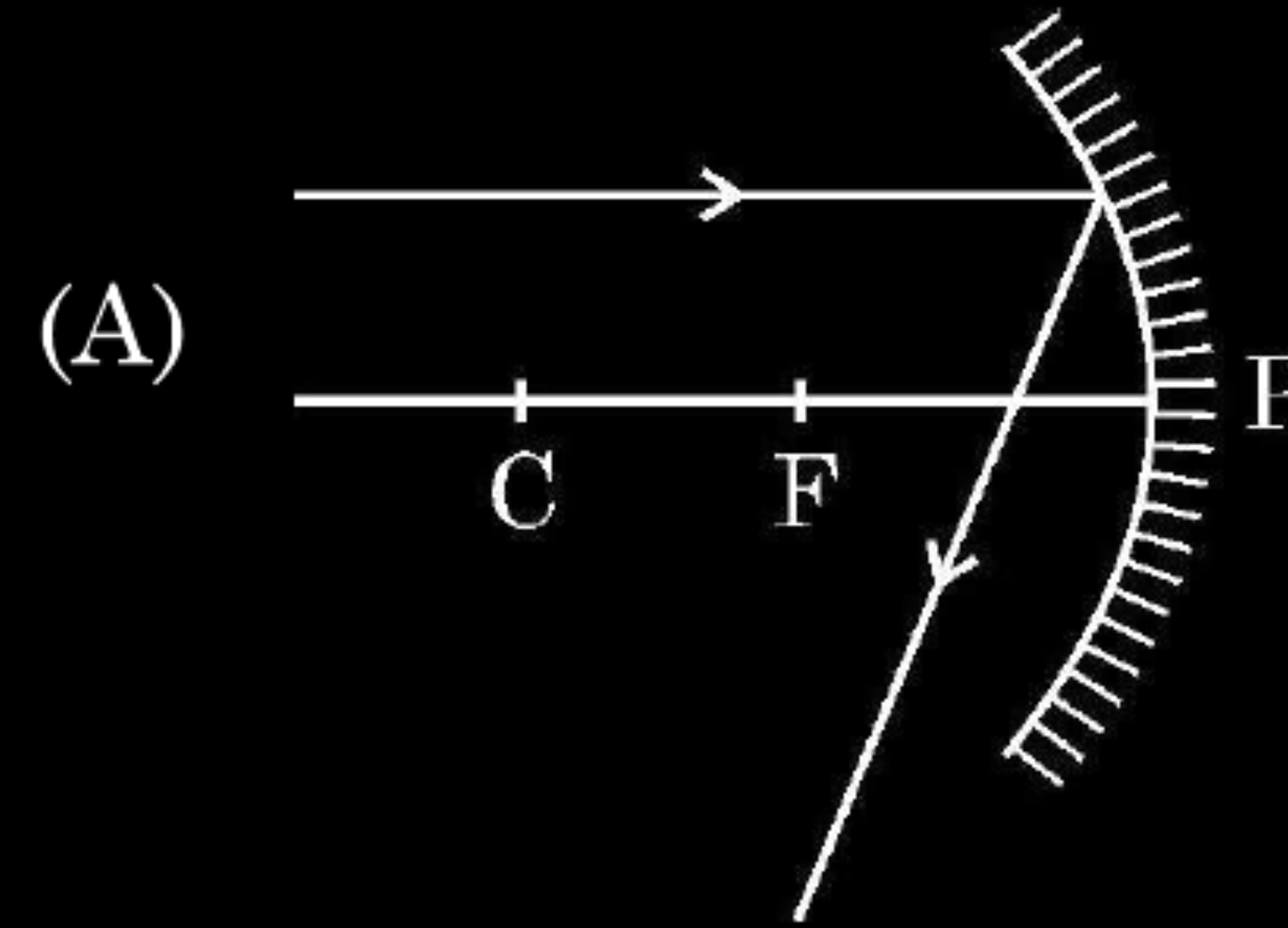
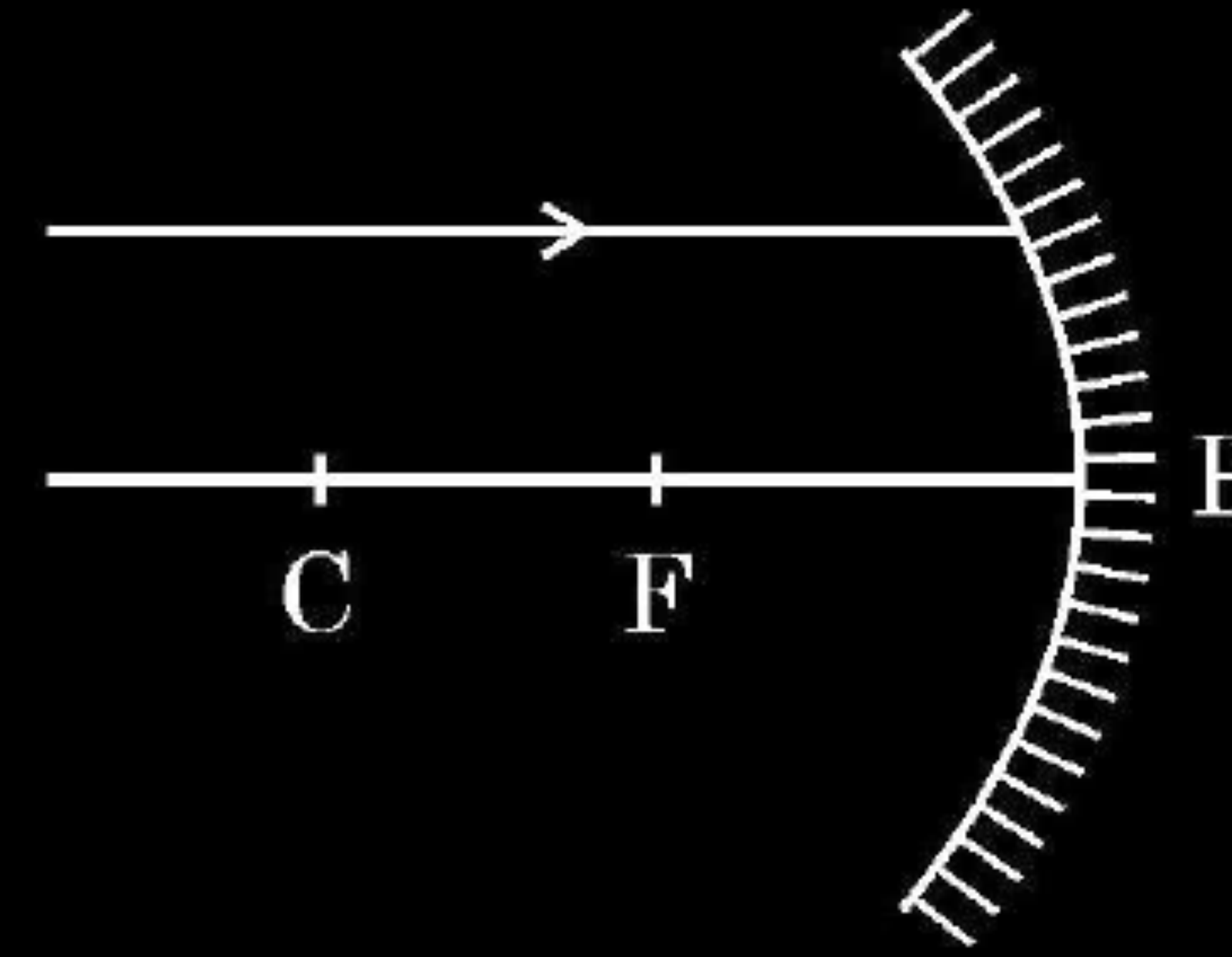
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14. Identify from the following the ray diagram which shows the correct path of the reflected ray for the ray incident on a concave mirror as shown :

1

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(C)

12. An object is placed at a distance of 30 cm from the reflecting surface of a concave mirror of radius of curvature 40 cm. The image formed is
- (A) Virtual and magnified (B) Virtual and diminished
(C) Real and magnified (D) Real and diminished

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12. An object is placed at a distance of 30 cm from the reflecting surface of a concave mirror of radius of curvature 40 cm. The image formed is

- | | |
|---------------------------|----------------------------|
| (A) Virtual and magnified | (B) Virtual and diminished |
| (C) Real and magnified | (D) Real and diminished |

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(C)/ Real and magnified

25. (a) List four properties of the images formed by plane mirrors. 2

OR

(b) List four properties of the images formed by convex mirrors when objects are placed anywhere in front of their reflecting surfaces.

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25. (a) List four properties of the images formed by plane mirrors. 2

OR

(b) List four properties of the images formed by convex mirrors when objects are placed anywhere in front of their reflecting surfaces.

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(a) Properties of the images formed by plane mirror: • Virtual • Erect • Same size as the object • behind the mirror at the same distance • laterally inverted (Any Four)	1/2 x4
(b) • Virtual • Erect • Diminished • Formed behind the mirror	1/2 x4

16. In order to obtain large images of the teeth of patients, the dentist holds the concave mirror in such a manner that the teeth are positioned
- (A) at the focus of mirror.
 - (B) between pole and focus of the mirror.
 - (C) between focus and centre of curvature of the mirror.
 - (D) at the centre of curvature of the mirror.

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16. In order to obtain large images of the teeth of patients, the dentist holds the concave mirror in such a manner that the teeth are positioned
- (A) at the focus of mirror.
 - (B) between pole and focus of the mirror.
 - (C) between focus and centre of curvature of the mirror.
 - (D) at the centre of curvature of the mirror.

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(B) / Between pole and focus of the mirror

21. During an experiment, a student observes that the given mirror is forming an erect and diminished image for all positions of the object placed in front of it.

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Based on the this information, answer the following questions :

- (a) Identify the mirror.
- (b) Using new Cartesian sign convention, write the magnitude of the magnification in this case.
- (c) What will be the radius of curvature of the mirror if its focal length is 20 cm ?

21. During an experiment, a student observes that the given mirror is forming an erect and diminished image for all positions of the object placed in front of it.

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Based on the this information, answer the following questions :

- (a) Identify the mirror.
- (b) Using new Cartesian sign convention, write the magnitude of the magnification in this case.
- (c) What will be the radius of curvature of the mirror if its focal length is 20 cm ?

(a) Convex mirror

$\frac{1}{2}$

(b) Positive and less than 1

$\frac{1}{2}$ + $\frac{1}{2}$

(c) + 40 cm or 40 cm

$\frac{1}{2}$

13. The image formed by a concave mirror of focal length 50 cm is real and of magnification -1 . In this case the distance between the object from its own image is :

(A) 50 cm

(B) 100 cm

(C) 200 cm

(D) Zero

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13. The image formed by a concave mirror of focal length 50 cm is real and of magnification -1 . In this case the distance between the object from its own image is :

(A) 50 cm

(B) 100 cm

(C) 200 cm

(D) Zero

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- 22.** Draw a labelled ray diagram to show the formation of image due to a concave mirror when an object is placed between its pole and principal focus.

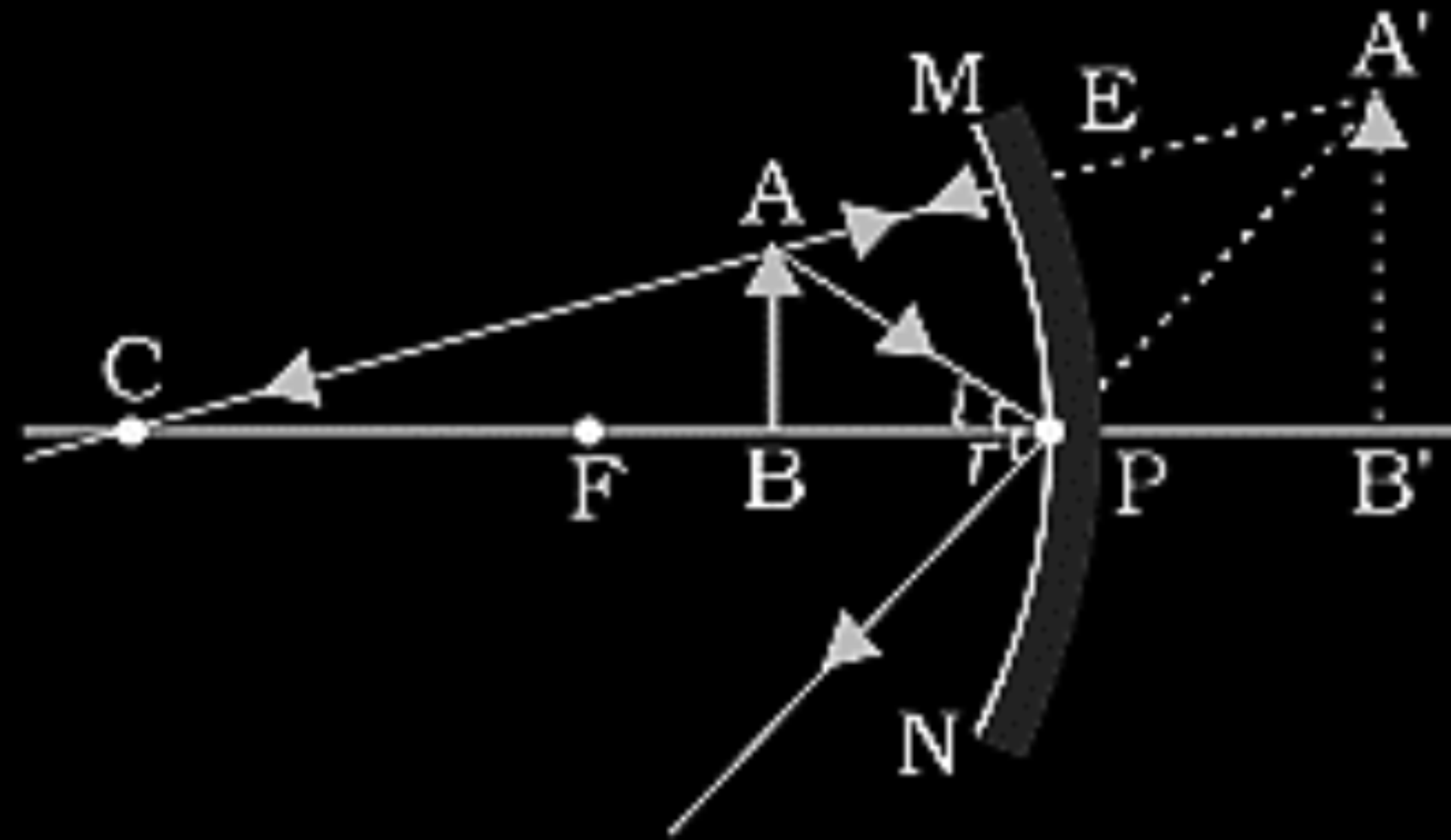
2

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22. Draw a labelled ray diagram to show the formation of image due to a concave mirror when an object is placed between its pole and principal focus.

2

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[P : Pole, F : Principal Focus, AB : Object, A'B' : Image]

[Diagram from NCERT fig 9.7(f) page 140]

Ray diagram

1

Labelling of pole and principal focus

$\frac{1}{2}$

Direction of rays

$\frac{1}{2}$

39. Under the guidance of his teacher on a sunny day, a student took a concave mirror of large aperture in his hand and directed its reflecting surface towards the sun. After that he directed the light reflected by the concave mirror on a thick sheet of white paper held close to the mirror. He observed a bright circular spot of light on the paper. Then he moved the mirror back and forth gradually until he got the brightest and smallest spot of light on the paper. The teacher said this sharp spot of light is, in fact, the real and inverted image of the sun and the distance between the mirror and this image is the focal length of this concave mirror.

- (a) Define the term principal focus of a concave mirror. 1
- (b) If the distance between the pole and focus of a concave mirror is 12 cm, what is the radius of curvature of the mirror ? 1

39. Under the guidance of his teacher on a sunny day, a student took a concave mirror of large aperture in his hand and directed its reflecting surface towards the sun. After that he directed the light reflected by the concave mirror on a thick sheet of white paper held close to the mirror. He observed a bright circular spot of light on the paper. Then he moved the mirror back and forth gradually until he got the brightest and smallest spot of light on the paper. The teacher said this sharp spot of light is, in fact, the real and inverted image of the sun and the distance between the mirror and this image is the focal length of this concave mirror.

- (a) Define the term principal focus of a concave mirror. 1
- (b) If the distance between the pole and focus of a concave mirror is 12 cm, what is the radius of curvature of the mirror ? 1

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(a) It is a point on the principal axis of the mirror where rays incident, parallel to principal axis will converge after reflection.	1
(b) $R = 2f = 2 \times 12 \text{ cm} = 24 \text{ cm}$	1

- (c) (i) List two properties of the image formed due to a concave mirror of focal length 15 cm in each of the following cases :

2

(1) object distance 20 cm

(2) object distance 10 cm

OR

- (c) (ii) A 3 cm long candle flame is placed in front of a concave mirror. If the distance between the pole and the candle flame is 40 cm and the image is formed at the same place where the candle flame is located, find the

2

- (c) (i) List two properties of the image formed due to a concave mirror of focal length 15 cm in each of the following cases :

2

(1) object distance 20 cm

(2) object distance 10 cm

OR

- (c) (ii) A 3 cm long candle flame is placed in front of a concave mirror. If the distance between the pole and the candle flame is 40 cm and the image is formed at the same place where the candle flame is located, find the

2

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(c) (i)

(1) Image is

- real
- inverted and
- magnified

(Any two)

(3) Image is

- virtual
- erect and
- magnified

(Any two)

(c)

(ii)

(1) In this case, the object is at C, therefore $R = 40$ cm

$$R = 2f$$

$$\therefore \text{Focal length } (f) = \frac{40}{2} \text{ cm} = 20 \text{ cm}$$

(2) Magnification(m) = -1 (Image of same size and inverted)

$\frac{1}{2}$

$\frac{1}{2}$

1

24. An object is placed at a distance of 10 cm from a convex mirror of focal length 15 cm. Find the position of the image formed by the mirror.

2

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24. An object is placed at a distance of 10 cm from a convex mirror of focal length 15 cm. Find the position of the image formed by the mirror.

2

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$$u = -10 \text{ cm}; f = +15 \text{ cm}$$

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$\frac{1}{15} = \frac{1}{v} + \frac{1}{-10 \text{ cm}}$$

$$\frac{1}{v} = \frac{1}{15 \text{ cm}} + \frac{1}{10 \text{ cm}}$$

$$v = +6 \text{ cm}$$

Image is formed behind the mirror.

$\frac{1}{2}$

$\frac{1}{2}$

1

39. Study the data given below showing the focal length of three concave mirrors A, B and C and the respective distances of objects placed in front of the mirrors :

Case	Mirror	Focal Length (cm)	Object Distance (cm)
1	A	20	45
2	B	15	30
3	C	30	20

- (i) In which one of the above cases the mirror will form a diminished image of the object ? Justify your answer. 1
- (ii) List two properties of the image formed in case 2. 1
- (iii) (A) What is the nature and size of the image formed by mirror C ? Draw ray diagram to justify your answer. 2

OR

- (iii) (B) An object is placed at a distance of 18 cm from the pole of a concave mirror of focal length 12 cm. Find the position of the image formed in this case. 2

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(i)

- Mirror A.
- as the object is placed beyond the centre of curvature of the mirror.

$\frac{1}{2}$

$\frac{1}{2}$

(ii) Same size/ Real / Inverted

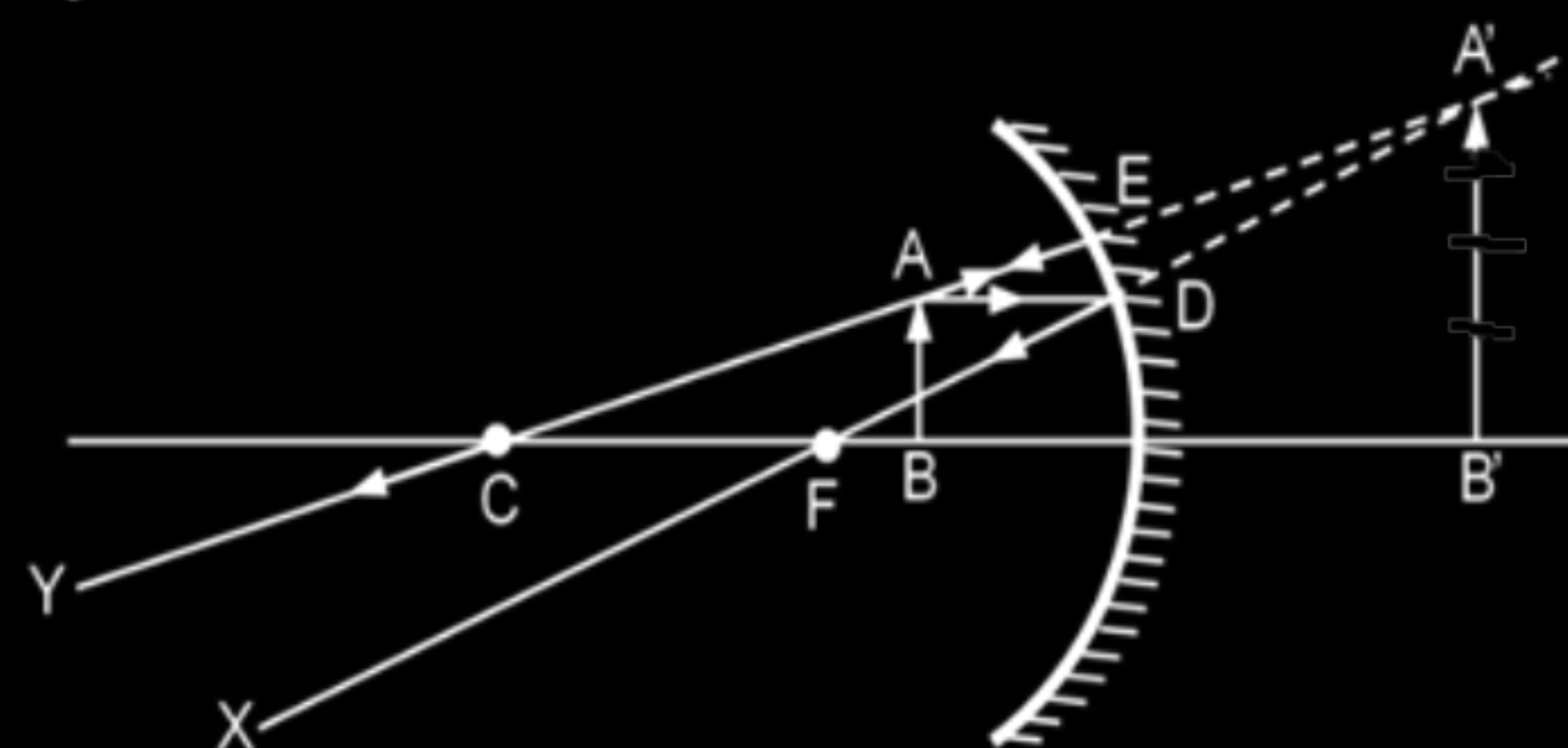
$\frac{1}{2} + \frac{1}{2}$

(Any two)

(iii) (A) Nature-Virtual and erect
Size-magnified

$\frac{1}{2}$

$\frac{1}{2}$



1

(Deduct $\frac{1}{2}$ mark if direction of rays are not marked)

$\frac{1}{2}$

(iii) (B) Here $f = -12$ cm, $u = -18$ cm, $v = ?$

Mirror formula $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$ or $\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$

$$\frac{1}{v} = \frac{1}{-12} - \frac{1}{-18}$$

$\frac{1}{2}$

1

$$v = -36\text{cm}$$

In front of the mirror at a distance of 36 cm from the pole of the mirror.

31. (a) An object is placed in front of a concave mirror of focal length 12 cm. If distance of the object from the pole of the mirror is 8 cm, then use mirror formula to determine the position of the image formed. Draw a labelled ray diagram to justify your answer in this case.

3

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31. (a) An object is placed in front of a concave mirror of focal length 12 cm. If distance of the object from the pole of the mirror is 8 cm, then use mirror formula to determine the position of the image formed. Draw a labelled ray diagram to justify your answer in this case.

3

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(a)

Here $f = -12$ cm, $u = -8$ cm, $v = ?$

Mirror formula $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

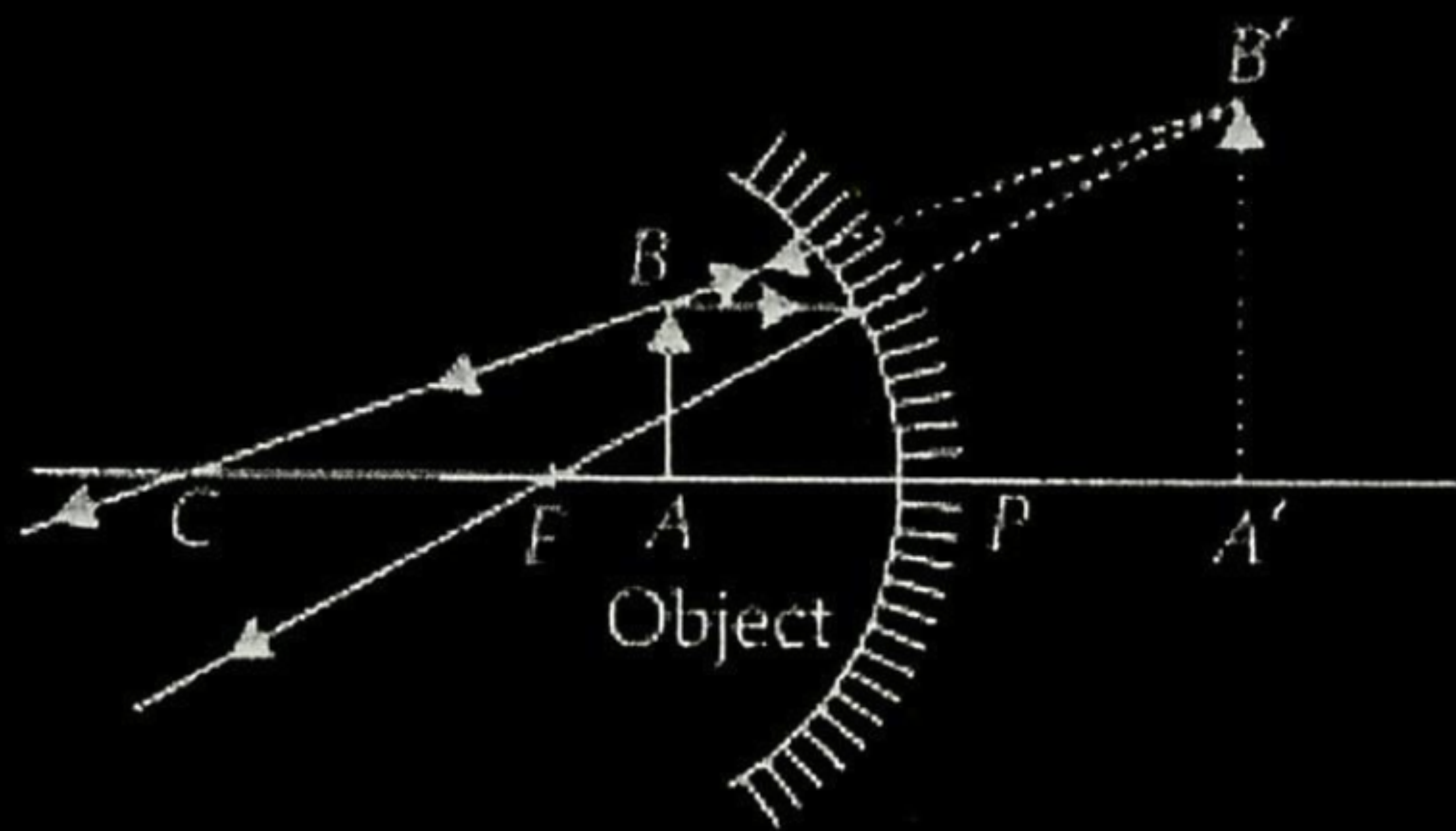
$$\therefore \frac{1}{v} = \frac{1}{f} - \frac{1}{u} = \frac{1}{-12 \text{ cm}} - \frac{1}{-8 \text{ cm}}$$

$$= + \frac{1}{24 \text{ cm}}$$

$$v = +24 \text{ cm}$$

Image is formed at 24 cm behind the mirror.

1



Note: Deduct $\frac{1}{2}$ mark if direction of rays is not shown.

31. (b) (i) The image of an object formed by a mirror is real, inverted and is of magnification-1. If the image is at a distance of 30 cm from the mirror, where is the object placed ? Give reason to justify your answer.
- (ii) Where would the image be if the object is moved 15 cm towards the mirror ? Draw ray diagram for the new position of the object to justify your answer.

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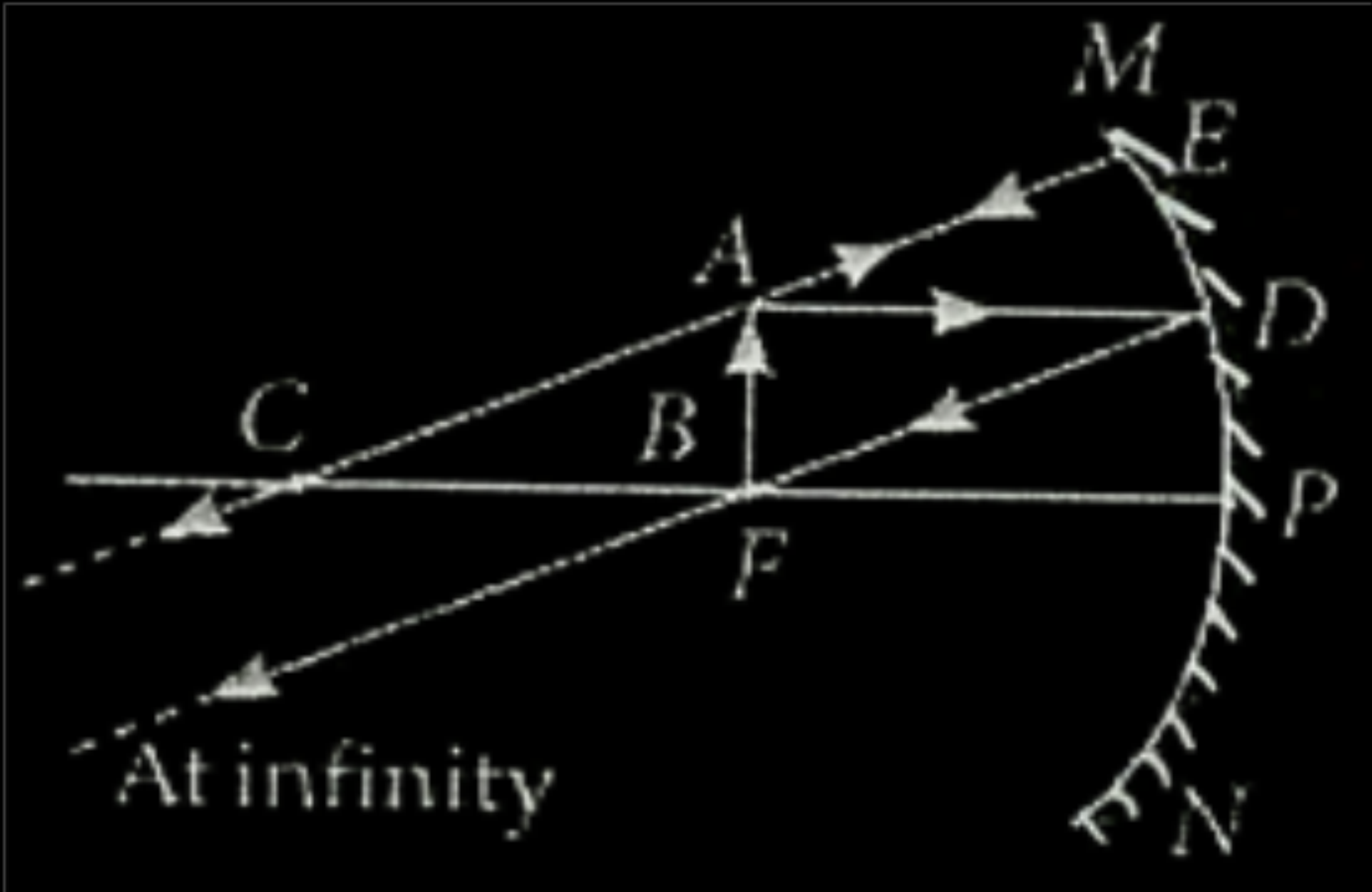
31. (b) (i) The image of an object formed by a mirror is real, inverted and is of magnification-1. If the image is at a distance of 30 cm from the mirror, where is the object placed ? Give reason to justify your answer.
- (ii) Where would the image be if the object is moved 15 cm towards the mirror ? Draw ray diagram for the new position of the object to justify your answer.

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(b)(i) Object is placed at 30 cm from the mirror.
Note: Award marks if expressed in the form of calculations.

Reason: Since magnification is -1, Image distance(v)= object distance(u)

(ii) Image is formed at infinity.



Note: Deduct ½ mark if direction of rays is not shown

36. (b) (i) Draw a ray diagram to show the path of the reflected ray in each of the following cases :

A ray of light incident on a convex mirror

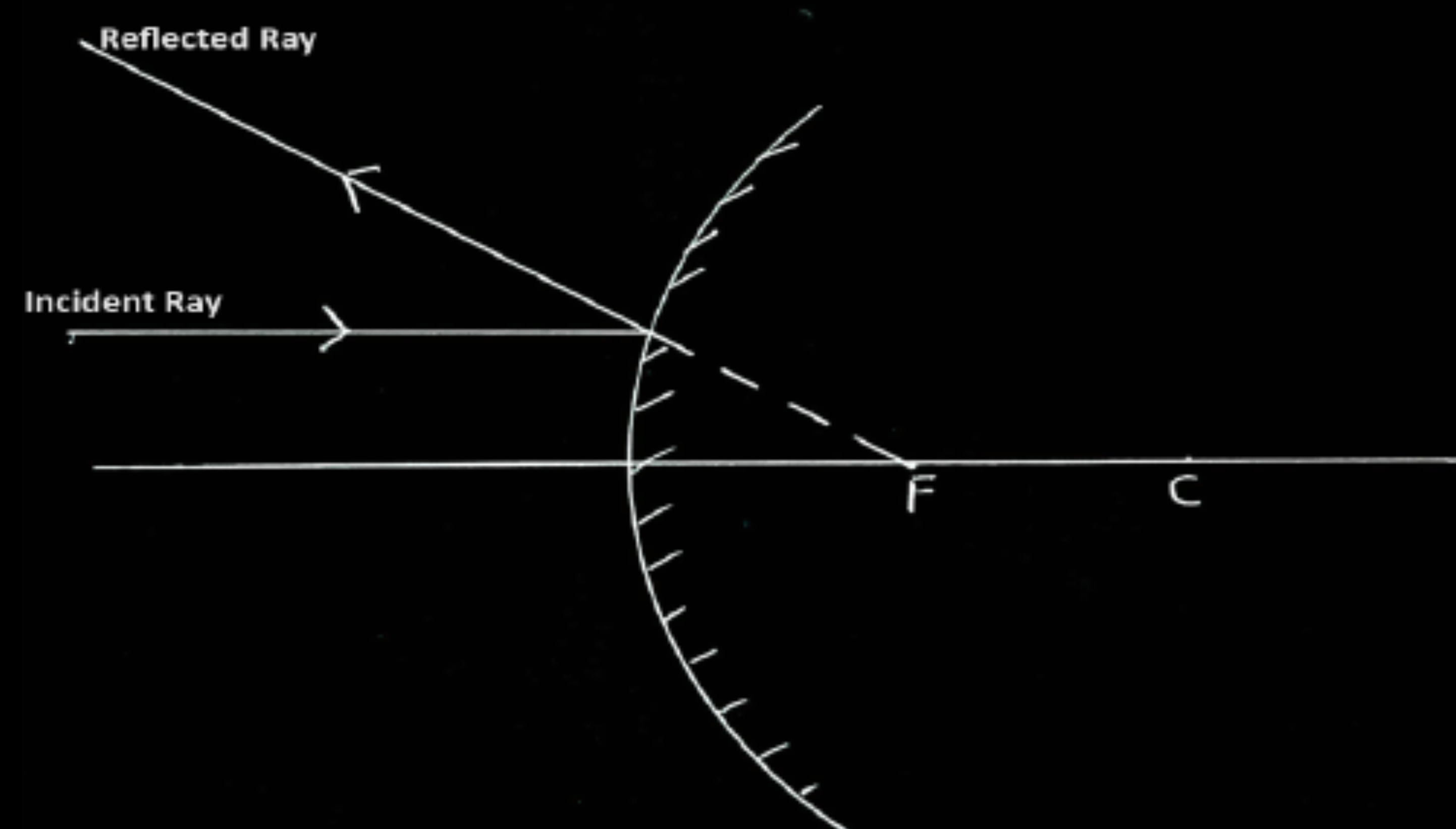
- (1) parallel to its principal axis, and
- (2) is directed towards its principal focus

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- (ii) A 1.5 cm tall candle flame is placed perpendicular to the principal axis of a concave mirror of focal length 12 cm. If the distance of the flame from the pole of the mirror is 18 cm, use mirror formula to determine the position and size of the image formed.

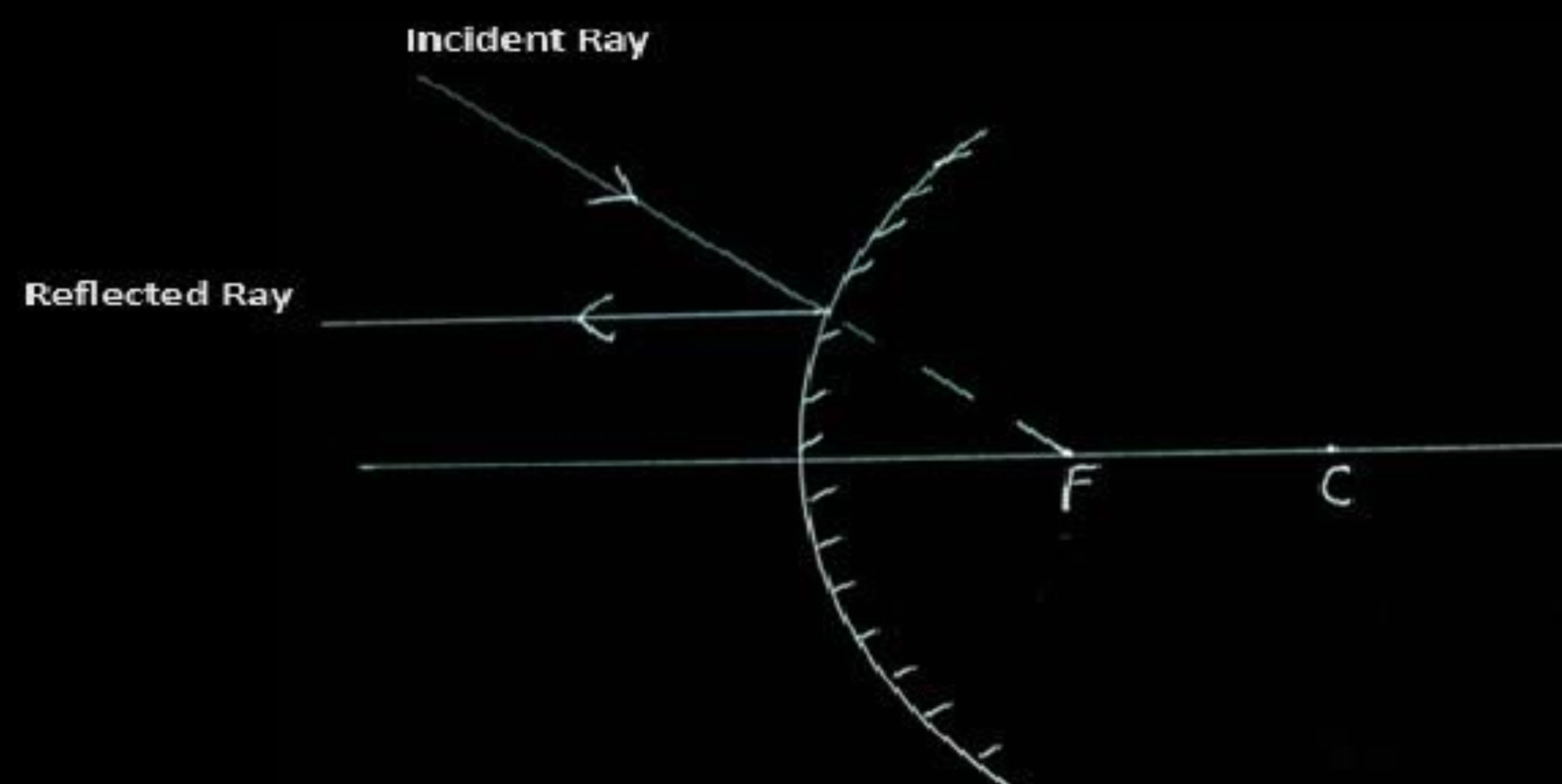
(b) (i)

(1)



1

(2)



1

(Note:- Deduct half mark if directions of rays are not shown)

(ii) Here $f = -12$ cm, $u = -18$ cm, $v = ?$, $h = 1.5$ cm, $h' = ?$

Mirror formula $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

$$\therefore \frac{1}{v} = \frac{1}{f} - \frac{1}{u}$$

$$= \frac{1}{-12 \text{ cm}} - \frac{1}{-18 \text{ cm}}$$

$$= \frac{-1}{36}$$

$$\therefore v = -36 \text{ cm}$$

$$m = \frac{h'}{h} = -\frac{v}{u}$$

$$\frac{h'}{1.5} = -\frac{(-36)}{(-18)}$$

$$h' = -3.0 \text{ cm}$$

$\frac{1}{2}$

$\frac{1}{2}$

1

$\frac{1}{2}$

$\frac{1}{2}$

39. A highly polished surface such as a mirror reflects most of the light falling on it. In our daily life we use two types of mirrors — plane and spherical. The reflecting surface of a spherical mirrors may be curved inwards or outwards. In concave mirrors, reflection takes place from the inner surface, while in convex mirrors reflection takes place from the outer surface.

- (a) Define the principal axis of a concave mirror. 1
- (b) A ray of light is incident on a concave mirror, parallel to its principal axis. If this ray after reflection from the mirror passes through the principal axis from a point at a distance of 10 cm from the pole of the mirror, find the radius of curvature of the mirror. 1
- (c) (i) An object is placed at a distance of 10 cm from the pole of a convex mirror of focal length 15 cm. Find the position of the image. 2

OR

- (c) (ii) A mirror forms a virtual, erect and diminished image of an object. Identify the type of this mirror. Draw a ray diagram to show the image formation in this case.

(a) It is straight line passing through the pole and centre of curvature of a concave mirror.

1

(b) Radius of curvature ,R= 20 cm

1

(c)

(i) $u = -10$ cm, $f = +15$ cm

$\frac{1}{2}$

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$\frac{1}{2}$

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u} = \frac{1}{15} - \frac{1}{-10}$$

$$\frac{1}{v} = \frac{1}{6}$$

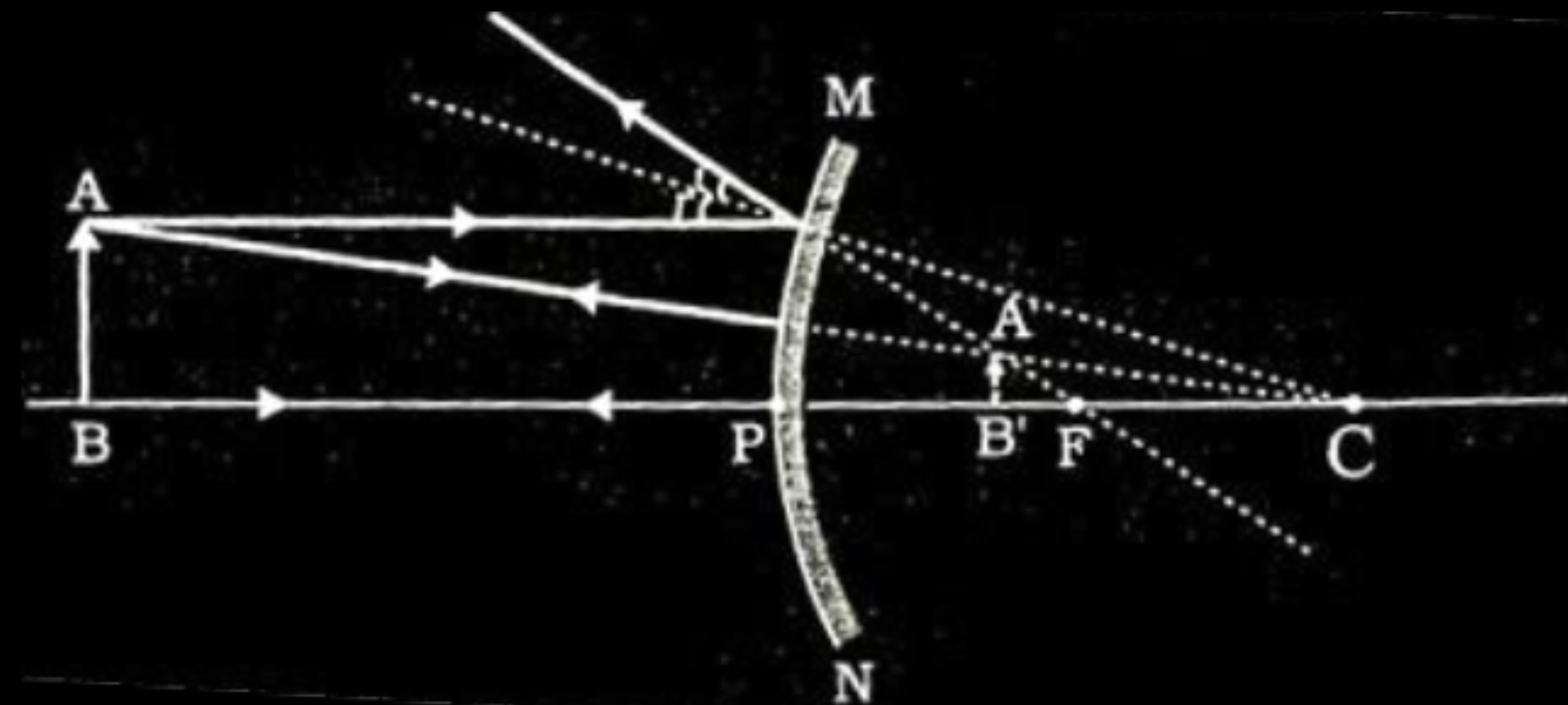
$$\Rightarrow v = +6 \text{ cm}$$

1

OR

(c) (ii) Convex mirror / Diverging mirror

$\frac{1}{2}$



1 $\frac{1}{2}$

[Note: Deduct $\frac{1}{2}$ mark if direction of rays is not shown]

25. “The linear magnification produced by a spherical mirror is $+3$.” Based on this statement answer the following questions :

2

- (a) What is the type of mirror ?
- (b) Where is the object located ?
- (c) List two properties of the image formed (other than the size/magnification).

25. “The linear magnification produced by a spherical mirror is + 3.” Based on this statement answer the following questions :

2

- (a) What is the type of mirror ?
- (b) Where is the object located ?
- (c) List two properties of the image formed (other than the size/magnification).

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(a) Concave mirror/ Converging mirror	$\frac{1}{2}$
(b) Between pole and focus	$\frac{1}{2}$
(c) • Virtual • Erect • Behind the mirror (Any two)	$\frac{1}{2} + \frac{1}{2}$

- 36.** (a) (i) State laws of reflection of light.
- (ii) An object of height 5.0 cm is placed at 15 cm in front of a concave mirror of focal length 10 cm. At what distance from the mirror should a screen be placed, so that a focussed image is obtained on it ? Find the height of the image.

36. (a) (i) State laws of reflection of light.

(ii) An object of height 5.0 cm is placed at 15 cm in front of a concave mirror of focal length 10 cm. At what distance from the mirror should a screen be placed, so that a focussed image is obtained on it ? Find the height of the image.

5

(a) (i)

- The angle of incidence is equal to the angle of reflection.
- The incident ray, the normal to the mirror at the point of incidence and the reflected ray, all lie in the same plane.

(ii) $u = -15$ cm, $f = -10$ cm (concave mirror)
 $h = 5.0$ cm

Mirror formula $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$

$$\frac{1}{v} = \frac{-1}{10 \text{ cm}} + \frac{1}{15 \text{ cm}} = \frac{-1}{30 \text{ cm}}$$

or $v = -30$ cm. The screen must be placed at a distance of 30 cm from the mirror in front of it

$$(m) = \frac{h'}{h} = -\frac{v}{u}$$

$$h = \frac{-v}{u} \times h = -\frac{-30 \text{ cm}}{-15 \text{ cm}} \times 5 \text{ cm} = -10 \text{ cm}$$

1+1

 $\frac{1}{2}$ $\frac{1}{2}$

1

1

2. For an object, a convex mirror always forms an image which is :

- (A) Virtual, erect and magnified
- (B) Virtual, erect and diminished
- (C) Virtual, inverted and diminished
- (D) Real, erect and diminished

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2. For an object, a convex mirror always forms an image which is :

- (A) Virtual, erect and magnified
- (B) Virtual, erect and diminished
- (C) Virtual, inverted and diminished
- (D) Real, erect and diminished

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(B)/ Virtual, erect and diminished

21. (a) List four properties of the images formed by a plane mirror.

2

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21. (a) List four properties of the images formed by a plane mirror.

2

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(a) Four properties:

- Virtual
- Erect
- Size of the image = Size of the object
- Image distance from the mirror = Object distance from the mirror
- Laterally inverted

(Any four)

$\frac{1}{2} \times 4$

(b) Write the definitions of the following terms in the context of a concave mirror :

2

- (i) Pole
- (ii) Centre of curvature
- (iii) Radius of curvature
- (iv) Principal axis

(b) Write the definitions of the following terms in the context of a concave mirror :

2

- (i) Pole
- (ii) Centre of curvature
- (iii) Radius of curvature
- (iv) Principal axis

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(b)		
(i)	Pole: Central point of the reflecting surface of a spherical mirror.	
(ii)	Centre of curvature: Centre of the sphere of which the reflecting surface of the spherical mirror forms a part.	
(iii)	Radius of curvature: Radius of the sphere of which the reflecting surface of the spherical mirror forms a part.	
(iv)	Principal axis: Straight line joining the pole and the centre of curvature of a spherical mirror	$\frac{1}{2} \times 4$

- 24.** (a) Find the magnification of the image formed by a spherical mirror from the following data :
 $u = -20 \text{ cm}$, $f = -15 \text{ cm}$.

CBSE 2023 C

24. (a) Find the magnification of the image formed by a spherical mirror from the following data :
 $u = -20$ cm, $f = -15$ cm.

CBSE 2023 C

2

(a)	$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$ $\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$ $= \frac{1}{-15} - \frac{1}{-20} = \frac{-1}{60}$ $v = -60 \text{ cm}$ $m = \frac{-v}{u}$ $= -\frac{-60}{-20} = -3$
-----	---

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

- (b) Draw a labelled ray diagram for the image formation by a concave mirror when an object is placed between its centre of curvature and focus.

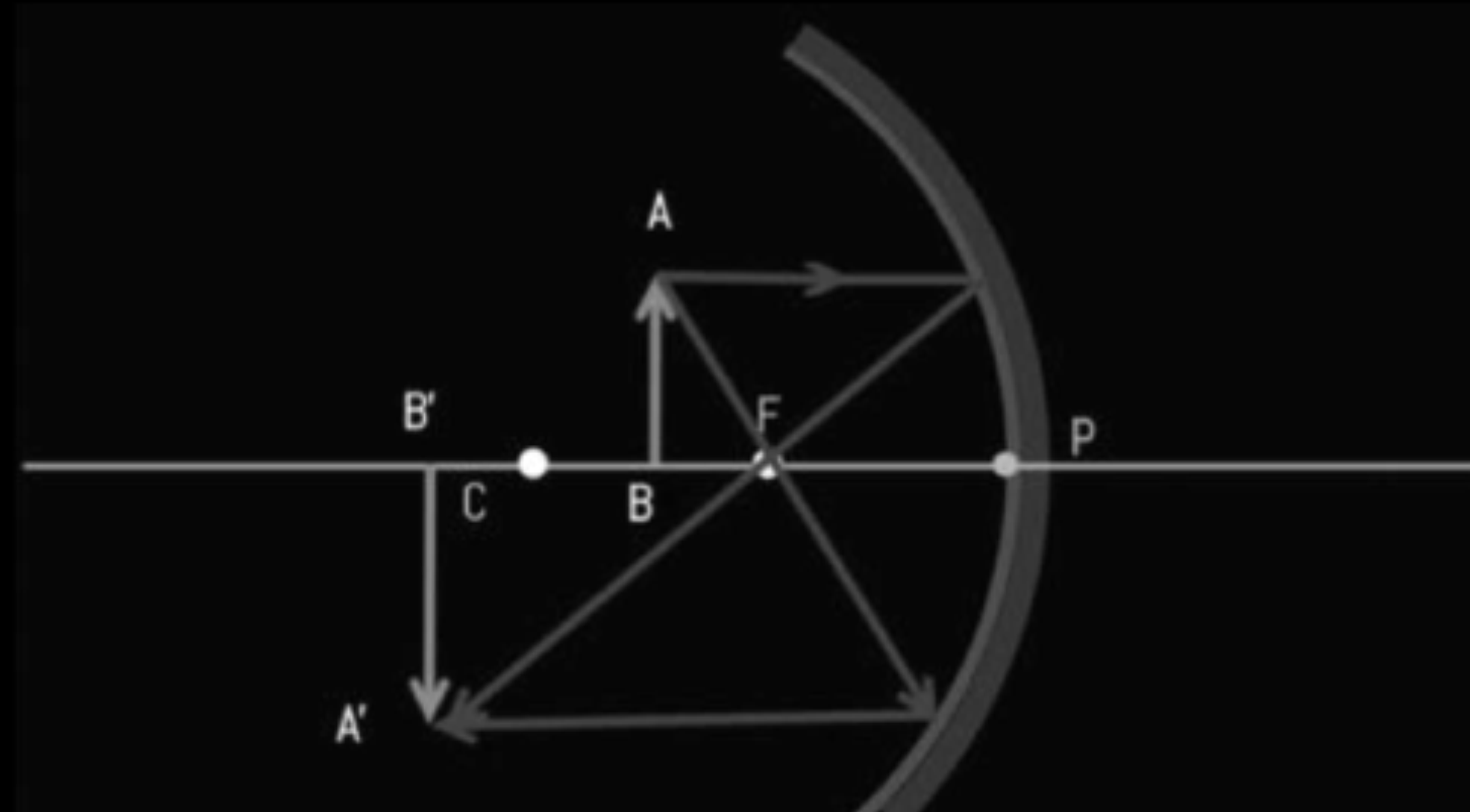
CBSE 2023 C

- (b) Draw a labelled ray diagram for the image formation by a concave mirror when an object is placed between its centre of curvature and focus.

CBSE 2023 C

2

(b)



Diagram

1½

Direction of the rays

½

31. (a) Define the following terms in the context of spherical mirrors :

(i) Pole

(ii) Centre of curvature

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(iii) Principal axis

(iv) Principal focus

(b) An object is placed at a distance of 60 cm from the pole of a concave mirror. If its real and inverted image is formed on a screen placed at a distance of 60 cm in front of the mirror, what is the focal length of the mirror ? Give justification for your answer.

31. (a) Define the following terms in the context of spherical mirrors :

- (i) Pole
- (ii) Centre of curvature
- (iii) Principal axis
- (iv) Principal focus

CBSE 2023 C

(b) An object is placed at a distance of 60 cm from the pole of a concave mirror. If its real and inverted image is formed on a screen placed at a distance of 60 cm in front of the mirror, what is the focal length of the mirror ? Give justification for your answer.

3

(a) (i) Centre of the reflecting surface of a spherical mirror (ii) Centre of the sphere of which mirror was a part (iii) Straight line passing through pole and centre of curvature of spherical mirror (iv) Point on principal axis where the reflected rays meet / appear to meet. (b) Focal length = $\frac{-60 \text{ cm}}{2} = -30 \text{ cm}$ Because the object is placed at the centre of curvature.	4 x 1/2 1/2 1/2
--	--------------------------------------

29. (a) A student has focussed the image of an object of height 3 cm on a white screen using a concave mirror of focal length 12 cm. If the distance of the object from the mirror is 18 cm, find the values of the following :

- (i) Distance of the image from the mirror
- (ii) Height of the image

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29. (a) A student has focussed the image of an object of height 3 cm on a white screen using a concave mirror of focal length 12 cm. If the distance of the object from the mirror is 18 cm, find the values of the following :

- (i) Distance of the image from the mirror
(ii) Height of the image

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3

(a) $h_1 = 3 \text{ cm}$
 $f = -12 \text{ cm}$
 $u = -18 \text{ cm}$

(i) Image Distance

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$\frac{1}{-12} = \frac{1}{v} + \frac{1}{-18}$$

$$\frac{1}{-12} + \frac{1}{18} = \frac{1}{v}$$

$$\frac{1}{v} = \frac{-3+2}{36} = \frac{-1}{36}$$

$$v = -36 \text{ cm}$$

(ii) Height of image

$$m = \frac{h_2}{h_1} = \frac{-v}{u}$$

$$\frac{h_2}{3} = \frac{-[-36]}{-18}$$

$$h_2 = \frac{-36}{18} \times 3$$

$$h_2 = -6 \text{ cm}$$

13. An object is placed in front of a convex mirror at infinity. According to the New Cartesian Sign Convention, the sign of the focal length and the sign of the image distance in this case are respectively :

(a) $+$, $-$

(b) $-$, $+$

(c) $-$, $-$

(d) $+$, $+$

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13. An object is placed in front of a convex mirror at infinity. According to the New Cartesian Sign Convention, the sign of the focal length and the sign of the image distance in this case are respectively :

(a) $+$, $-$

(b) $-$, $+$

(c) $-$, $-$

(d) $+$, $+$

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(d)

31. (a) A student has focussed the image of an object of height 3 cm on a white screen using a concave mirror of focal length 12 cm. If the distance of the object from the mirror is 18 cm, find the values of the following :

(i) Distance of the image from the mirror

(ii) Height of the image

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31. (a) A student has focussed the image of an object of height 3 cm on a white screen using a concave mirror of focal length 12 cm. If the distance of the object from the mirror is 18 cm, find the values of the following :

(i) Distance of the image from the mirror

(ii) Height of the image

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3

(a) Here $h = 3$ cm; $f = -12$ cm, $u = -18$ cm,

(Award full marks if data not written but calculations are correct)

(i)

$$v = ?, h' = ?$$

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u} = \frac{1}{-12 \text{ cm}} - \frac{1}{-18 \text{ cm}}$$

$$\therefore v = -36 \text{ cm}$$

(ii)

$$h' = -\frac{v}{u} \times h$$

$$h' = (-) \frac{-36 \text{ cm}}{-18 \text{ cm}} \times 3 \text{ cm} = -6 \text{ cm}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

- 36.** (a) Name the type of mirror that should be used to obtain the following types of images :
- (i) A magnified and virtual image of an object
 - (ii) A diminished and virtual image of an object
- (b) Draw labelled ray diagrams to justify your answers in each case mentioned above.
- (c) Which of these mirrors can also be used to obtain a magnified and real image of an object ? State the position of the object in this case.

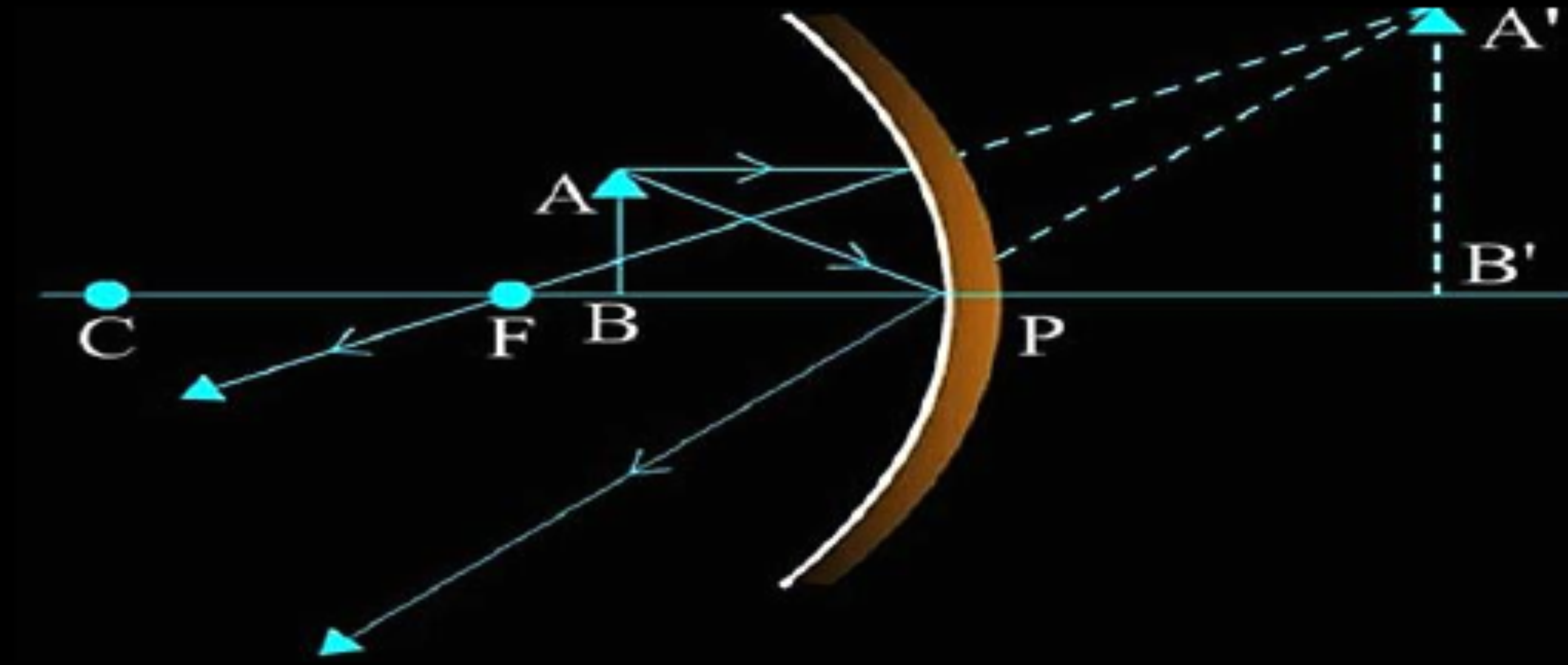
(a) (i) Concave mirror / Converging mirror

1

(ii) Convex mirror / Diverging mirror

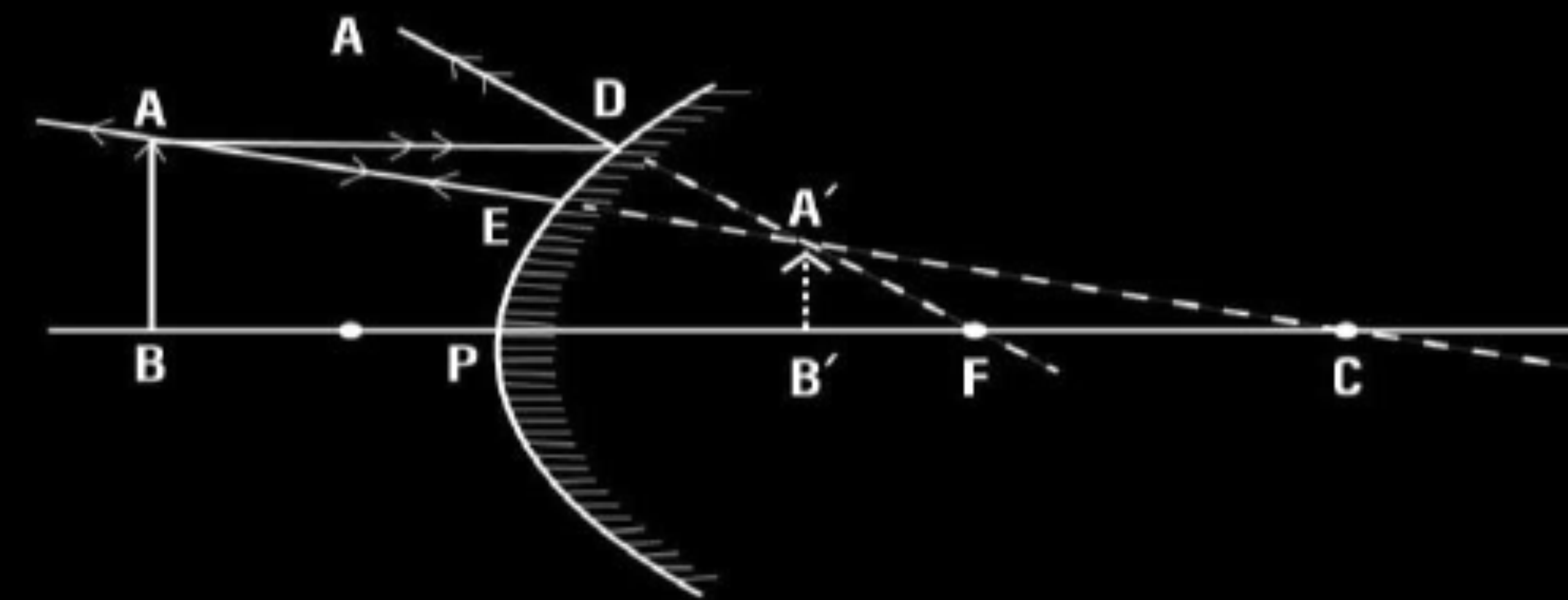
1

(b) (i)



1

(ii)



1

(Overall deduct $\frac{1}{2}$ mark for not drawing the arrows)

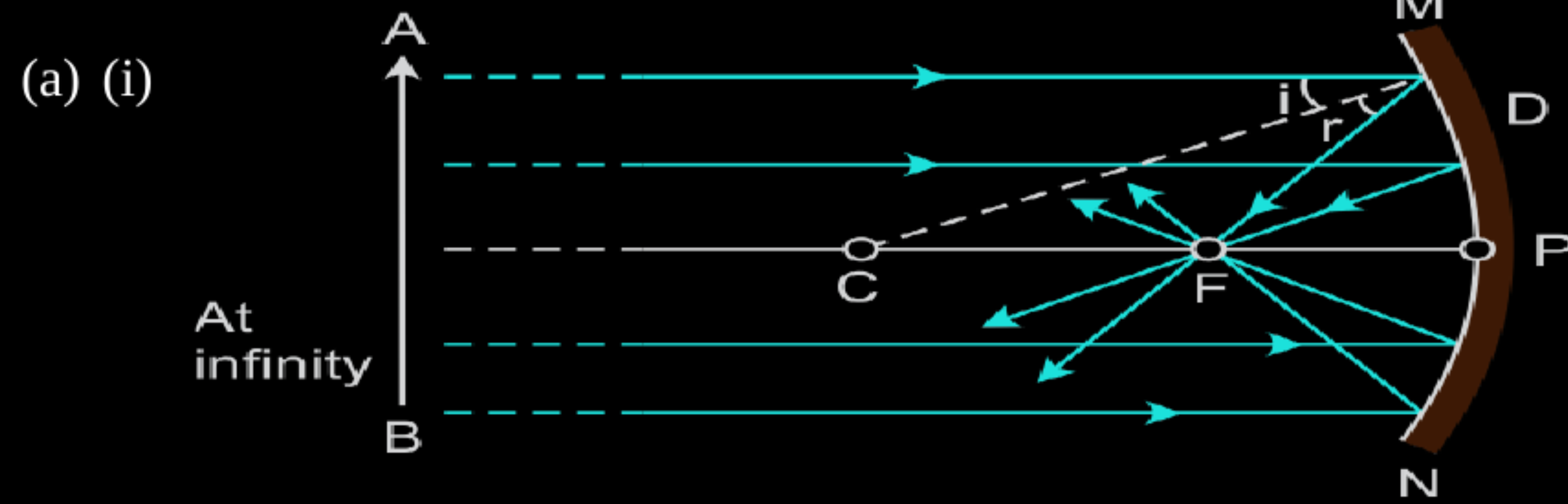
(c) • Concave mirror / Converging mirror

$\frac{1}{2}$

• Between focus and centre of curvature of the mirror / between F & C

$\frac{1}{2}$

- 36.** (a) Draw a ray diagram for showing the convergence of a parallel beam of light by a concave mirror and mark the positions of pole and the centre of curvature in it.
- (b) An object 4 cm in size is placed at 25 cm in front of a concave mirror of focal length 15 cm. At what distance from the mirror should a screen be placed in order to obtain a sharp image ? Find the nature and size of the image.
- (c) List any two uses of a concave mirror.



(P, F, C should be labelled correctly.)

(b) $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

$$\Rightarrow \frac{1}{v} = \frac{1}{f} - \frac{1}{u} = \frac{1}{-15} - \frac{1}{-25} = -\frac{1}{15} + \frac{1}{25} = \frac{-2}{75}$$

or $v = -37.5 \text{ cm}$

The screen should be placed at 37.5 cm from the mirror. The image is real and inverted.

Also, magnification, $m = \frac{h'}{h} = -\frac{v}{u}$

$$h' = \frac{vh}{u} = -\frac{(-37.5 \text{ cm})(+4.0 \text{ cm})}{(-25.0 \text{ cm})}$$

Height of the image = $h' = -6.0 \text{ cm}$. The image is inverted and enlarged.

(c) Torches, search lights

(or Any other)

1

$\frac{1}{2}$

$\frac{1}{2}$

1

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}, \frac{1}{2}$

30. A student wants to draw a ray diagram to show the formation of image due to a concave mirror when an object is placed in between its focus and centre of curvature.

- (a) Write the minimum number of incident rays required.
- (b) State the paths of the incident rays.
- (c) State the paths of the reflected rays.
- (d) Write the sign of magnification according to the New Cartesian Sign Convention.

30. A student wants to draw a ray diagram to show the formation of image due to a concave mirror when an object is placed in between its focus and centre of curvature.

- (a) Write the minimum number of incident rays required.
- (b) State the paths of the incident rays.
- (c) State the paths of the reflected rays.
- (d) Write the sign of magnification according to the New Cartesian Sign Convention.

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3

(a)	Two / 2	$\frac{1}{2}$
(b)	• Parallel to the principal axis • Passing through the principal focus • Ray passing through centre of curvature • Ray incident obliquely to principal axis. (any two points)	$\frac{1}{2}, \frac{1}{2}$
(c)	• After reflection passes through the principal focus. • After reflection becomes parallel to the principal axis. • After reflection, reflected back along same path. • Reflected obliquely (any two points)	$\frac{1}{2}, \frac{1}{2}$
(d)	negative	$\frac{1}{2}$

7. A student wants to obtain an erect image of an object using a concave mirror of 10 cm focal length. What will be the distance of the object from mirror ?

1

(a) Less than 10 cm

(b) 10 cm

(c) between 10 cm and 20 cm

(d) more than 20 cm

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7. A student wants to obtain an erect image of an object using a concave mirror of 10 cm focal length. What will be the distance of the object from mirror ?

1

(a) Less than 10 cm

(b) 10 cm

(c) between 10 cm and 20 cm

(d) more than 20 cm

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(a)

39. A student took three concave mirrors of different focal lengths and performed the experiment to see the image formation by placing an object at different distances with these mirrors as shown in the following table. 4

Case No.	Object-distance	Focal length
I	45 cm	20 cm
II	30 cm	15 cm
III	20 cm	30 cm

Now answer the following questions :

- (a) List two properties of the image formed in Case I.
- (b) In which one of the cases given in the table, the mirror will form real image of same size and why ?
- (c) Name the type of mirror used by dentists. Give reason why do they use such type of mirrors.

OR

- (c) Look at the table and identify the situation (object distance and focal length) which resembles the situation in which concave mirrors are used as shaving mirrors ? Draw a ray diagram to show the image formation in this case.

(a) Real, inverted, diminished (Any two)

$\frac{1}{2}$, $\frac{1}{2}$

(b) Case II

$\frac{1}{2}$

Because focal length of mirror is 15 cm, object distance is 30cm which means the object is placed at C.

$\frac{1}{2}$

(c) Dentists use concave mirrors

1

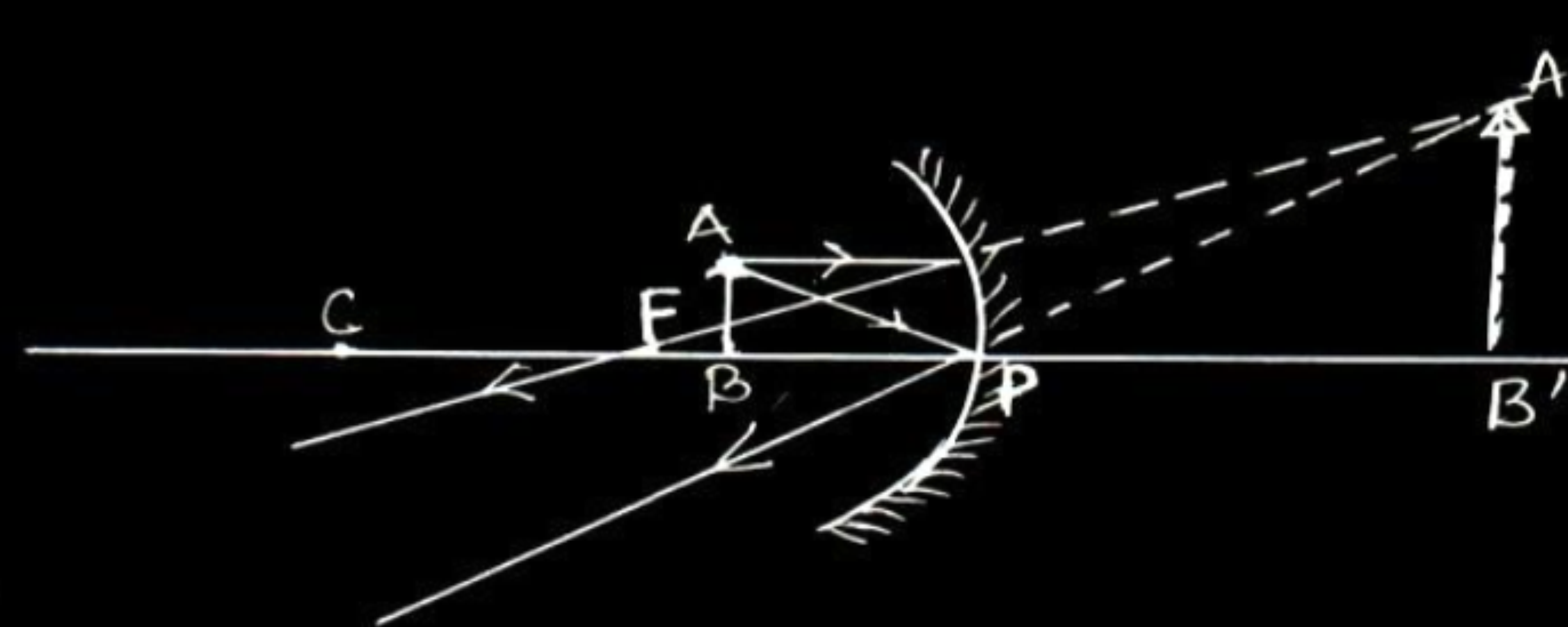
Because concave mirror forms erect and enlarged image when object is very close to the mirror.

1

OR

1

(c) Case III



1

(Deduct $\frac{1}{2}$ mark if direction of ray is not marked.)

7. To obtain a magnification of $+2$ with a concave mirror of radius of curvature 60 cm the object distance must be

1

(a) -90 cm

(b) -45 cm

(c) -30 cm

(d) -15 cm

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7. To obtain a magnification of + 2 with a concave mirror of radius of curvature 60 cm the object distance must be

1

(a) – 90 cm

(b) – 45 cm

(c) – 30 cm

(d) – 15 cm

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(d)

7. In torch lights and head lights of vehicles, the bulb is placed

1

- (a) between the pole and the focus of the reflector.
- (b) very near to the focus of the reflector.
- (c) between the focus and centre of curvature of the reflector.
- (d) at the centre of curvature of the reflector.

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7. In torch lights and head lights of vehicles, the bulb is placed

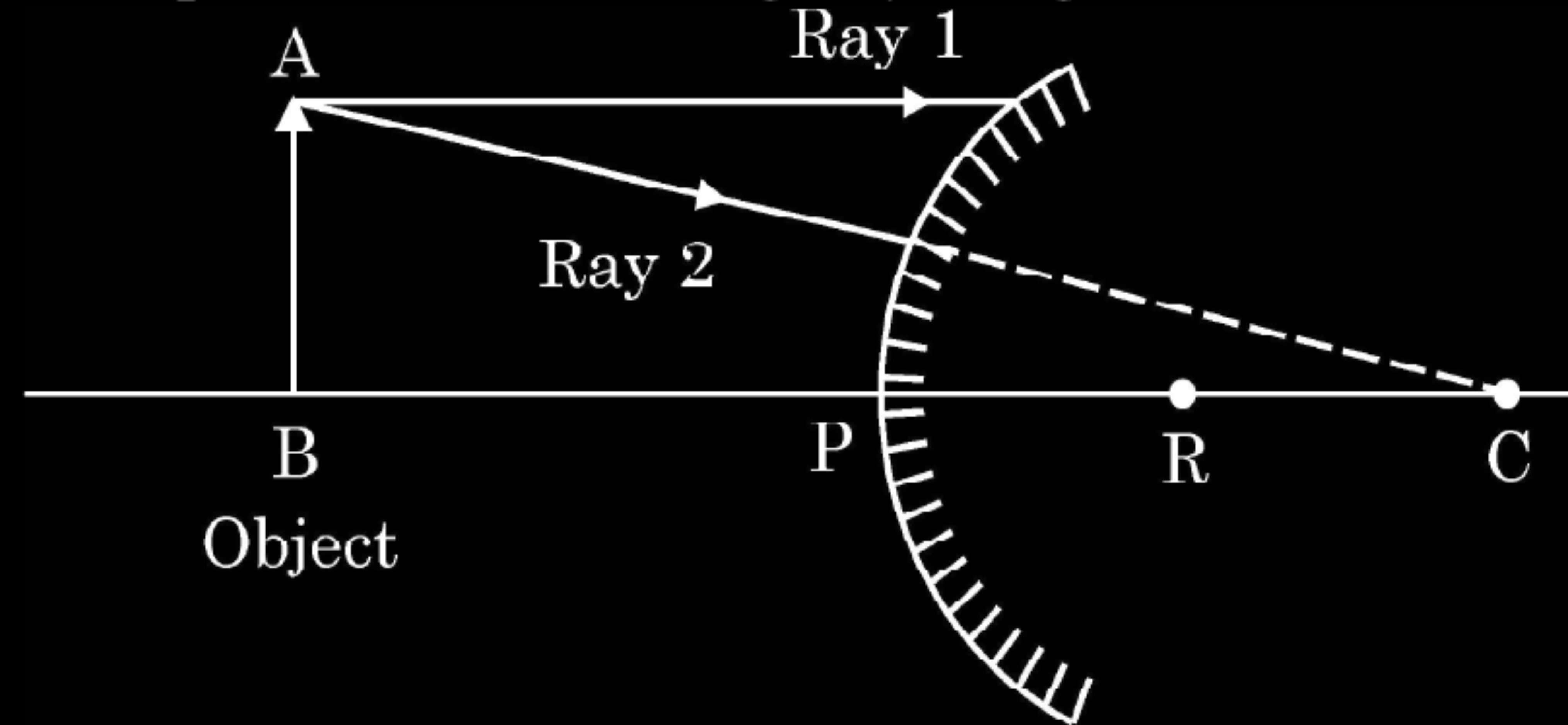
1

- (a) between the pole and the focus of the reflector.
- (b) very near to the focus of the reflector.
- (c) between the focus and centre of curvature of the reflector.
- (d) at the centre of curvature of the reflector.

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(b)

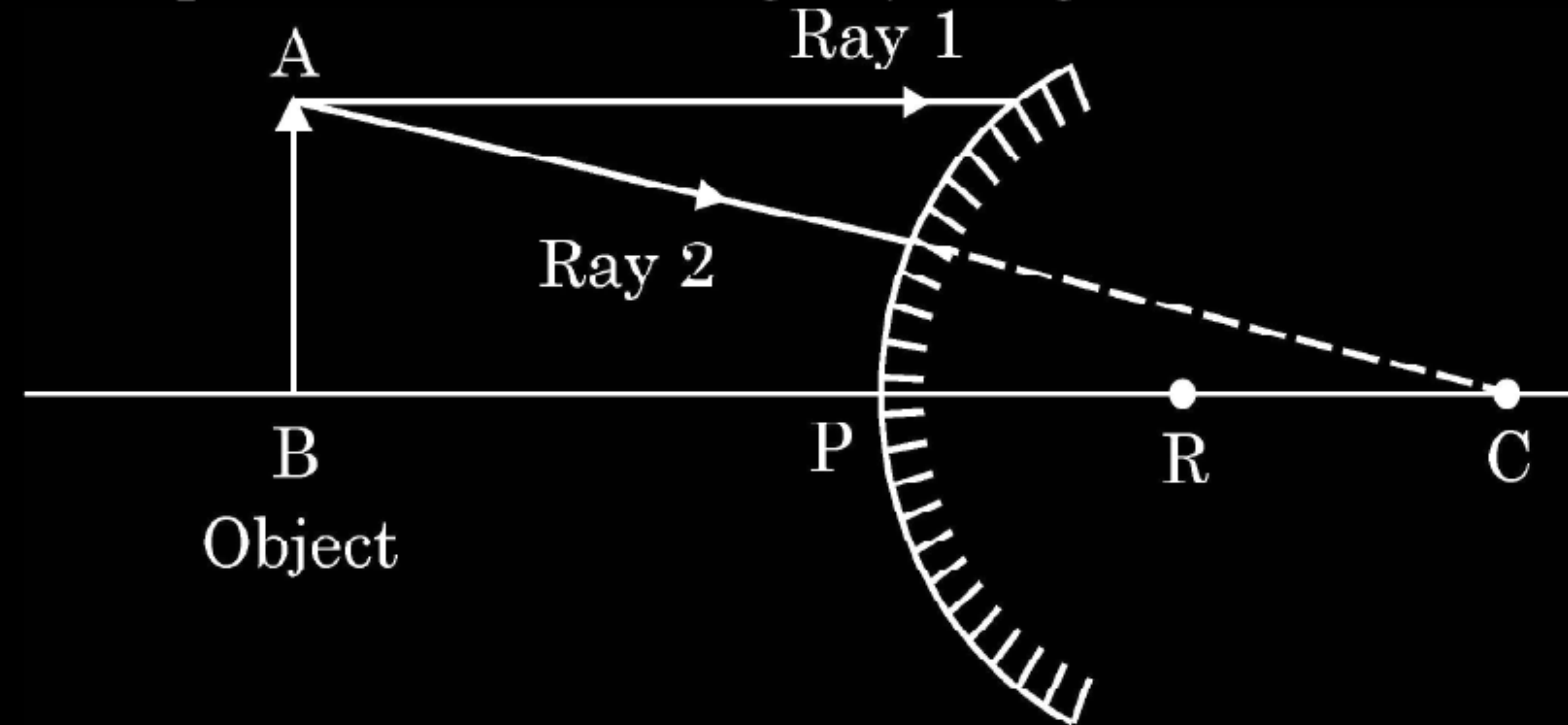
30. (a) Complete the following ray diagram to show the formation of image : 3



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- (b) Mention the nature, position and size of the image formed in this case.
- (c) State the sign of the image distance in this case using the Cartesian sign convention.

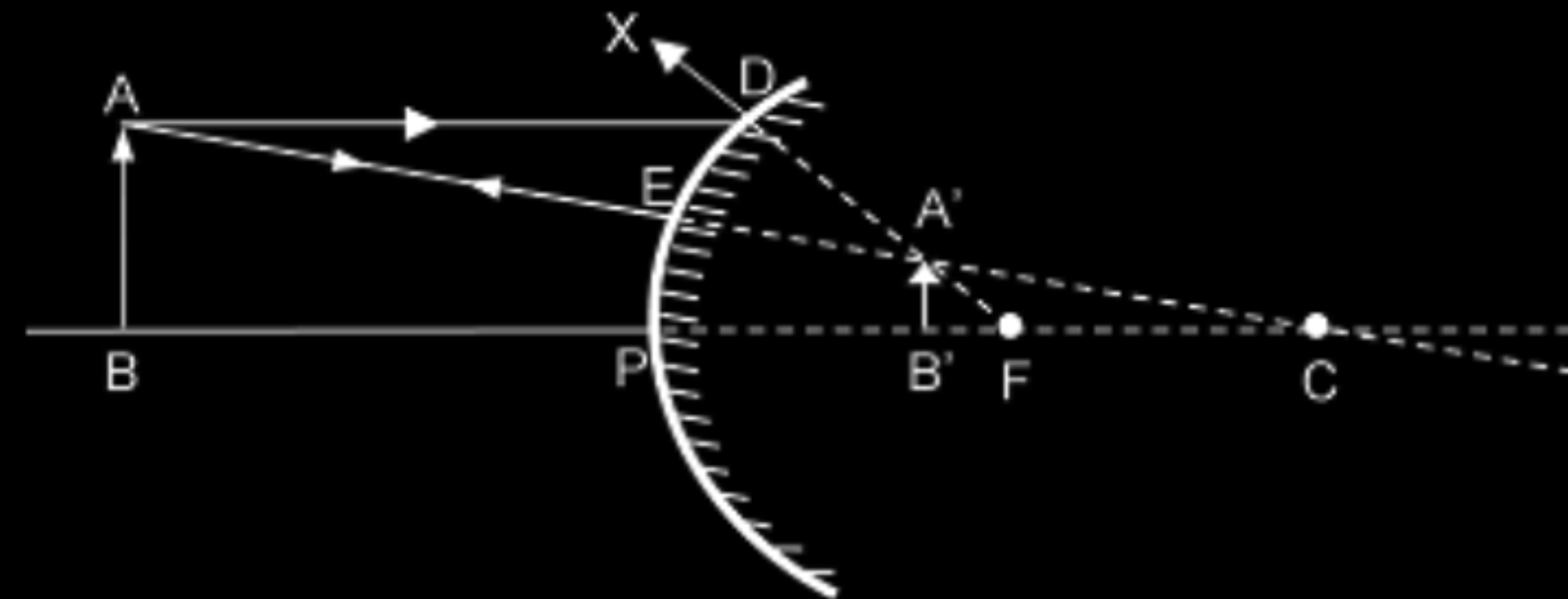
30. (a) Complete the following ray diagram to show the formation of image : 3



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- (b) Mention the nature, position and size of the image formed in this case.
- (c) State the sign of the image distance in this case using the Cartesian sign convention.

(a)



$A'B'$ is the image formed.

Credit full mark if attempted.

(b)

Nature: Virtual and erect

Position: Behind the mirror (between P and F)

1

$\frac{1}{2}$

$\frac{1}{2}$

Size: Diminished

$\frac{1}{2}$

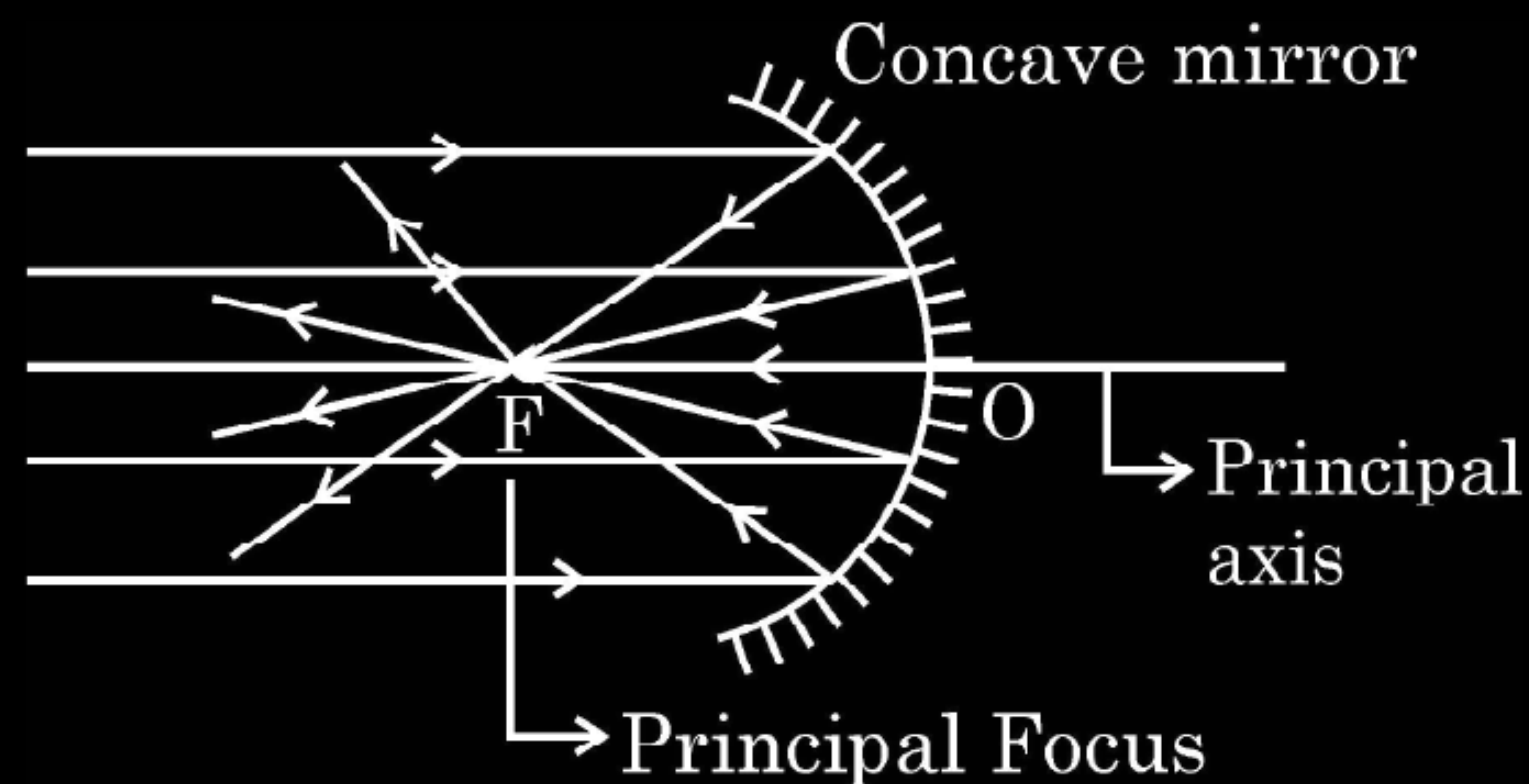
(c)

Positive

$\frac{1}{2}$

39. Hold a concave mirror in your hand and direct its reflecting surface towards the sun. Direct the light reflected by the mirror on to a white card-board held close to the mirror. Move the card-board back and forth gradually until you find a bright, sharp spot of light on the board. This spot of light is the image of the sun on the sheet of paper; which is also termed as “Principal Focus” of the concave mirror.

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- (a) List two applications of concave mirror. 1
- (b) If the distance between the mirror and the principal focus is 15 cm, find the radius of curvature of the mirror. 1

- (c) Draw a ray diagram to show the type of image formed when an object is placed between pole and focus of a concave mirror. **2**

OR

- (c) An object 10 cm in size is placed at 100 cm in front of a concave mirror. If its image is formed at the same point where the object is located, find : **2**
- (i) focal length of the mirror, and
 - (ii) magnification of the image formed with sign as per Cartesian sign convention.

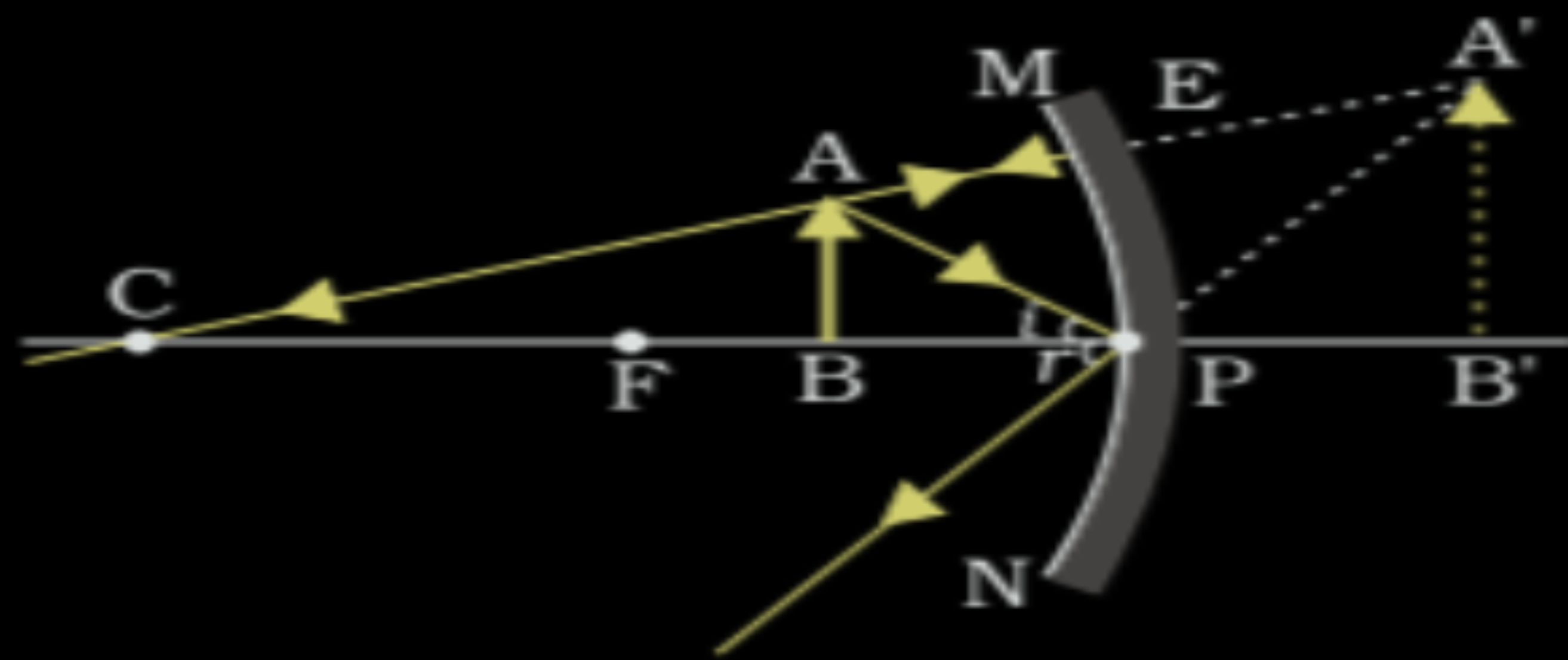
(a) Torches, search light, vehicles head lights, shaving mirrors,
dentist's mirror, Solar furnaces. (any two)

(b) $f = 15\text{cm}$

$$R = 2f$$

$$R = 2 \times 15\text{ cm} = 30\text{ cm}$$

(c)



(Note: $\frac{1}{2}$ mark to be deducted for not drawing the arrows.)

$\frac{1}{2}, \frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

2

(c)

(i) $h = + 10\text{cm}$

$$u = - 100\text{ cm}$$

$$v = - 100\text{ cm}$$

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{100} - \frac{1}{100} = \frac{1}{f}$$

$$\frac{-2}{100} = \frac{1}{f}$$

$$f = -50\text{ cm}$$

Alternate answer for (i)

$$\text{Since } u = v$$

Therefore, object is placed at centre of curvature (C)

$$f = \frac{R}{2}$$

$$f = \frac{-100}{2}$$

$$f = -50\text{ cm}$$

$$(ii) \quad m = \frac{-v}{u} = \frac{-(-100)}{100} = -1$$

29. A student wants to project the image of a candle flame on a screen 60 cm in front of a mirror by keeping the candle at a distance of 15 cm from its pole.

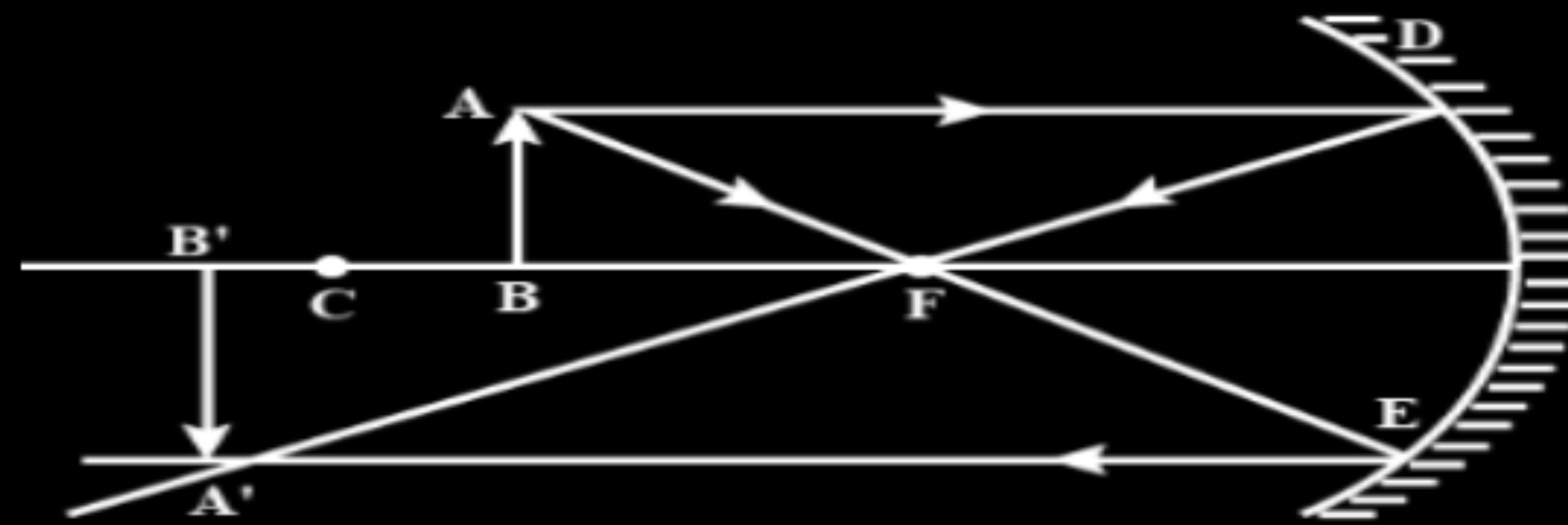
- (a) Name the type of mirror used.
- (b) Also calculate :
 - (i) Magnification of the image produced
 - (ii) Distance between object and its image
- (c) Draw a ray diagram to show the image formation.

CBSE 2023

29. A student wants to project the image of a candle flame on a screen 60 cm in front of a mirror by keeping the candle at a distance of 15 cm from its pole.

- (a) Name the type of mirror used.
- (b) Also calculate :
 - (i) Magnification of the image produced
 - (ii) Distance between object and its image
- (c) Draw a ray diagram to show the image formation.

CBSE 2023

(a) Concave Mirror / Converging Mirror	$\frac{1}{2}$
(b) (i). $m = \frac{-v}{u} = - \frac{(-60)}{(-15)} = -4$	$\frac{1}{2} + \frac{1}{2}$
(ii). 45 cm from the object	$\frac{1}{2}$
(c)  (Note: $\frac{1}{2}$ mark to be deducted for not drawing the arrows.)	1

30. (A) Define the following terms in the context of a diverging mirror :

3

(i) Principal focus

(ii) Focal length

Draw a labelled ray diagram to illustrate your answer.

CBSE 2023

30. (A) Define the following terms in the context of a diverging mirror :

3

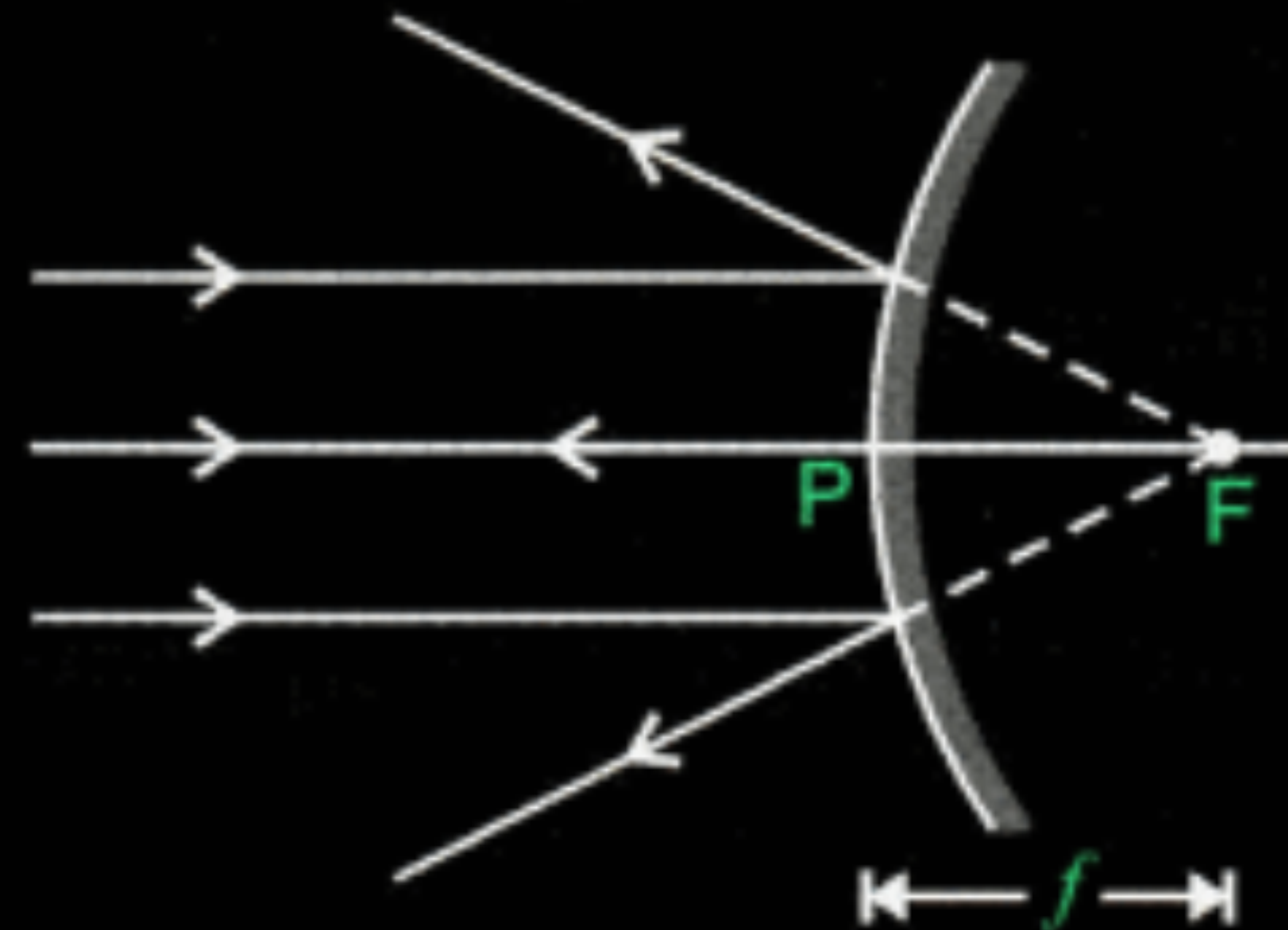
- (i) Principal focus
- (ii) Focal length

Draw a labelled ray diagram to illustrate your answer.

CBSE 2023

(A)

- (i) It is a point on the principal axis of a diverging mirror from where the rays parallel to principal axis appear to diverge after reflection.
- (ii) The distance between the pole and the principal focus of a mirror.



1

1

1

29. The magnification produced when an object is placed at a distance of 20 cm from a spherical mirror is $+1/2$. Where should the object be placed to reduce the magnification to $+1/3$.

CBSE 2023

29. The magnification produced when an object is placed at a distance of 20 cm from a spherical mirror is $+1/2$. Where should the object be placed to reduce the magnification to $+1/3$.

CBSE 2023

3

Given,

Distance of object, $u = -20$ cm

Magnification $m = \frac{1}{2}$

$$\therefore m = \frac{-v}{u} \Rightarrow v = -\frac{1}{2} \times -20 = 10 \text{ cm}$$

Using mirror formula

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$\frac{1}{f} = \frac{1}{10} - \frac{1}{20} = \frac{1}{20}$$

$f = 20$ cm

If $m = \frac{1}{3}$ and v' and u' are the corresponding image distance and object

$$\text{distance. } \frac{1}{3} = \frac{-v'}{u'} \Rightarrow v' = \frac{-u'}{3}$$

$$\frac{1}{f} = \frac{1}{v'} + \frac{1}{u'}$$

$$\frac{1}{20} = \frac{-3}{u'} + \frac{1}{u'} \Rightarrow \frac{1}{20} = \frac{-2}{u'}$$

$u' = -40$ cm.

$\frac{1}{2}, \frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

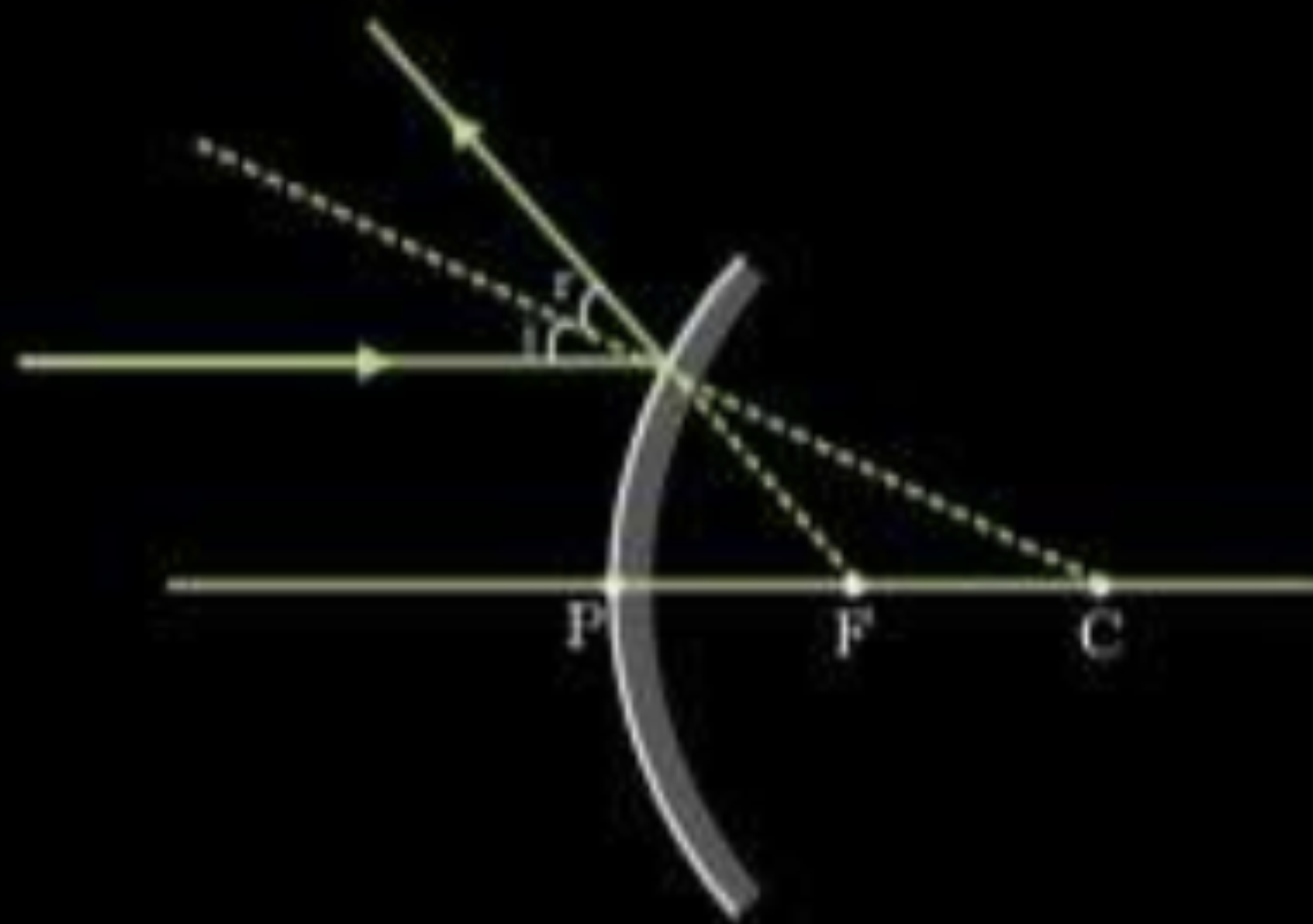
$\frac{1}{2}$

- 22.** Draw ray diagram in each of the following cases to show what happens after reflection to the incident ray when
- (a) it is parallel to the principal axis and falling on a convex mirror.
 - (b) it is falling on a concave mirror while passing through its principal focus.
 - (c) it is coming oblique to the principal axis and falling on the pole of a convex mirror.

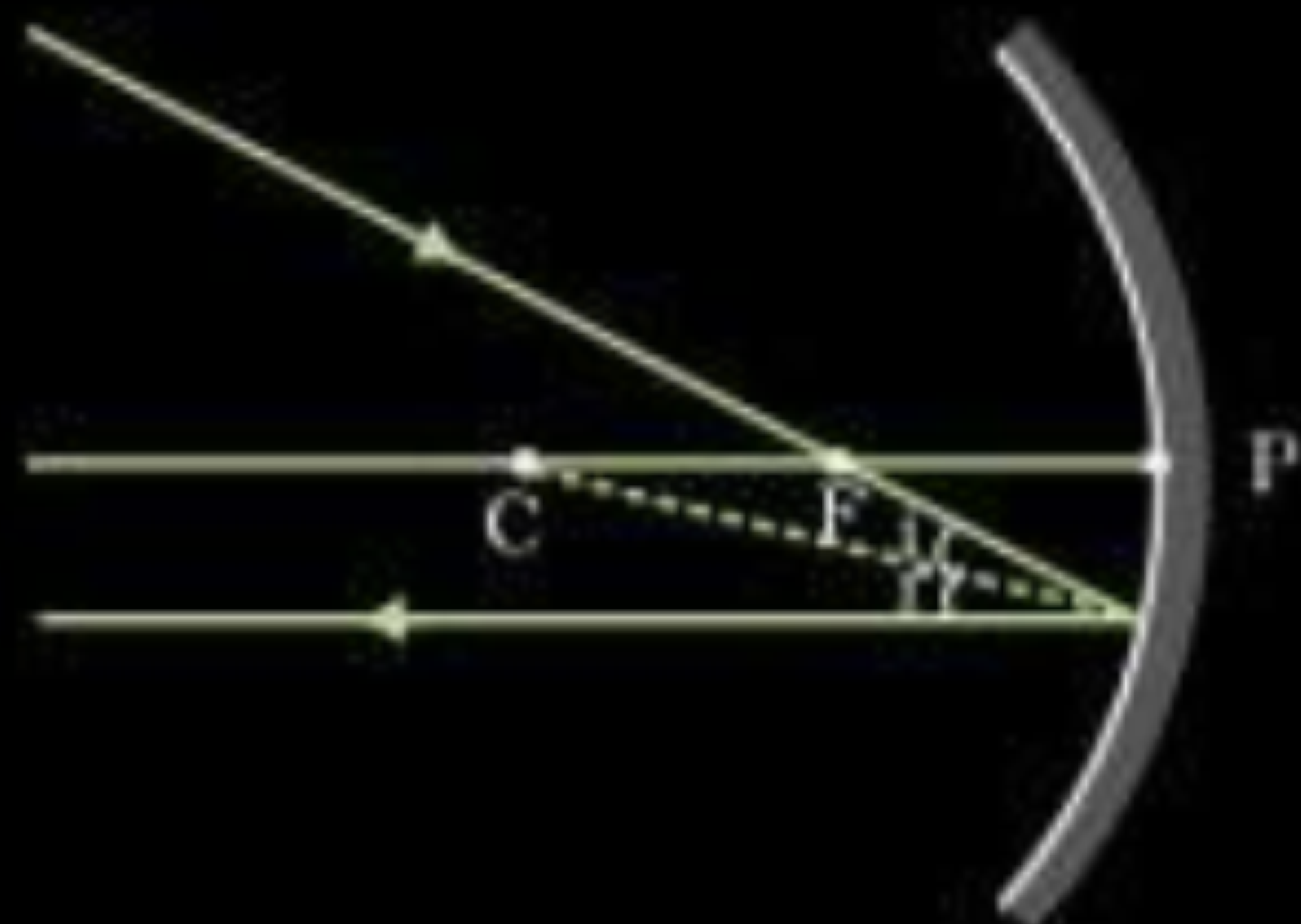
22. Draw ray diagram in each of the following cases to show what happens after reflection to the incident ray when
- (a) it is parallel to the principal axis and falling on a convex mirror.
 - (b) it is falling on a concave mirror while passing through its principal focus.
 - (c) it is coming oblique to the principal axis and falling on the pole of a convex mirror.

CBSE 2020

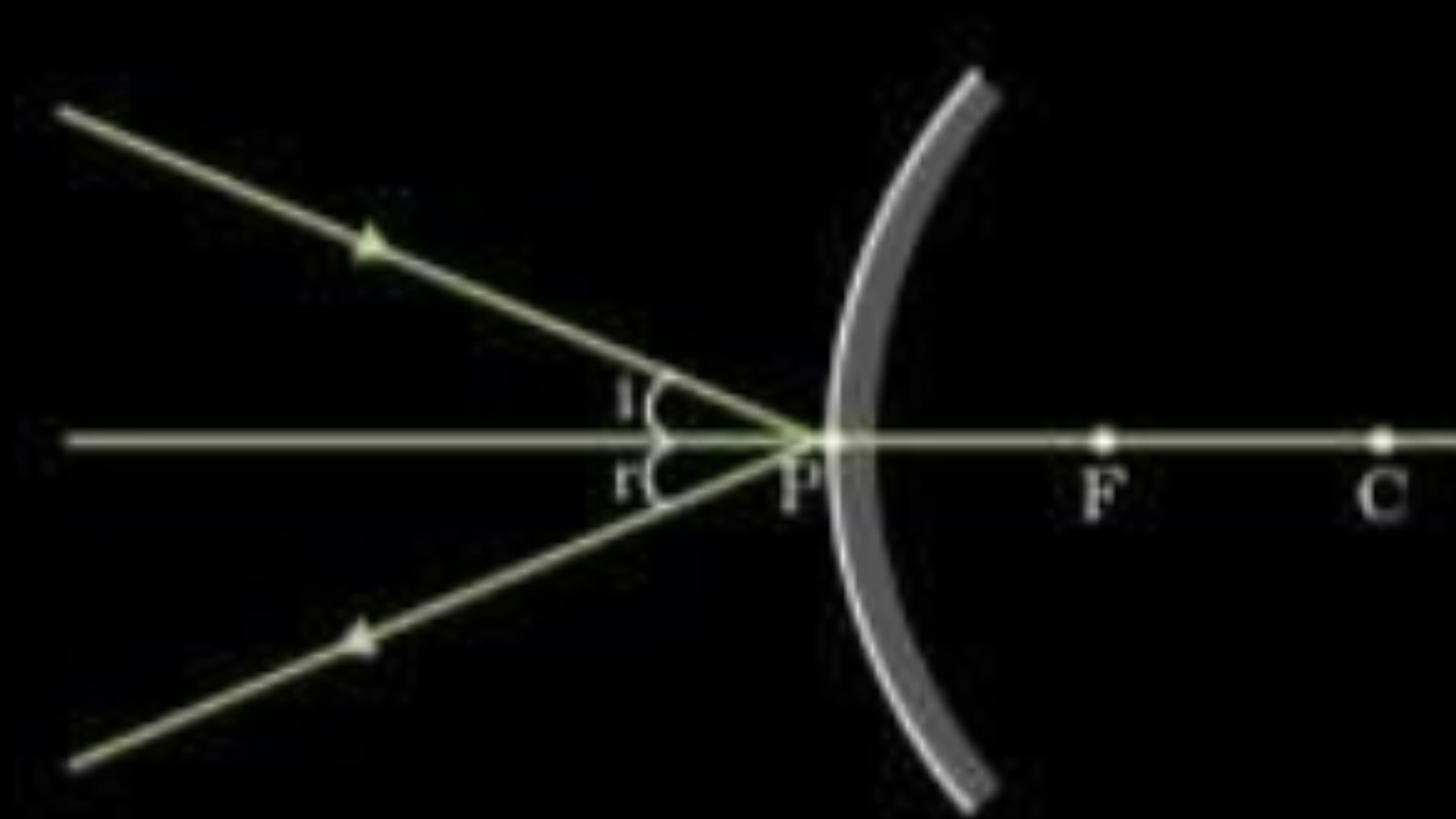
3



1



1



1

(Note : Deduct $\frac{1}{2}$ marks overall if no arrows are shown)

21. A concave mirror is used for image formation for different positions of an object. What inferences can be drawn about the following when an object is placed at a distance of 10 cm from the pole of a concave mirror of focal length 15 cm ?

- (a) Position of the image
- (b) Size of the image
- (c) Nature of the image

Draw a labelled ray diagram to justify your inferences.

CBSE 2020

21. A concave mirror is used for image formation for different positions of an object. What inferences can be drawn about the following when an object is placed at a distance of 10 cm from the pole of a concave mirror of focal length 15 cm ?

- (a) Position of the image
- (b) Size of the image
- (c) Nature of the image

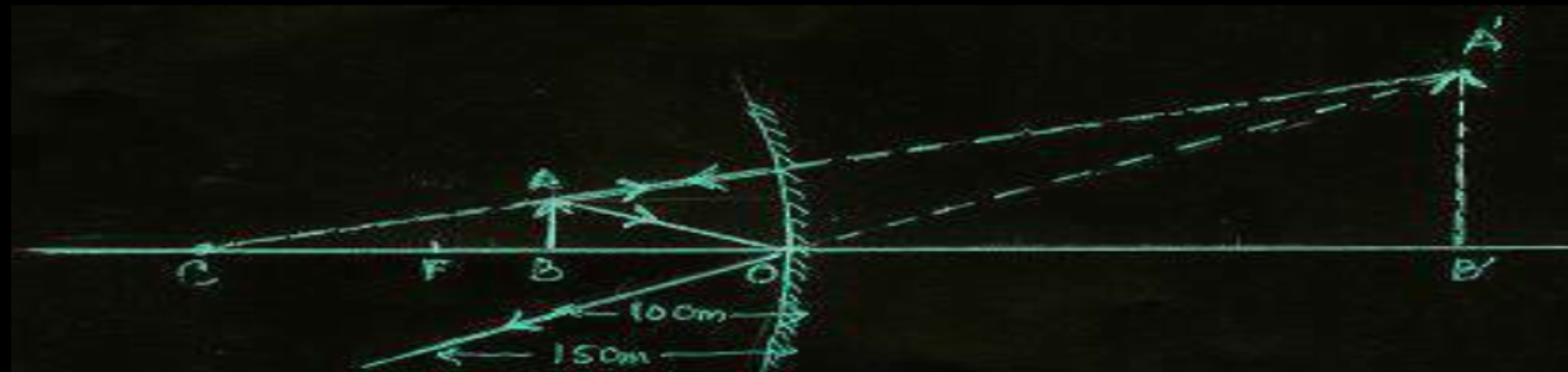
Draw a labelled ray diagram to justify your inferences.

CBSE 2020

3

- (a) Behind the mirror
- (b) Magnified
- (c) Virtual and erect

Labelled ray diagram



$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$1\frac{1}{2}$

30. (a) A security mirror used in a big showroom has radius of curvature 5 m. If a customer is standing at a distance of 20 m from the cash counter, find the position, nature and size of the image formed in the security mirror.
- (b) Neha visited a dentist in his clinic. She observed that the dentist was holding an instrument fitted with a mirror. State the nature of this mirror and reason for its use in the instrument used by dentist.

5

CBSE 2020

30. (a) A security mirror used in a big showroom has radius of curvature 5 m. If a customer is standing at a distance of 20 m from the cash counter, find the position, nature and size of the image formed in the security mirror.
- (b) Neha visited a dentist in his clinic. She observed that the dentist was holding an instrument fitted with a mirror. State the nature of this mirror and reason for its use in the instrument used by dentist.

5

CBSE 2020

(a) It is a convex mirror. So focal length should be positive. Radius of curvature $R = +5 \text{ m}$ $\therefore$ focal length $f = \frac{5}{2} = +2.5 \text{ m}$ Object distance $u = -20 \text{ m}$ Mirror formula $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ $\frac{1}{v} + \frac{1}{-20} = \frac{1}{2.5}$ $\frac{1}{v} = \frac{1}{20} + \frac{1}{2.5}$ $\frac{1}{v} = \frac{1}{20} + \frac{10}{25}$	$\frac{1}{v} = \frac{5+40}{100} = \frac{45}{100}$ $v = \frac{100}{45} = \frac{20}{9} = +2.2 \text{ m}$ - Nature of image = virtual and erect image - Size of image : diminished image (b) Concave Mirror Reason : to obtain erect and enlarged image of teeth
---	---

30. (a) A concave mirror of focal length 10 cm can produce a magnified real as well as virtual image of an object placed in front of it. Draw ray diagrams to justify this statement.
- (b) An object is placed perpendicular to the principal axis of a convex mirror of focal length 10 cm. The distance of the object from the pole of the mirror is 10 cm. Find the position of the image formed.
- CBSE 2020

 5

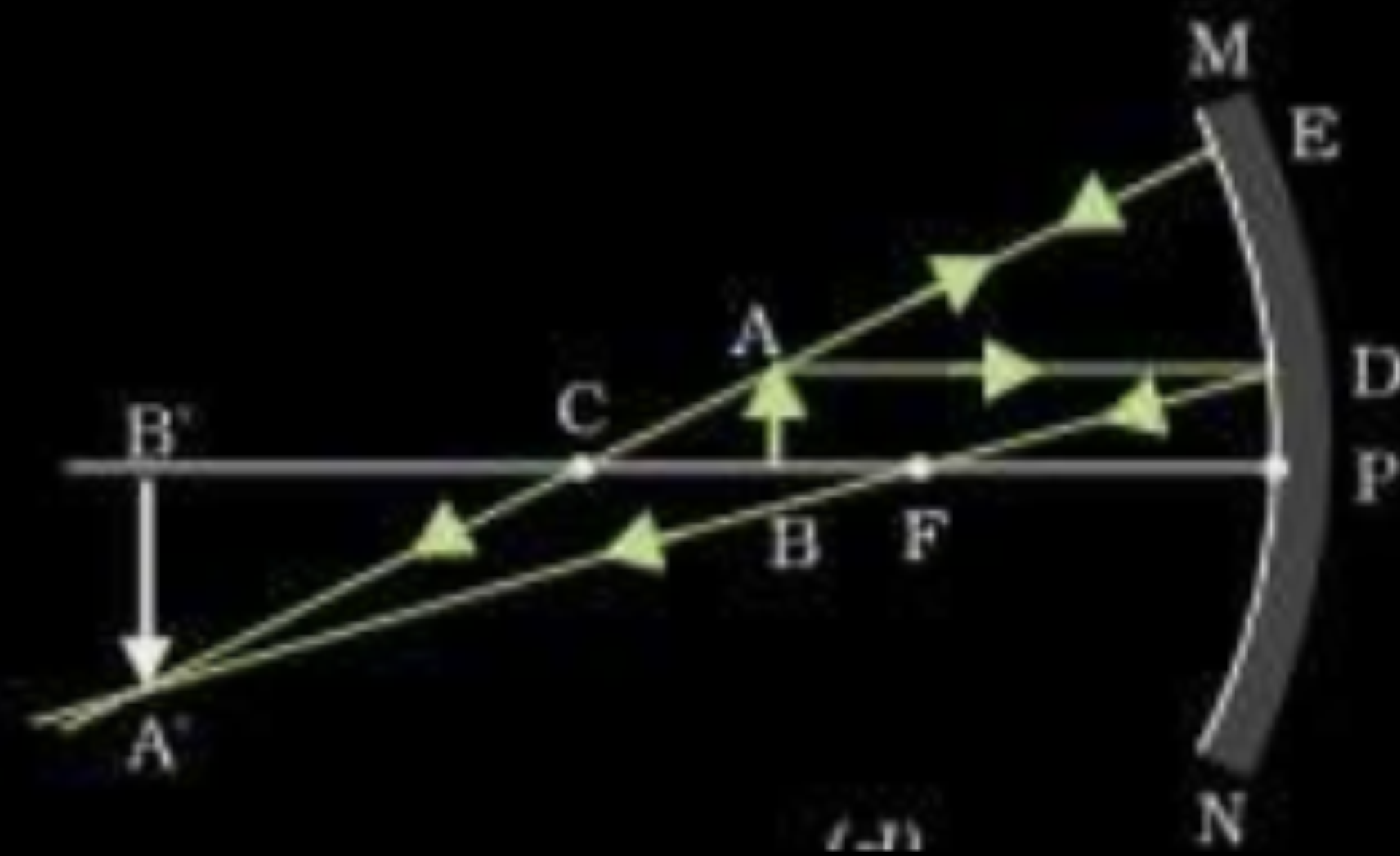
30. (a) A concave mirror of focal length 10 cm can produce a magnified real as well as virtual image of an object placed in front of it. Draw ray diagrams to justify this statement.

(b) An object is placed perpendicular to the principal axis of a convex mirror of focal length 10 cm. The distance of the object from the pole of the mirror is 10 cm. Find the position of the image formed.

CBSE 2020

5

(a)
(i) Real and magnified



Object distance must be between 10 to 20 cm

(ii) Virtual and magnified



Object distance must be less than 10 cm

(b) $f = 10 \text{ cm}$; $u = -10 \text{ cm}$

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$$

$$\frac{1}{v} = \frac{1}{10} - \frac{1}{-10}$$

$$\frac{1}{v} = \frac{1}{10} + \frac{1}{10}$$

$$\frac{1}{v} = \frac{1}{5} \therefore v = 5 \text{ cm}$$

An object 4.0 cm in size, is placed 25.0 cm in front of a concave mirror of focal length 15.0 cm.

- (i) At what distance from the mirror should a screen be placed in order to obtain a sharp image?
- (ii) Find the size of the image.
- (iii) Draw a ray diagram to show the formation of image in this case.

CBSE 2020

Given $h = +4.0$ cm, $u = -25.0$ cm, $f = -15.0$ cm

$\frac{1}{2}$

i) image distance $v = ?$; mirror formula : $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

$\frac{1}{2}$

$$\text{or } \frac{1}{v} = \frac{1}{f} - \frac{1}{u} ; = -\frac{1}{15} - \left(-\frac{1}{25}\right)$$

$$= \frac{-1}{15} + \frac{1}{25} = \frac{-5+3}{75}$$

$$= \frac{-2}{75}$$

$\frac{1}{2}$

$$v = -37.5 \text{ cm}$$

The screen should be placed 37.5 cm in front of the mirror.

ii) $m = \frac{h^1}{h} = -\frac{v}{u}$

$\frac{1}{2}$

$$: h^1 = -\frac{v}{u} \cdot h$$

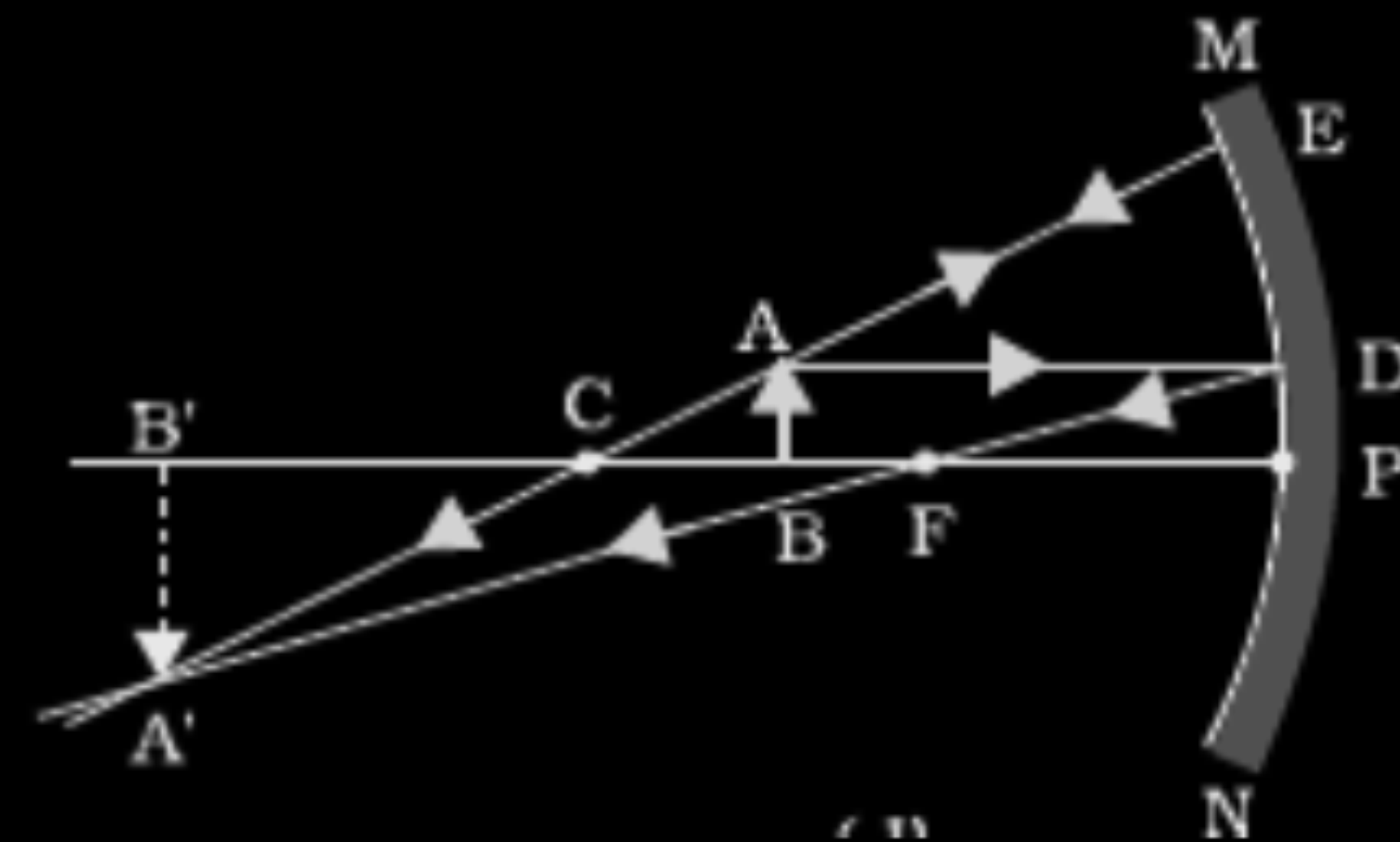
$\frac{1}{2}$

$$= -\frac{(-37.5 \times 4)}{-25}$$

$\frac{1}{2}$

$$h^1 = -6.0 \text{ cm (size of the image).}$$

iii)



1

Note: Deduct half mark for not showing arrows in ray diagrams.

A student holding a mirror in his hand, directed the reflecting surface of the mirror towards the Sun. He then directed the reflected light on to a sheet of paper held close to the mirror.

- (a) What should he do to burn the paper ?
- (b) Which type of mirror does he have ?
- (c) Will he be able to determine the approximate value of focal length of this mirror from this activity ? Give reason and draw ray diagram to justify your answer in this case.

CBSE 2019

A student holding a mirror in his hand, directed the reflecting surface of the mirror towards the Sun. He then directed the reflected light on to a sheet of paper held close to the mirror.

- (a) What should he do to burn the paper ?
- (b) Which type of mirror does he have ?
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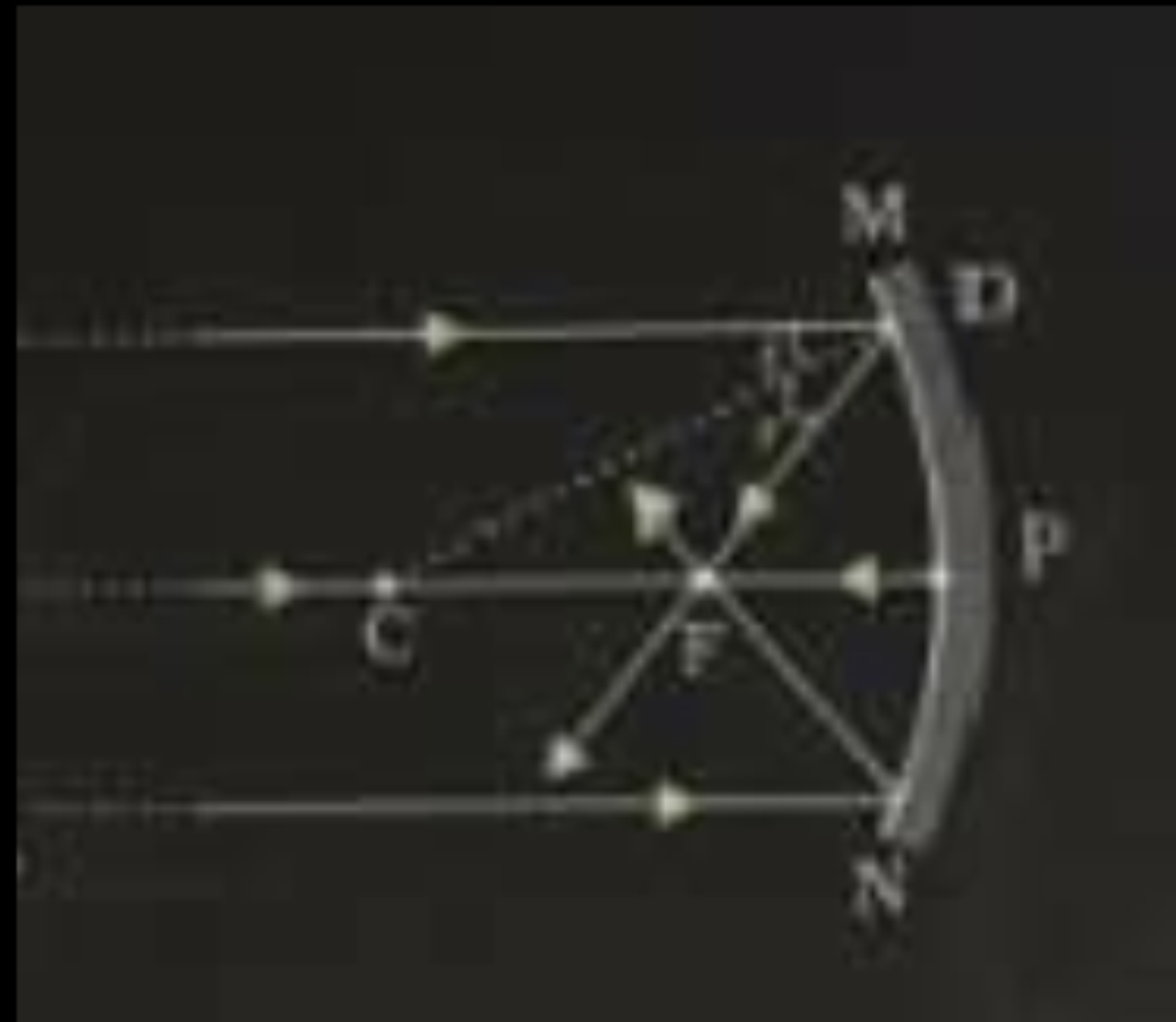
CBSE 2019

- (a) Move the mirror/paper to focus the rays at one point
- (b) Concave mirror
- (c) Yes, distance between mirror and focal point gives approximate focal length.

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2} + \frac{1}{2}$$



$$\frac{1}{2} + \frac{1}{2}$$

A 6 cm tall object is placed perpendicular to the principal axis of a concave mirror of focal length 30 cm. The distance of the object from the mirror is 45 cm. Use mirror formula to determine the position, nature and size of the image formed. Also draw labelled ray diagram to show the image formation in this case.

CBSE 2019

A 6 cm tall object is placed perpendicular to the principal axis of a concave mirror of focal length 30 cm. The distance of the object from the mirror is 45 cm. Use mirror formula to determine the position, nature and size of the image formed. Also draw labelled ray diagram to show the image formation in this case.

CBSE 2019

Given –

$$h = 6 \text{ cm}$$

$$f = -30 \text{ cm}$$

$$u = -45 \text{ cm}$$

mirror formula

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$$

$$= -\frac{1}{30} - \frac{1}{(-45)}$$

$$= -\frac{1}{30} + \frac{1}{45} = -\frac{1}{90}$$

$f = -90 \text{ cm}$ from the pole of mirror

Size of the image

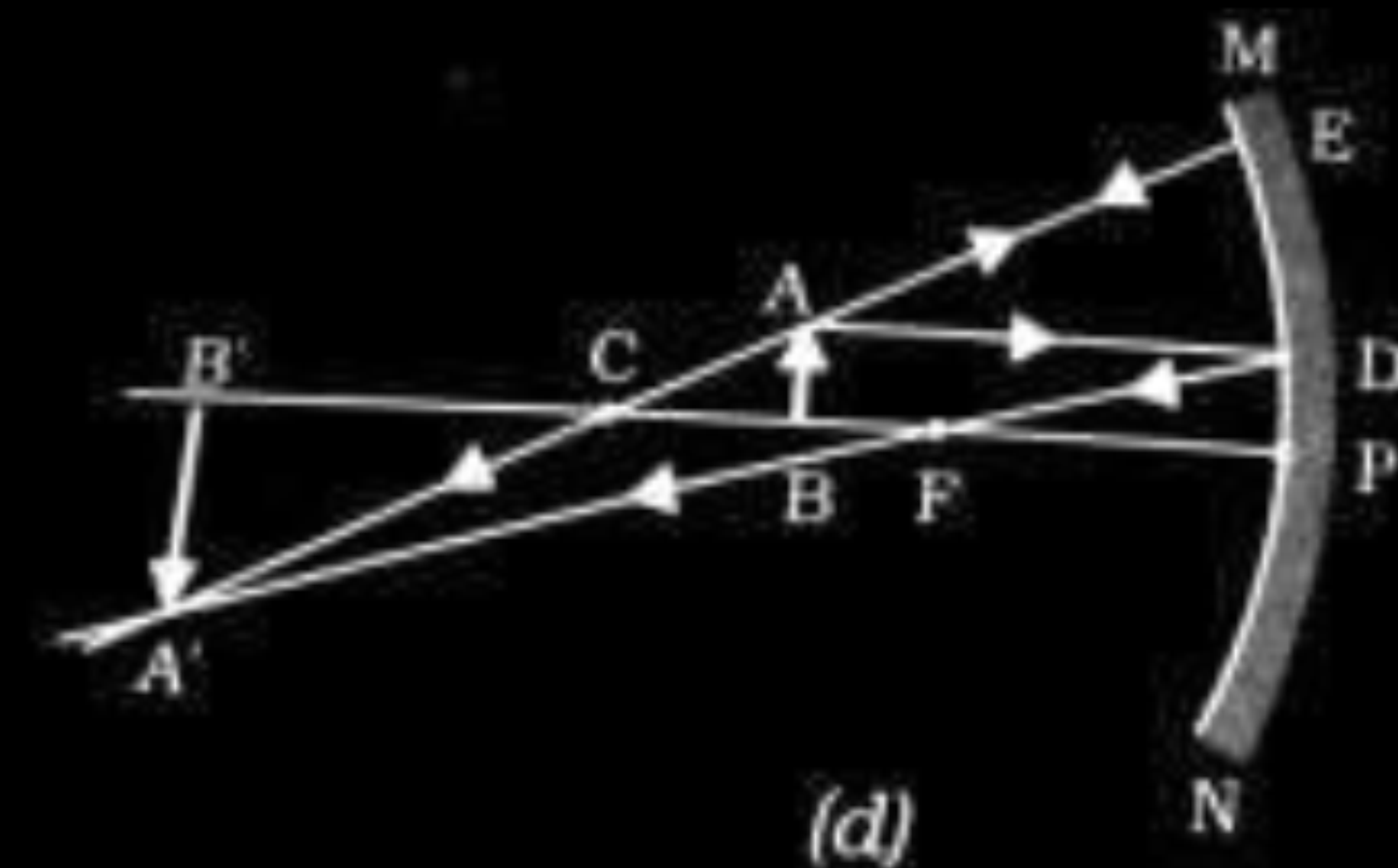
$$m = \frac{-v}{u}$$

$$= -\frac{90}{45} = -2$$

$$h^1 = -2 \times 6 \text{ cm}$$

$$= -12 \text{ cm}$$

Image formed will be real, inverted and enlarged.



Well labelled diagram

- (a) List four characteristics of the images formed by plane mirrors.
- (b) A 5 cm tall object is placed at a distance of 20 cm from a concave mirror of focal length 30 cm. Use mirror formula to determine the position and size of the image formed.

CBSE 2019

- (a) List four characteristics of the images formed by plane mirrors.
- (b) A 5 cm tall object is placed at a distance of 20 cm from a concave mirror of focal length 30 cm. Use mirror formula to determine the position and size of the image formed.

CBSE 2019

(a) Characteristics:

- i) The image is same size as the object.
- ii) The image is erect and virtual.
- iii) The image is laterally inverted.
- iv) The distance between the object and mirror is same as the distance between image and mirror.

$\frac{1}{2} \times 4$

(b)

$$h = 5\text{cm}$$

$$u = -20\text{cm}$$

$$f = -30$$

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$\frac{1}{2}$

$$\frac{1}{v} = \frac{1}{-30} + \frac{1}{-20}$$

1

$$v = -60\text{cm}$$

$\frac{1}{2}$

$$\frac{h'}{h} = \frac{v}{u}$$

$\frac{1}{2}$

$$h' = 15\text{cm}$$

$\frac{1}{2}$

Size - enlarged

List in proper sequence the steps of the experiment for determining the approximate focal length of a given concave mirror by obtaining the image of a distant object.

CBSE 2019

List in proper sequence the steps of the experiment for determining the approximate focal length of a given concave mirror by obtaining the image of a distant object.

CBSE 2019

- | | |
|---|------------------------|
| <ul style="list-style-type: none">i) Fix a concave mirror on a stand and place it near a source of bright lightii) Place a screen fitted on a stand in front of the mirroriii) Move the screen back and forth, until a sharp and clear image of a distant object like a tree is obtained on the screeniv) Mark the position of mirror and screen on the scale and note the distance between them | $\frac{1}{2} \times 4$ |
|---|------------------------|

State laws of reflection of light.

CBSE 2019

- | | |
|--|---|
| • First law: The incident ray, the reflected ray and the normal to the surface at the point of incidence all lie in the same plane. | 1 |
| • Second law: Angle of incidence is equal to angle of reflection. | 1 |

List four characteristics of the image formed by a concave mirror of focal length 40 cm when the object is placed in front of it at a distance of 20 cm from its pole.

List four characteristics of the image formed by a concave mirror of focal length 40 cm when the object is placed in front of it at a distance of 20 cm from its pole.

(i) Erect (ii) virtual (iii) magnified (iv) Behind the mirror

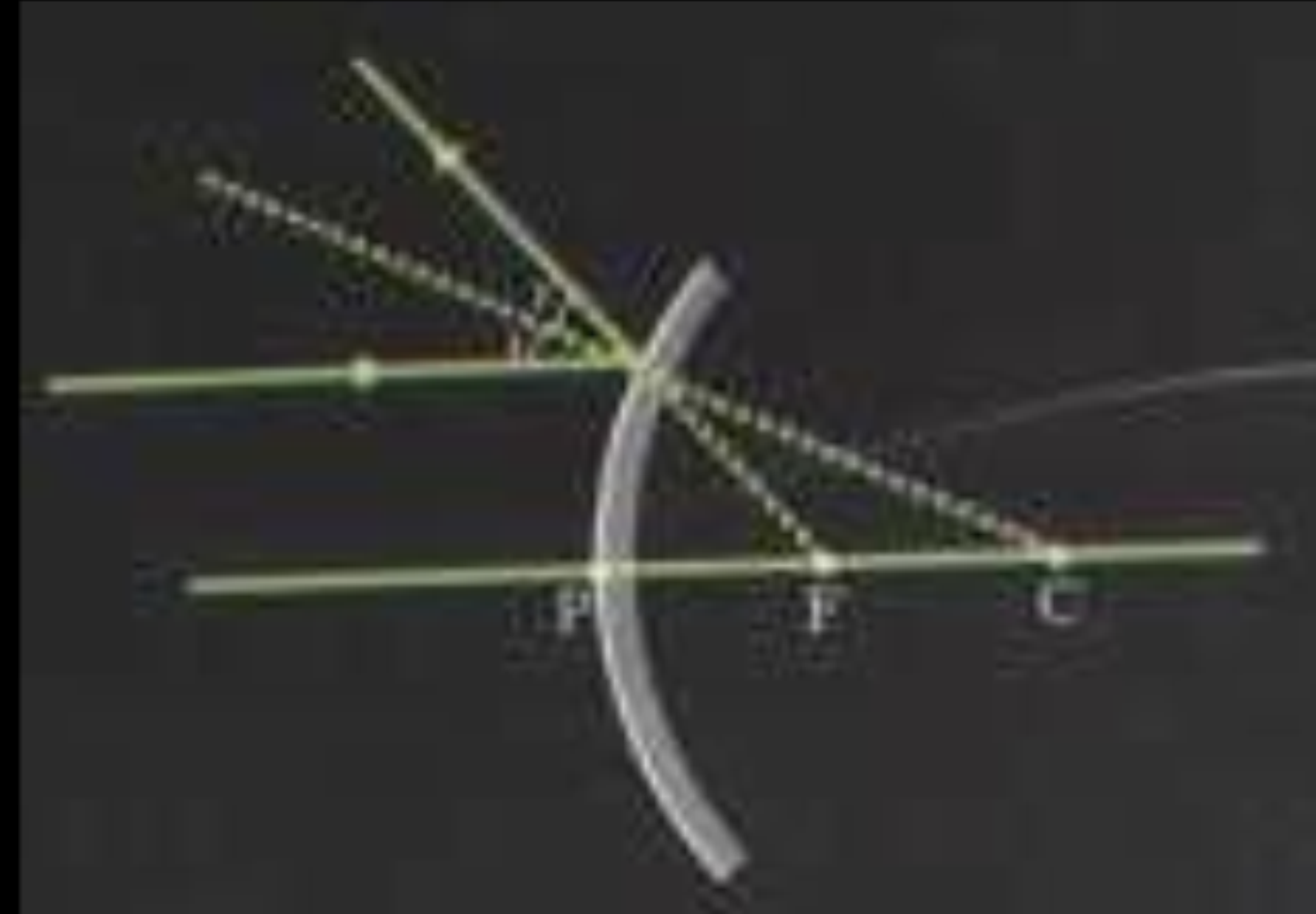
$\frac{1}{2} \times 4$

Draw a labelled ray diagram to show the path of the reflected ray corresponding to an incident ray of light parallel to the principal axis of a convex mirror. Mark the angle of incidence and angle of reflection on it.

CBSE 2019

Draw a labelled ray diagram to show the path of the reflected ray corresponding to an incident ray of light parallel to the principal axis of a convex mirror. Mark the angle of incidence and angle of reflection on it.

CBSE 2019



- (i) draw mirror.
- (ii) Complete ray diagram
- (iii) $\angle i$ and $\angle r$ islabelled
- (iv) Arrows marked

$\frac{1}{2}$
 $\frac{1}{2}$
 $\frac{1}{2}$
 $\frac{1}{2}$

A concave mirror has a focal length of 20 cm. At what distance from the mirror should a 4 cm tall object be placed so that it forms an image at a distance of 30 cm from the mirror ? Also calculate the size of the image formed.

CBSE 2019

A concave mirror has a focal length of 20 cm. At what distance from the mirror should a 4 cm tall object be placed so that it forms an image at a distance of 30 cm from the mirror ? Also calculate the size of the image formed.

CBSE 2019

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

1/2

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$$

$$\frac{1}{u} = \frac{1}{-20} + \frac{1}{30}$$

$$u = -60\text{cm (without unit (cm) deduct } \frac{1}{2} \text{ mark)}$$

1

$$m = \frac{-v}{u}$$

1/2

$$m = -\left[\frac{-30}{-60}\right]$$

$$m = -\frac{1}{2}$$

$$\frac{h_2}{h_1} = m$$

$$h_2 = h_1 \times m$$

$$h_2 = 4 \times \left[-\frac{1}{2}\right]$$

$$h_2 = -2 \text{ cm}$$

1

A teacher gives a convex lens and a concave mirror of focal length of 20 cm each to his student and asks him to find their focal lengths by obtaining the image of a distant object. The student uses a distant tree as the object and obtains its sharp image, one by one, on a screen. The distances d_1 and d_2 between the lens/mirror and the screen in the two cases and the nature of their respective sharp images are likely to be

- (a) (20 cm, 40 cm) and (erect and erect)
- (b) (20 cm, 40 cm) and (inverted and inverted)
- (c) (20 cm, 20 cm) and (inverted and inverted)
- (d) (20 cm, 40 cm) and (erect and inverted)

Give reason for your answer.

A teacher gives a convex lens and a concave mirror of focal length of 20 cm each to his student and asks him to find their focal lengths by obtaining the image of a distant object. The student uses a distant tree as the object and obtains its sharp image, one by one, on a screen. The distances d_1 and d_2 between the lens/mirror and the screen in the two cases and the nature of their respective sharp images are likely to be

- (a) (20 cm, 40 cm) and (erect and erect)
- (b) (20 cm, 40 cm) and (inverted and inverted)
- (c) (20 cm, 20 cm) and (inverted and inverted)
- (d) (20 cm, 40 cm) and (erect and inverted)

Give reason for your answer.

CBSE 2019	
(c) (20 cm, 20 cm) and (inverted and inverted)	1
Reason: Only real and inverted image can be obtained on the screen and in both cases the image is formed at the principal focus.	1

Draw a labelled ray diagram to show the path of the reflected ray corresponding to the ray which is incident obliquely to the principal axis, towards the pole of a convex mirror. Mark the angle of incidence and angle of reflection on it.

CBSE 2019

Draw a labelled ray diagram to show the path of the reflected ray corresponding to the ray which is incident obliquely to the principal axis, towards the pole of a convex mirror. Mark the angle of incidence and angle of reflection on it.

CBSE 2019

Section-B



- Correct ray diagram
- labelling of $\angle i$ and $\angle r$

1
 $\frac{1}{2} \times 2 = 1$

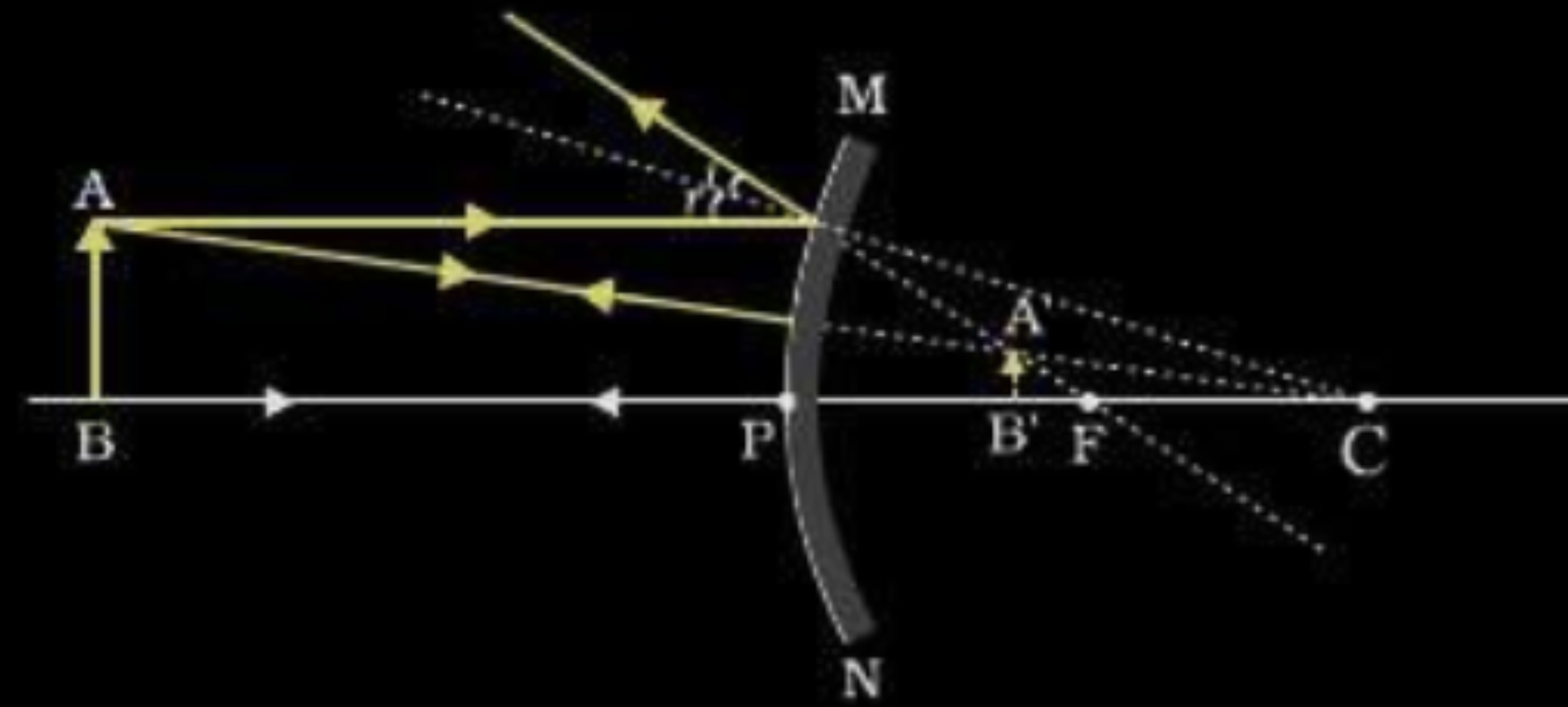
If the image formed by a spherical mirror for all positions of the object placed in front of it is always erect and diminished, what type of mirror is it ? Draw a labelled ray diagram to support your answer.

CBSE 2018

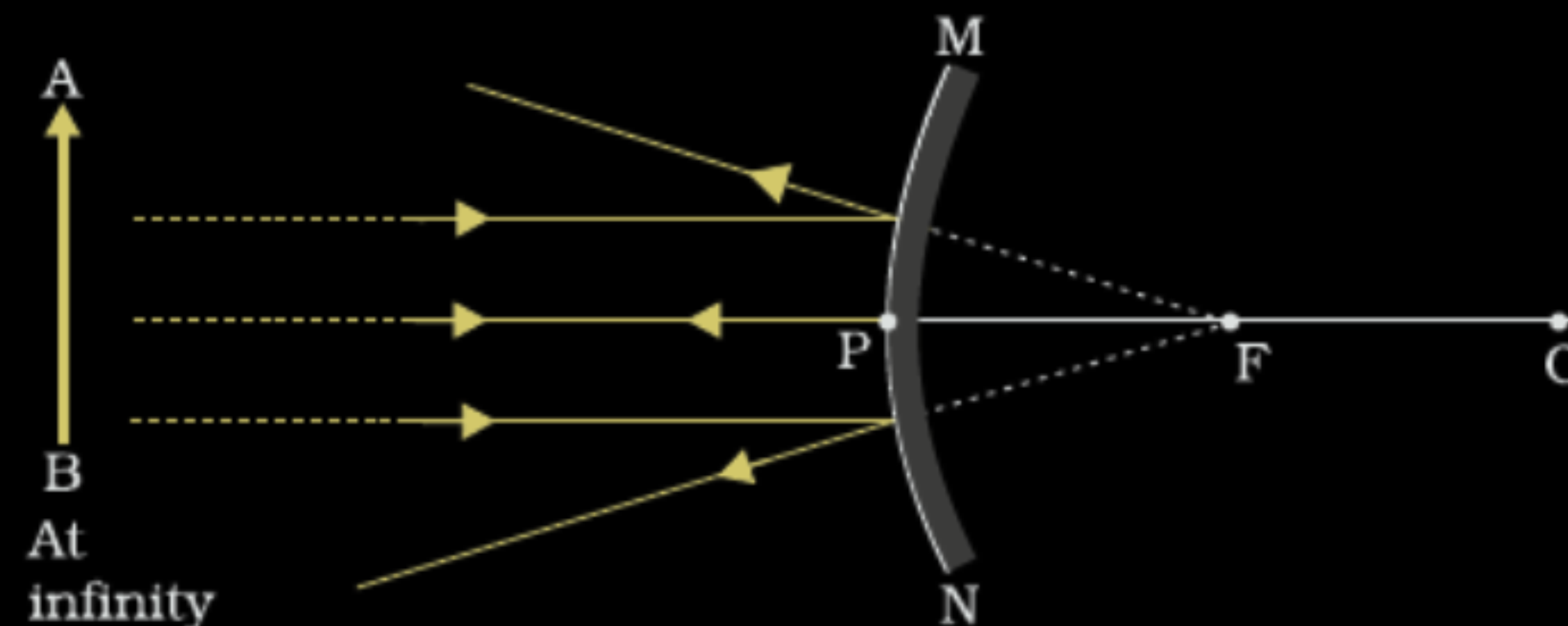
If the image formed by a spherical mirror for all positions of the object placed in front of it is always erect and diminished, what type of mirror is it ? Draw a labelled ray diagram to support your answer.

CBSE 2018

- Convex Mirror
- Labelled Ray diagram for any position of object



OR



Note: If arrows not marked, $\frac{1}{2}$ mark to be deducted.

$\frac{1}{2}$

$1 \frac{1}{2}$

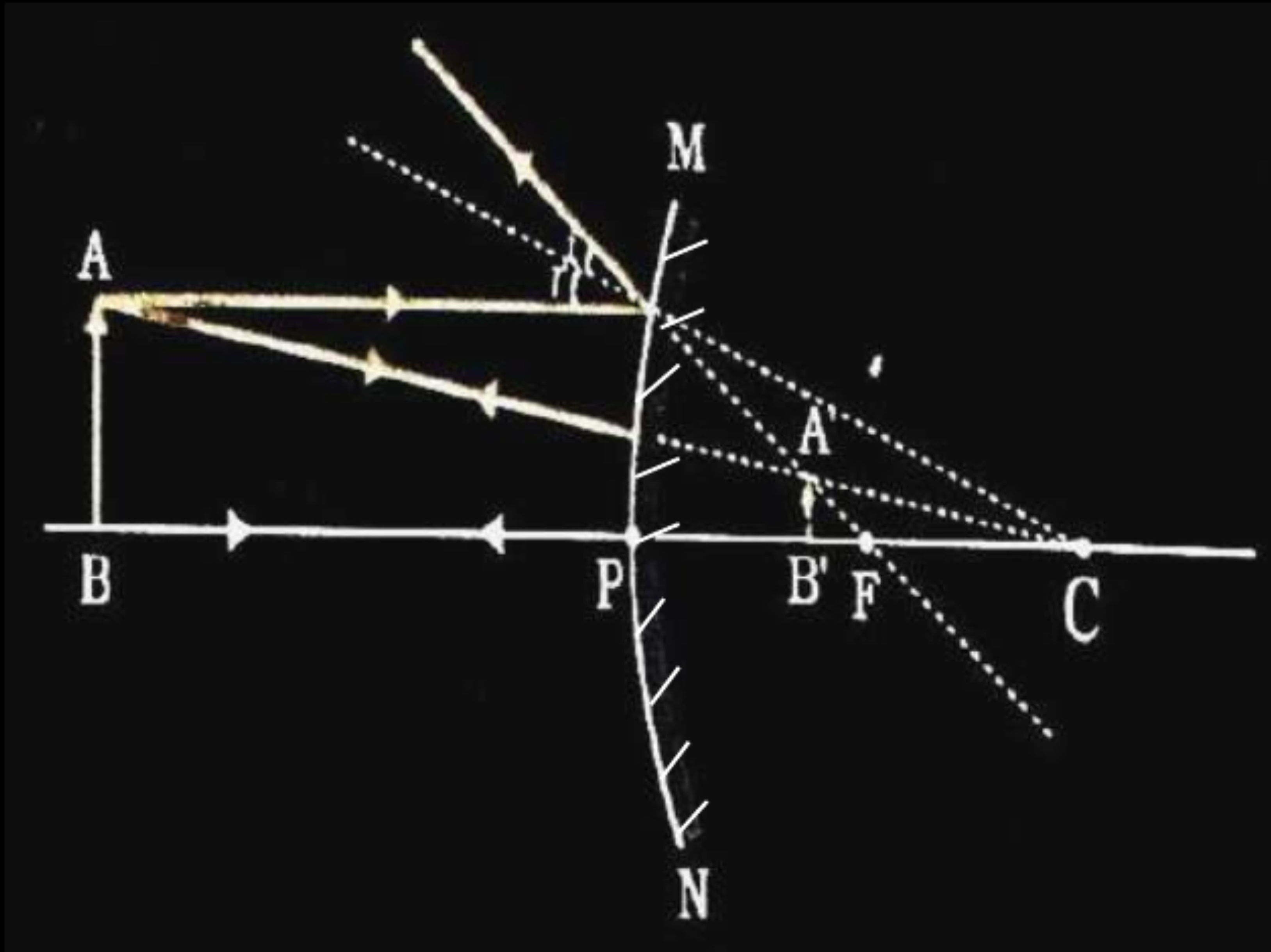
- (a) If the image formed by a mirror for all positions of the object placed in front of it is always diminished, erect and virtual, state the type of the mirror and also draw a ray diagram to justify your answer. Write one use such mirrors are put to and why.
- (b) Define the radius of curvature of spherical mirrors. Find the nature and focal length of a spherical mirror whose radius of curvature is + 24 cm.

CBSE 2017

- (a) If the image formed by a mirror for all positions of the object placed in front of it is always diminished, erect and virtual, state the type of the mirror and also draw a ray diagram to justify your answer. Write one use such mirrors are put to and why.
- (b) Define the radius of curvature of spherical mirrors. Find the nature and focal length of a spherical mirror whose radius of curvature is + 24 cm.

CBSE 2017

a) Convex / diverging mirror)



Direction of rays

Use:- As a rear view mirror / any other use

Reason :- Always give erect and diminished image / Large field of view

$\frac{1}{2}$

1

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

(b) The radius of the sphere of which the mirror forms a part / The distance between pole and center of curvature of a mirror.

Nature of the mirror – convex / diverging mirror

$$R = 2f = 24 \text{ cm}$$

$$\therefore f = +12 \text{ cm}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

“The magnification produced by a spherical mirror is -3 ”. List four informations you obtain from this statement about the mirror/image.

CBSE 2016

“The magnification produced by a spherical mirror is -3 ”. List four informations you obtain from this statement about the mirror/image.

CBSE 2016

1) Inverted image; 2) magnified; 3) concave mirror ; 4) real image

4 x $\frac{1}{2}$

The image of an object formed by a lens is of magnification -1 . If the distance between the object and its image is 60 cm, what is the focal length of the lens ? If the object is moved 20 cm towards the lens, where would the image be formed ? State reason and also draw a ray diagram in support of your answer.

CBSE 2016

The image of an object formed by a lens is of magnification -1 . If the distance between the object and its image is 60 cm, what is the focal length of the lens ? If the object is moved 20 cm towards the lens, where would the image be formed ? State reason and also draw a ray diagram in support of your answer.

CBSE 2016

Image with magnification -1 means image is inverted and of the same size.

Therefore, object is at $2F$ and the image is also at $2F$ on the other side of the lens.

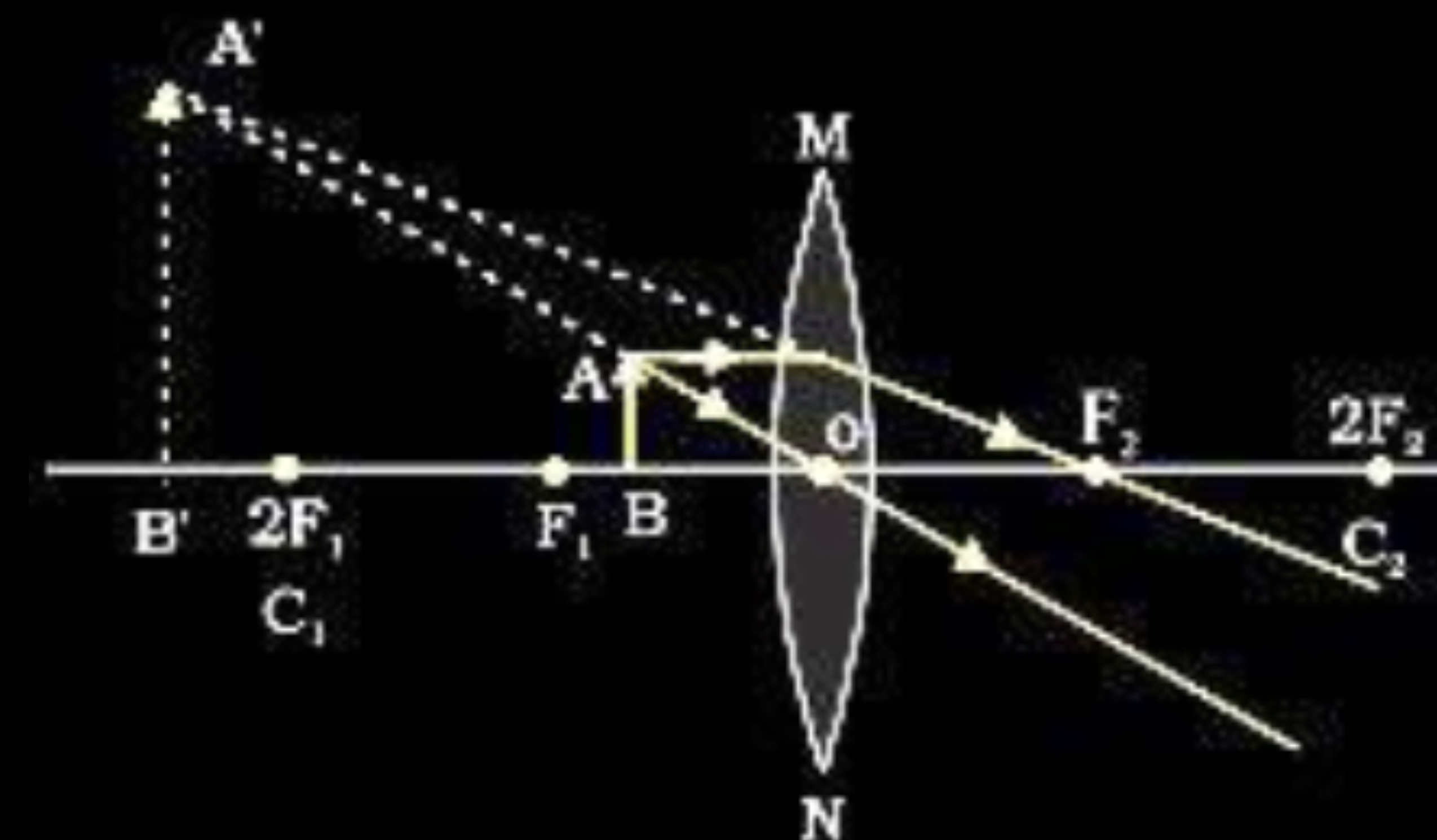
Therefore, distance between the object and its image is $4f = 60$ cm 1

$$\Rightarrow f = 15$$
 cm $\frac{1}{2}$

Object distance $2f = 30$ cm. if the object is shifted towards the lens by 20 cm, the new object distance = 30 cm $-$ 20 cm = 10 cm. $\frac{1}{2}$

This distance is less than the focal length, and the image formed in this case would be virtual, erect and will form on the same side as the object

1



- (a) Define focal length of a spherical lens.
- (b) A divergent lens has a focal length of 30 cm. At what distance should an object of height 5 cm from the optical centre of the lens be placed so that its image is formed 15 cm away from the lens ? Find the size of the image also.
- (c) Draw a ray diagram to show the formation of image in the above situation.

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a) Distance between optical centre and focus of the lens.

1

b) $f = -30$ cm; $u = ?$; $h_1 = 5$ cm; $h_2 = ?$; $v = -15$ cm

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u} \implies \frac{1}{u} = \frac{1}{v} - \frac{1}{f} \implies u = \frac{vf}{f-v}$$

$\frac{1}{2}$

$$\implies u = \frac{-15 \text{ cm} \times -30 \text{ cm}}{-30 \text{ cm} - (-15 \text{ cm})} = -30 \text{ cm}$$

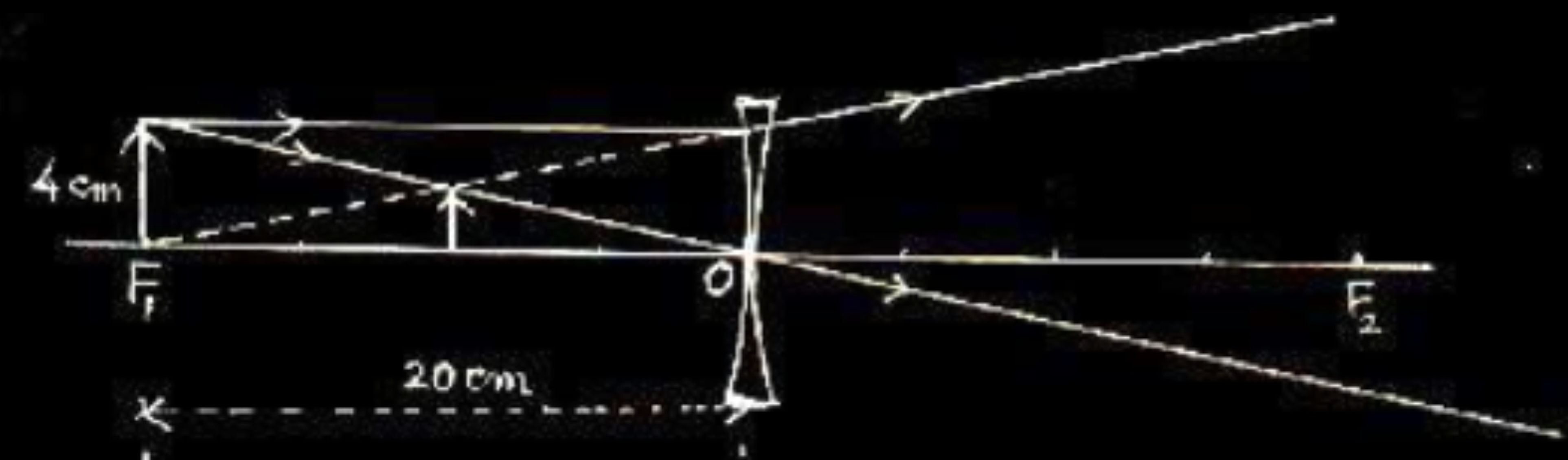
$1 \frac{1}{2}$

$$m = \frac{v}{u} = \frac{h_2}{h_1} \implies h_2 = \frac{v}{u} \times h_1$$

$$= \frac{-15 \text{ cm}}{-30 \text{ cm}} \times 5 \text{ cm} = 2.5 \text{ cm}$$

1

1



It is desired to obtain an erect image of an object, using concave mirror of focal length of 12 cm.

- (i) What should be the range of distance of an object placed in front of the mirror ?
- (ii) Will the image be smaller or larger than the object. Draw ray diagram to show the formation of image in this case.
- (iii) Where will the image of this object be, if it is placed 24 cm in front of the mirror ? Draw ray diagram for this situation also to justify your answer.

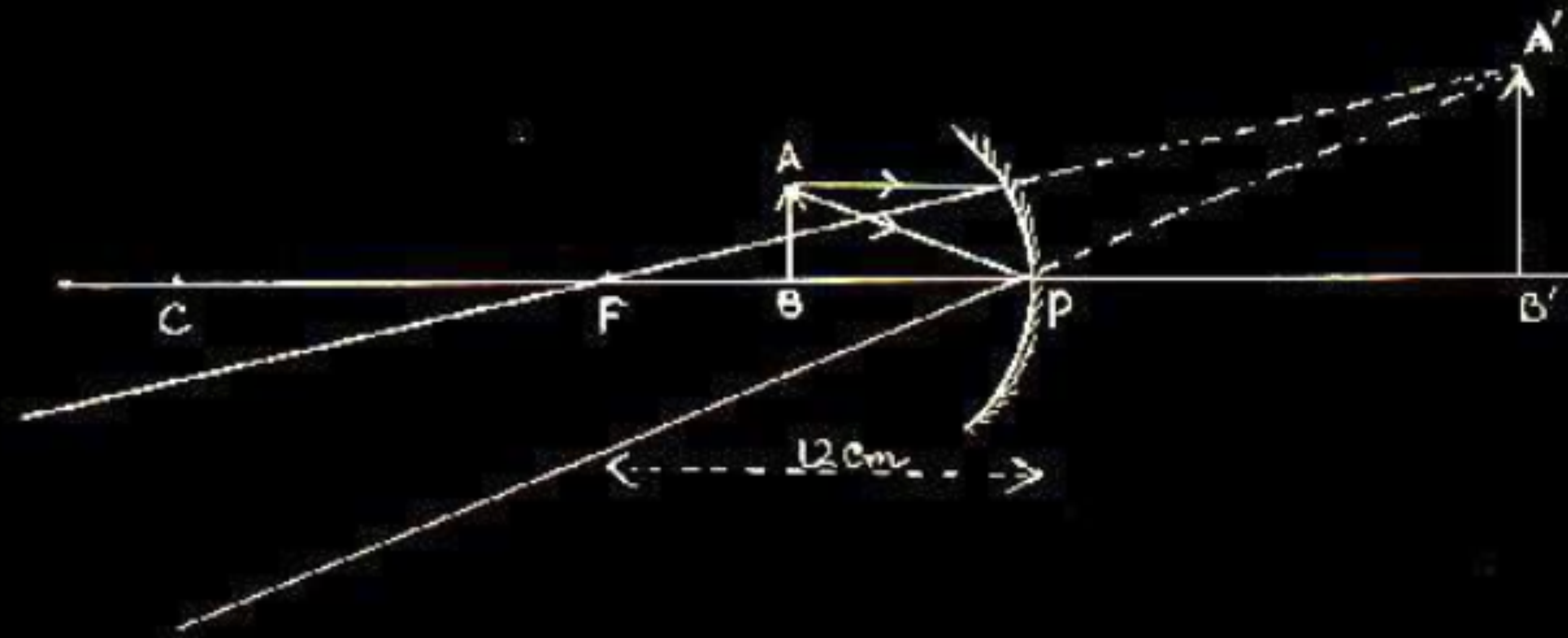
Show the positions of pole, principal focus and the centre of curvature in the above ray diagrams.

(i) Range of distance – between 0 cm - < 12 cm

1

ii) larger than the object

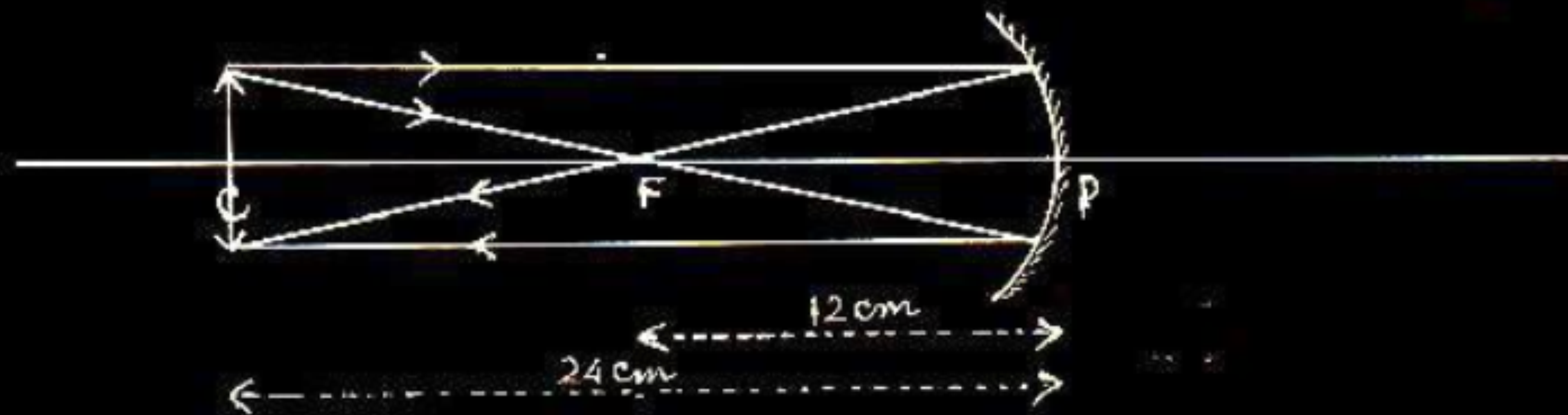
$\frac{1}{2}$



1 $\frac{1}{2}$

iii) Image also at 24 cm in front of the mirror

$\frac{1}{2}$



1 $\frac{1}{2}$

To determine the approximate value of the focal length of a given concave mirror, you focus the image of a distant object formed by the mirror on a screen. The image obtained on the screen, as compared to the object is always :

- (a) Laterally inverted and diminished
- (b) Inverted and diminished
- (c) Erect and diminished
- (d) Erect and highly diminished

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(b)

The image formed by a spherical mirror is real, inverted and is of magnification -2 . If the image is at a distance of 30 cm from the mirror, where is the object placed ? Find the focal length of the mirror. List two characteristics of the image formed if the object is moved 10 cm towards the mirror.

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$$m = -2 \quad \frac{v}{u} = 2 \quad v = -30 \text{ cm}$$

$$u = -15$$

1

$$f = \frac{uv}{u+v} = \frac{-15 \text{ cm} \times -30 \text{ cm}}{-15 \text{ cm} + (-30) \text{ cm}} = \frac{450}{-45} = -10 \text{ cm}$$

1

If the object is shifted 10 cm towards the mirror $u = -5 \text{ cm}$

Therefore the object is between pole and focus and the image formed is

i) Virtual ii) Erect iii) magnified (any two) $\frac{1}{2}, \frac{1}{2}$

Name the type of mirrors used in the design of solar furnaces. Explain how high temperature is achieved by this device.

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Name the type of mirrors used in the design of solar furnaces. Explain how high temperature is achieved by this device.

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- Concave Mirrors / Converging Mirrors 1/2
- When a solar furnace is placed at the focus of a large concave mirror/ reflector, it focuses a parallel beam of light on the furnace, consequently a high temperature is achieved after some time. 3 x 1/2

(a) Define the following terms in the context of spherical mirrors :

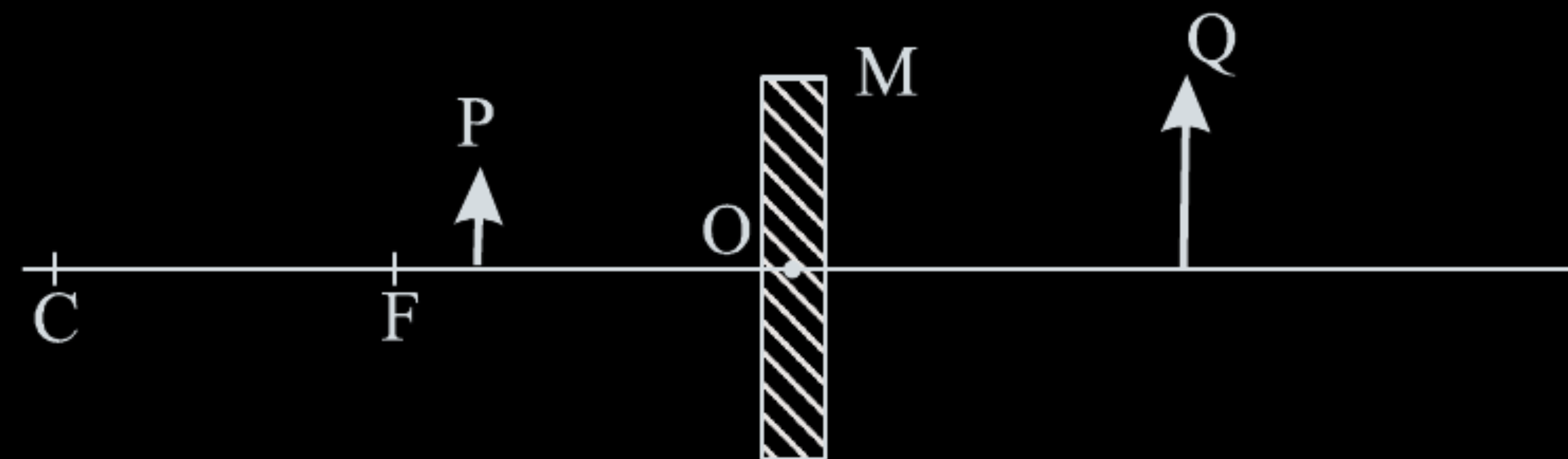
- (i) Pole
- (ii) Centre of curvature
- (iii) Principal axis
- (iv) Principal focus

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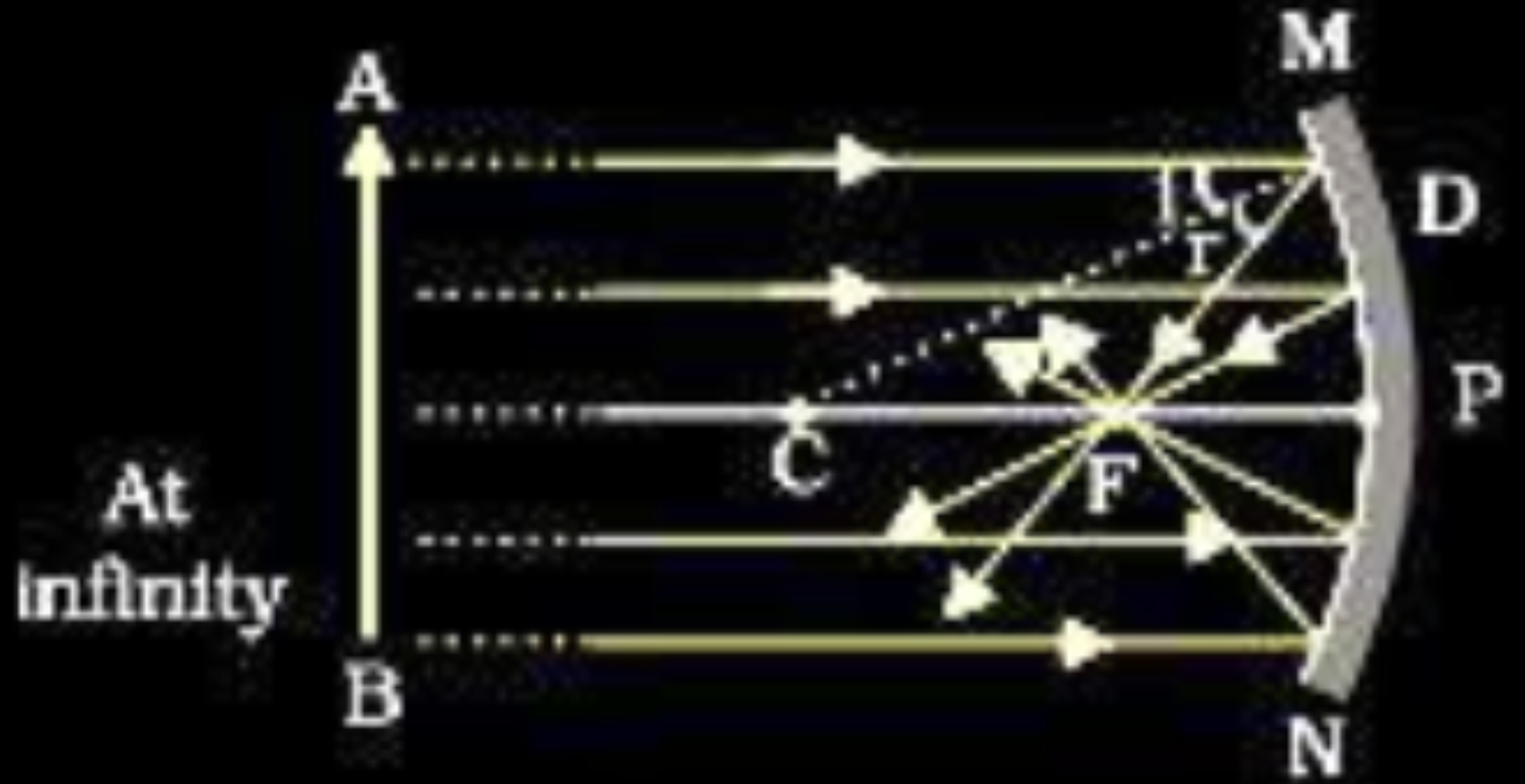
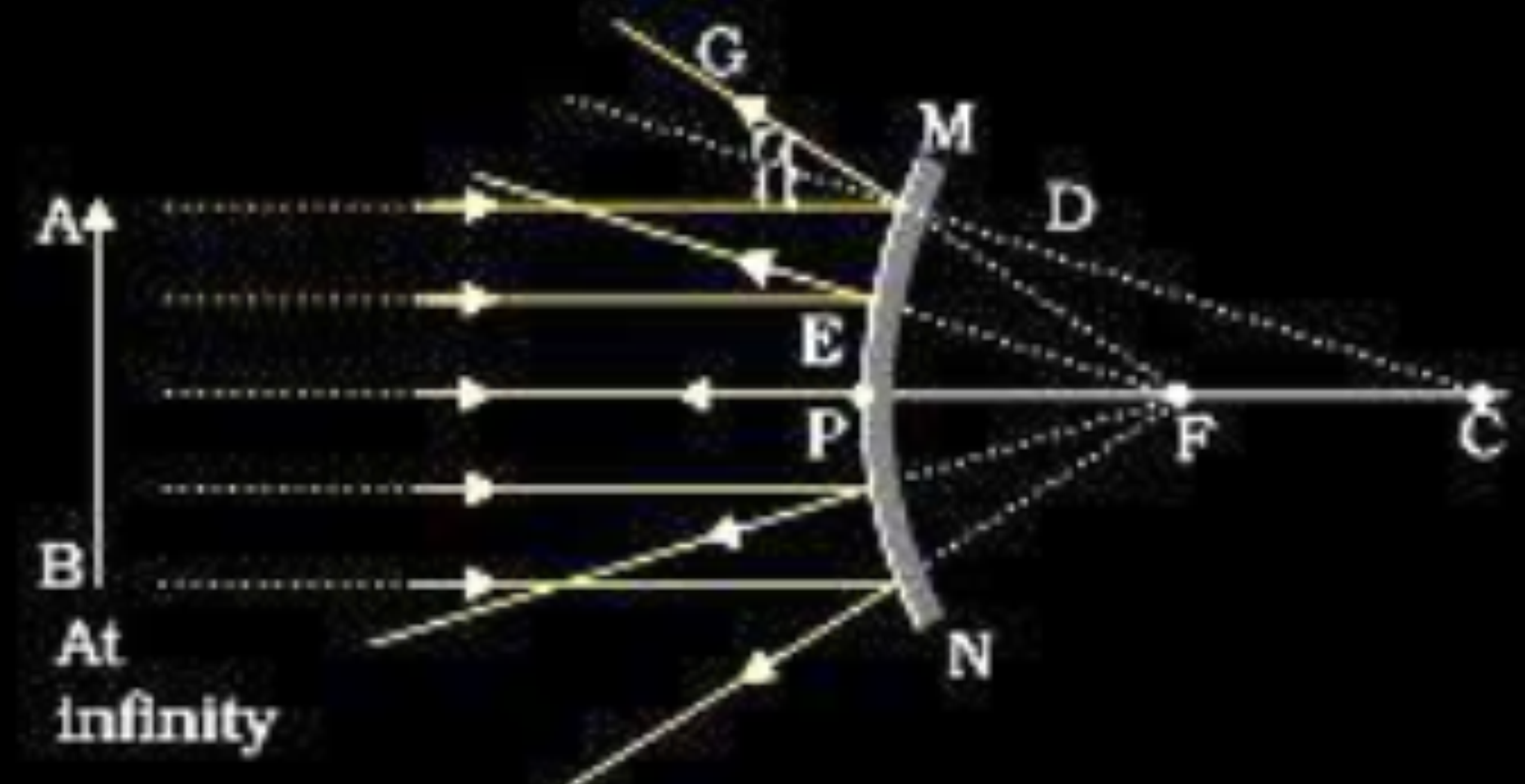
(b) Draw ray diagrams to show the principal focus of a

- (i) Concave mirror
- (ii) Convex mirror

(c) Consider the following diagram in which M is a mirror and P is an object and Q is its magnified image formed by the mirror.



State the type of the mirror M and one characteristic property of the image Q.

- a)
- i) Pole – Centre of the reflecting surface of the mirror.
 - ii) Centre of curvature – The centre of the hollow sphere of which the reflecting surface of mirror forms a part.
 - iii) Principal axis – Straight-line passing through the pole and the centre of curvature of a spherical mirror.
 - iv) Principal focus – Incident rays parallel to principal axis, after reflection, either converge to or appear to diverge from a fixed point on the principal axis called principal focus of the spherical mirror. 4x ½
- b)
- i)  1
 - ii)  1
- c) Concave mirror ½
 Image formed is virtual ½

A student obtained a sharp image of a candle flame placed at the distant end of the laboratory table on a screen using a concave mirror to determine its focal length. The teacher suggested him to focus a distant building about 1 km far from the laboratory, for getting more correct value of the focal length. In order to focus the distant building on the same screen the student should slightly move the :

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- (a) mirror away from the screen
- (b) screen away from the mirror
- (c) screen towards the mirror
- (d) screen towards the building

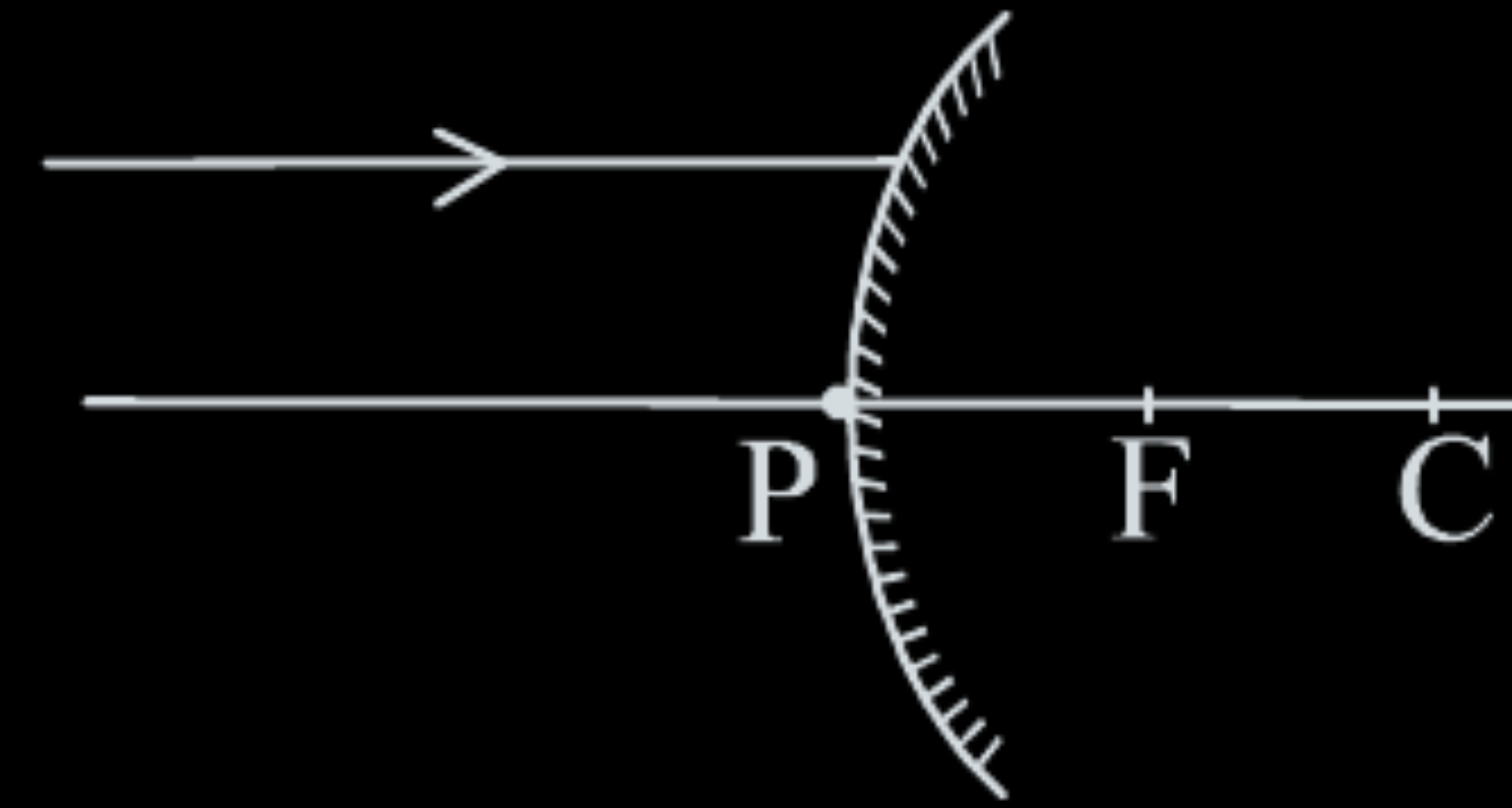
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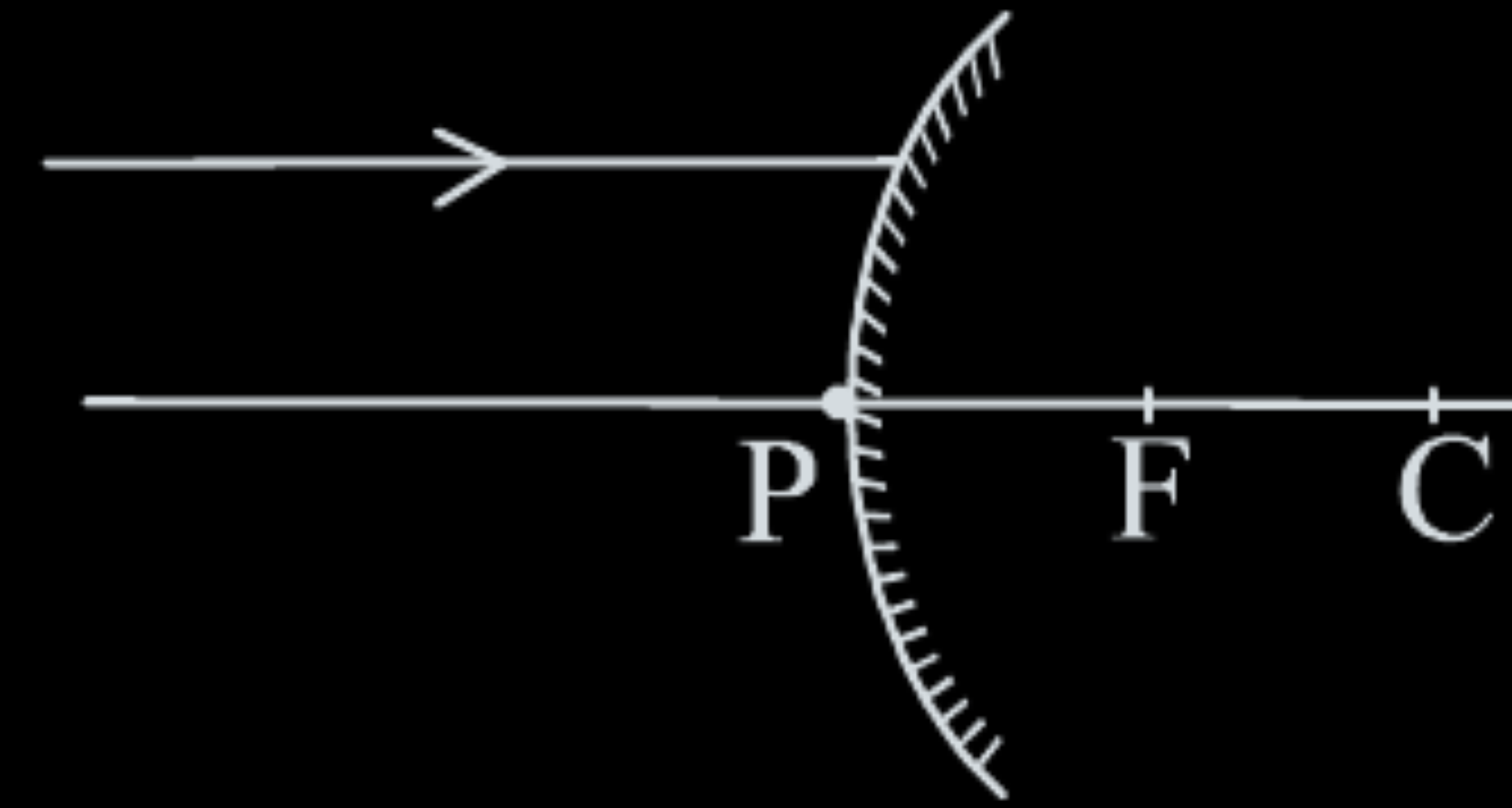
(c)

A ray of light is incident on a convex mirror as shown. Redraw the diagram and complete the path of this ray after reflection from the mirror. Mark angle of incidence and angle of reflection on it.

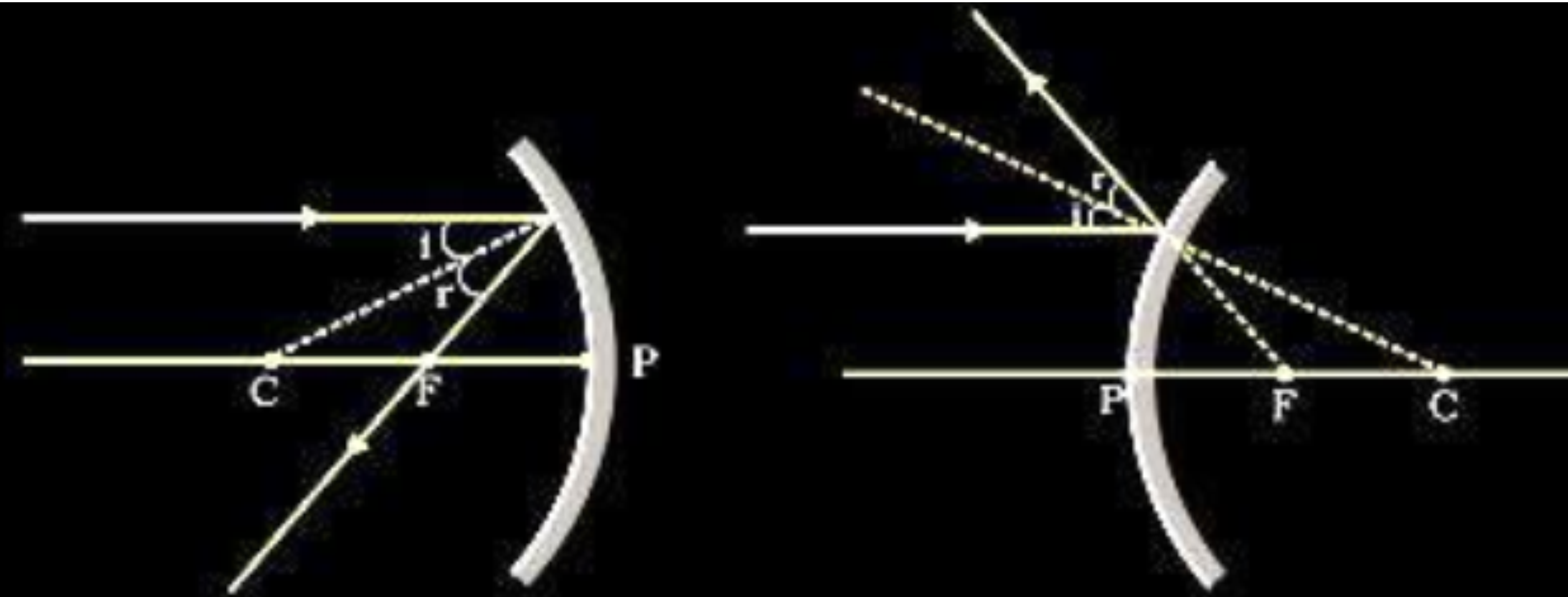


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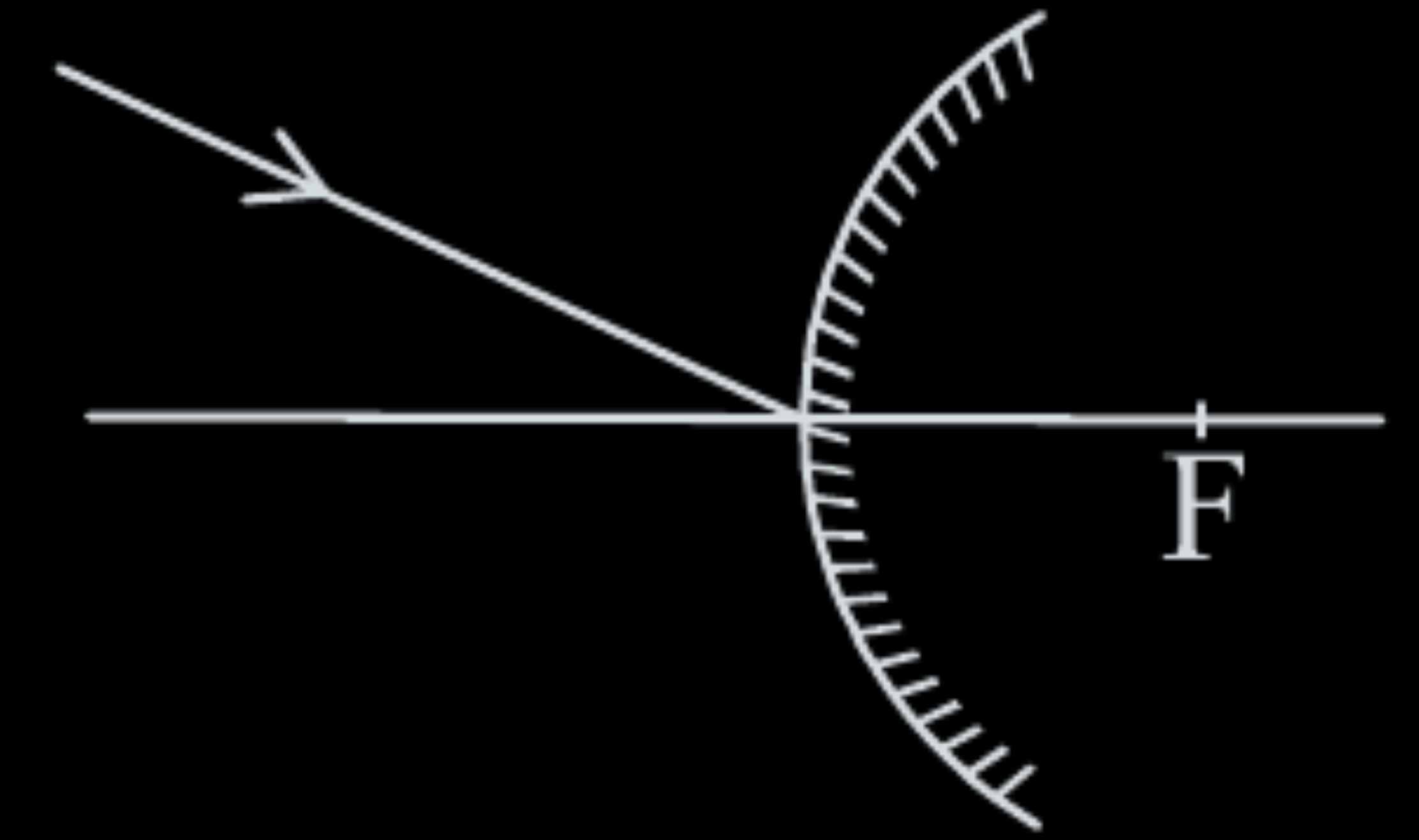
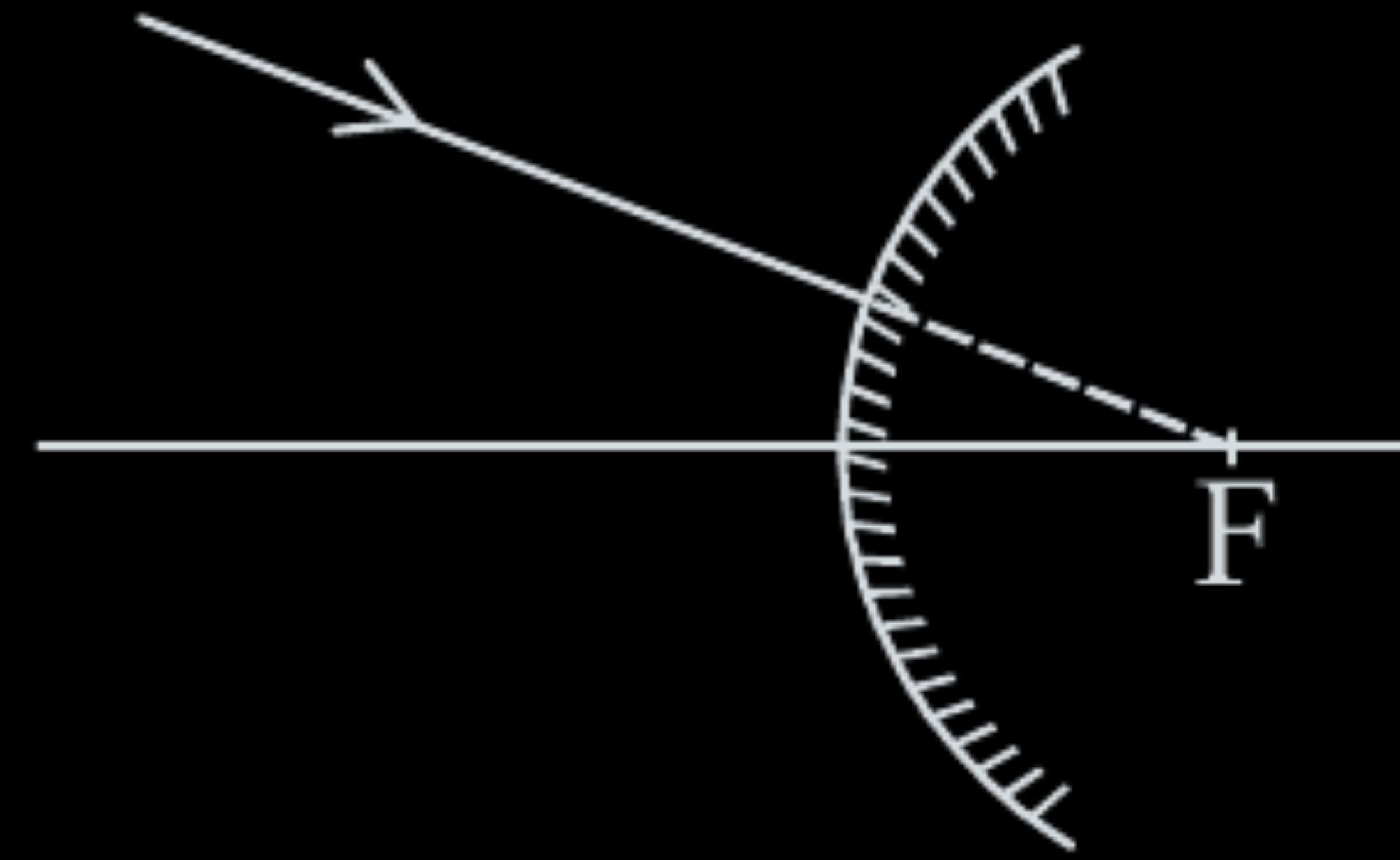
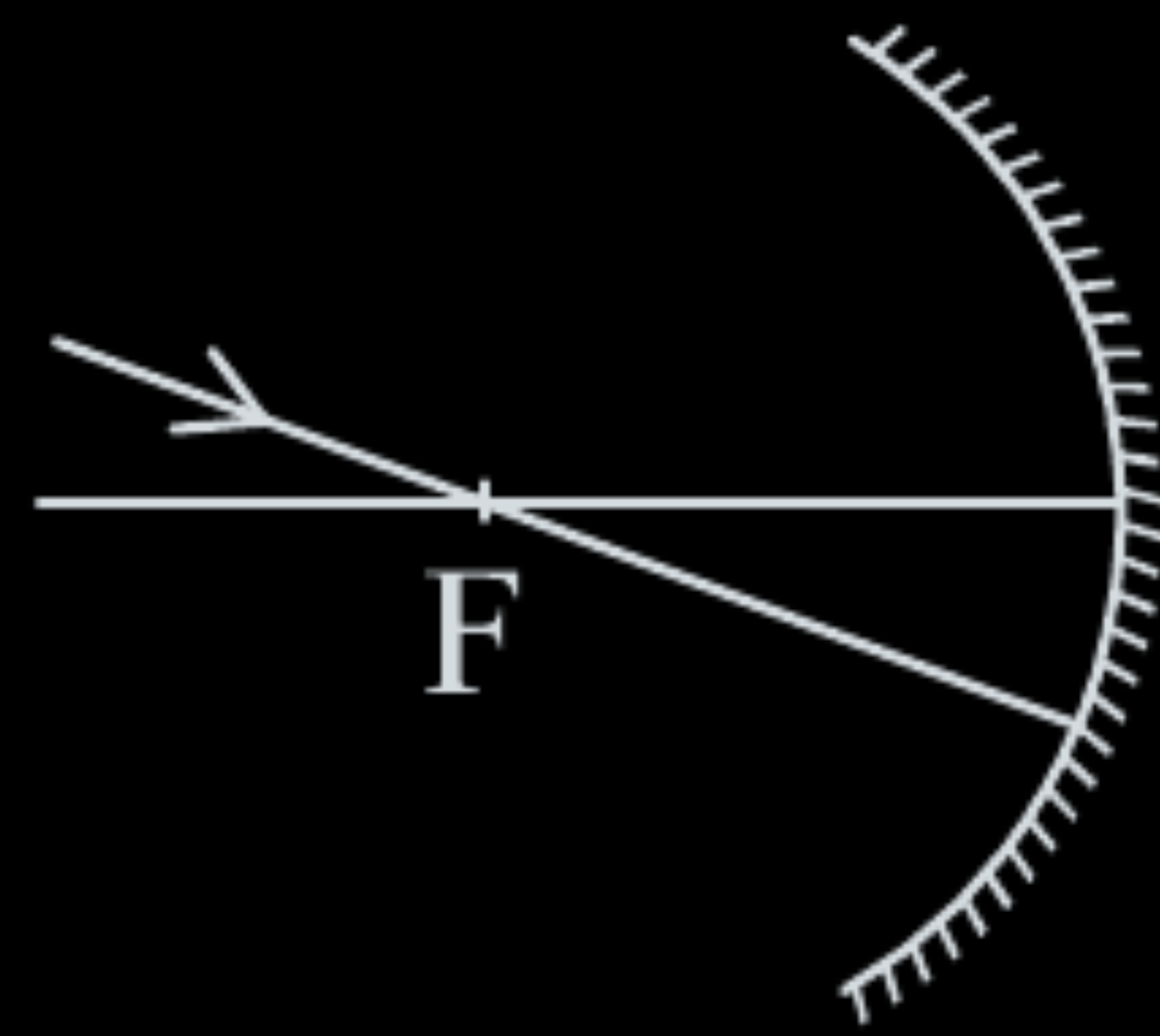


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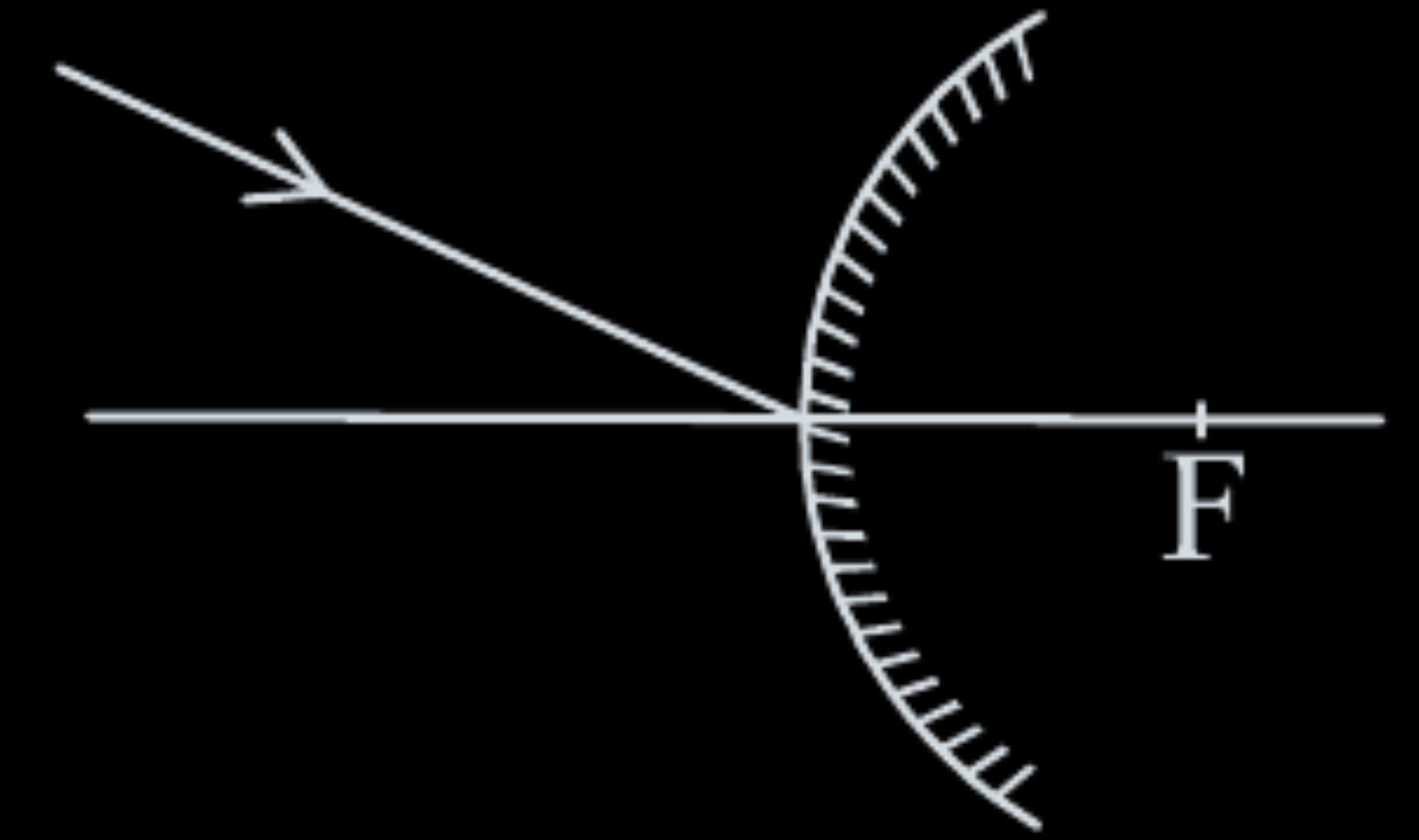
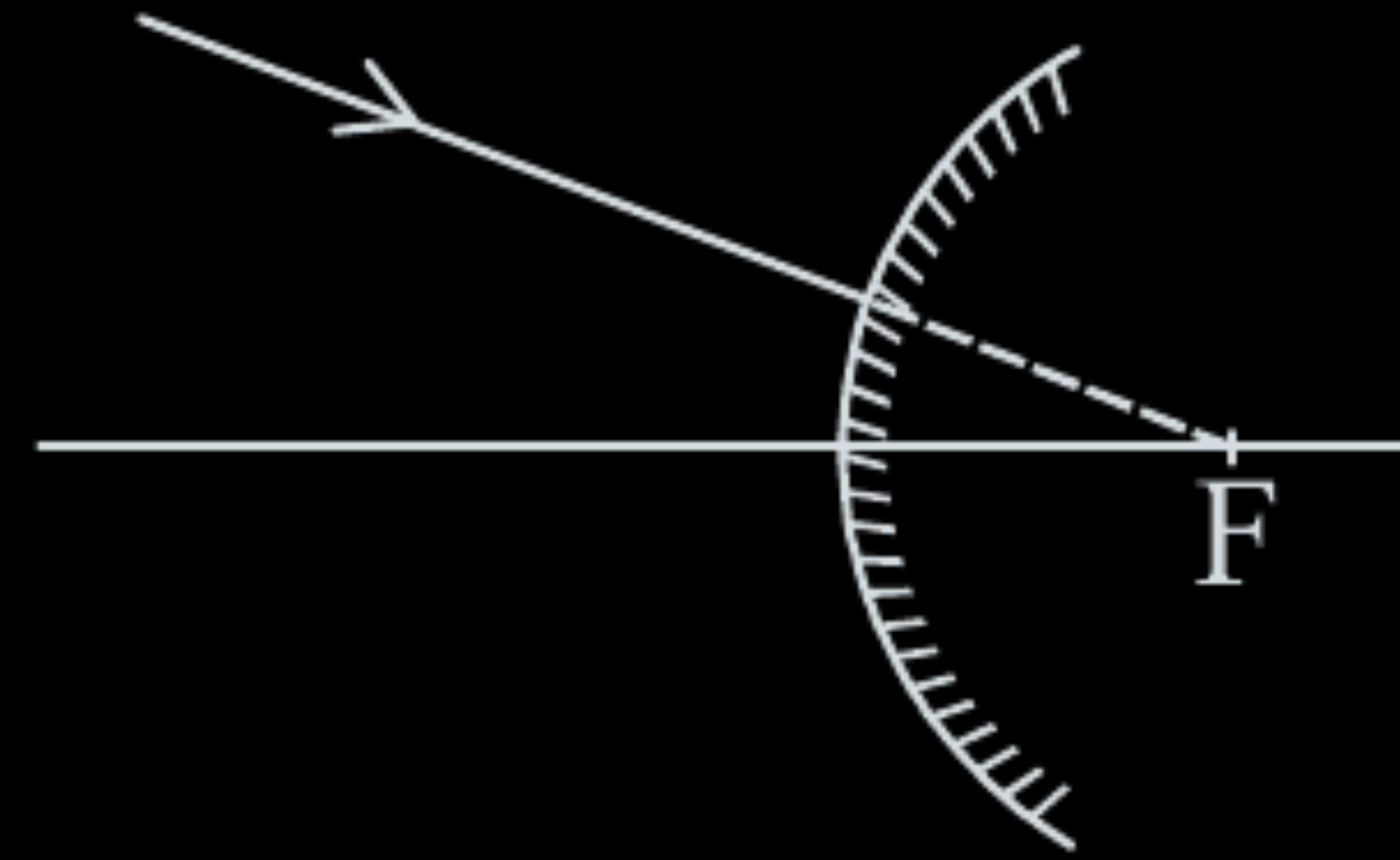
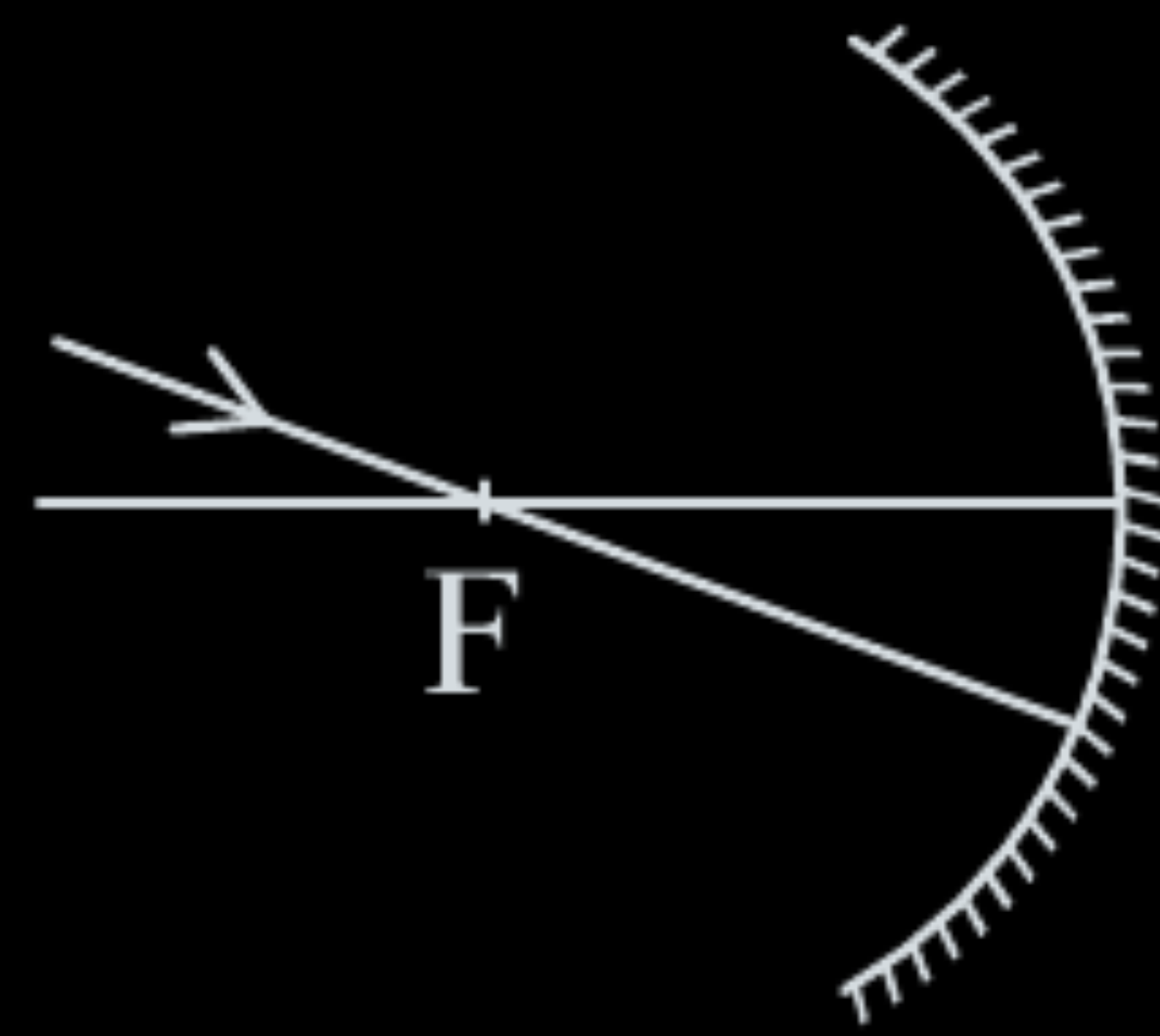


Tracing the reflected ray	1
Marking $\angle i$ & $\angle r$	$\frac{1}{2}, \frac{1}{2}$

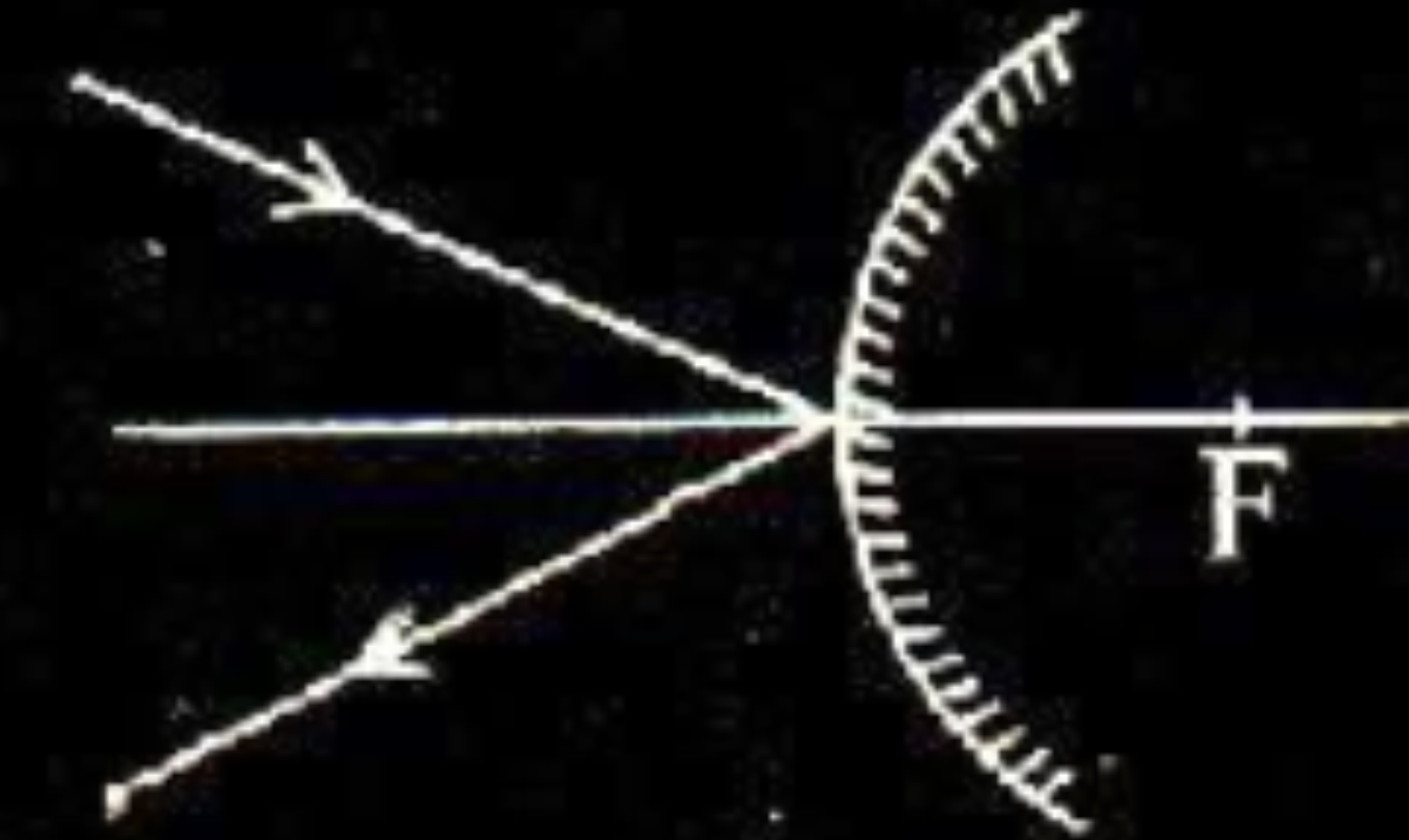
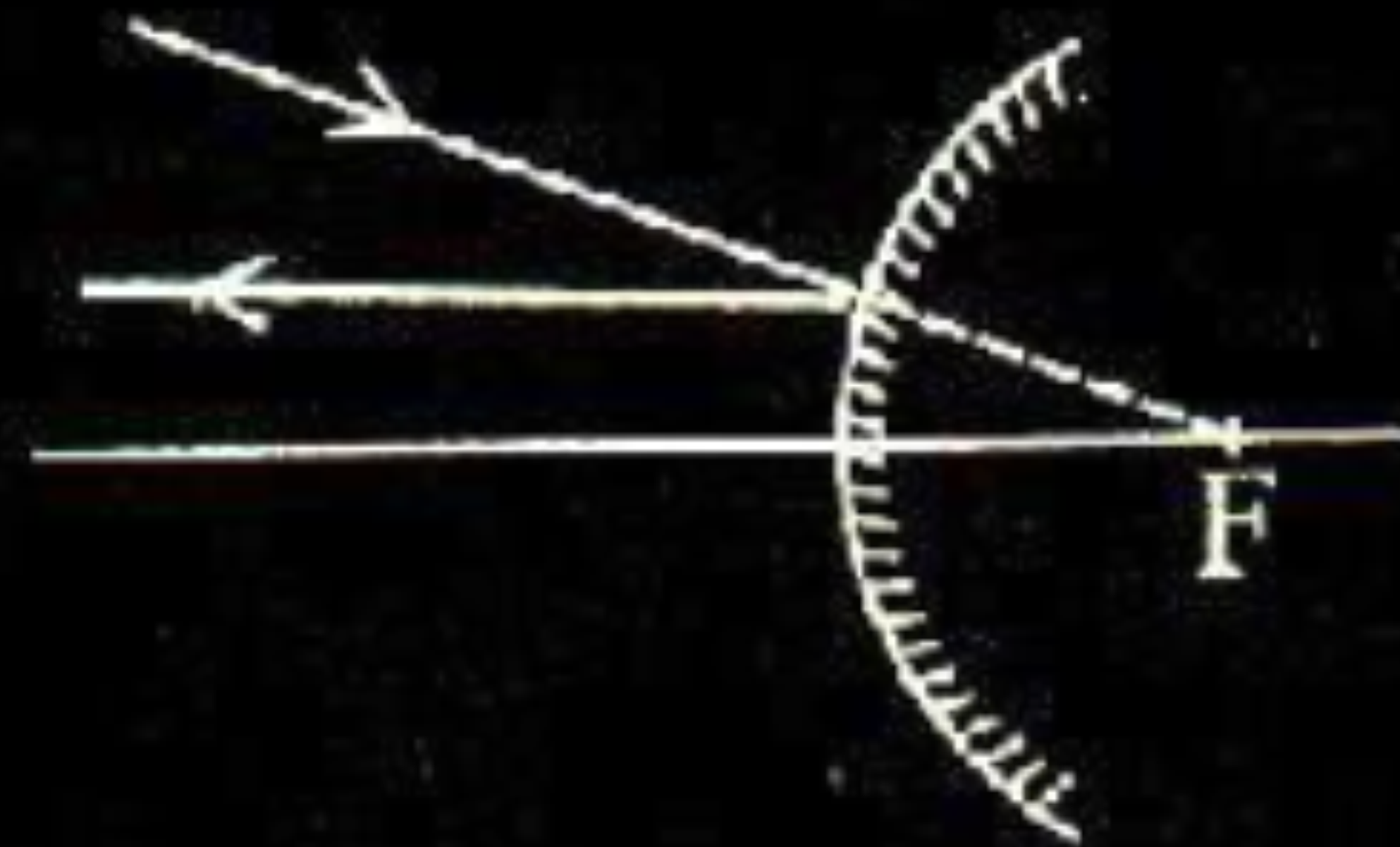
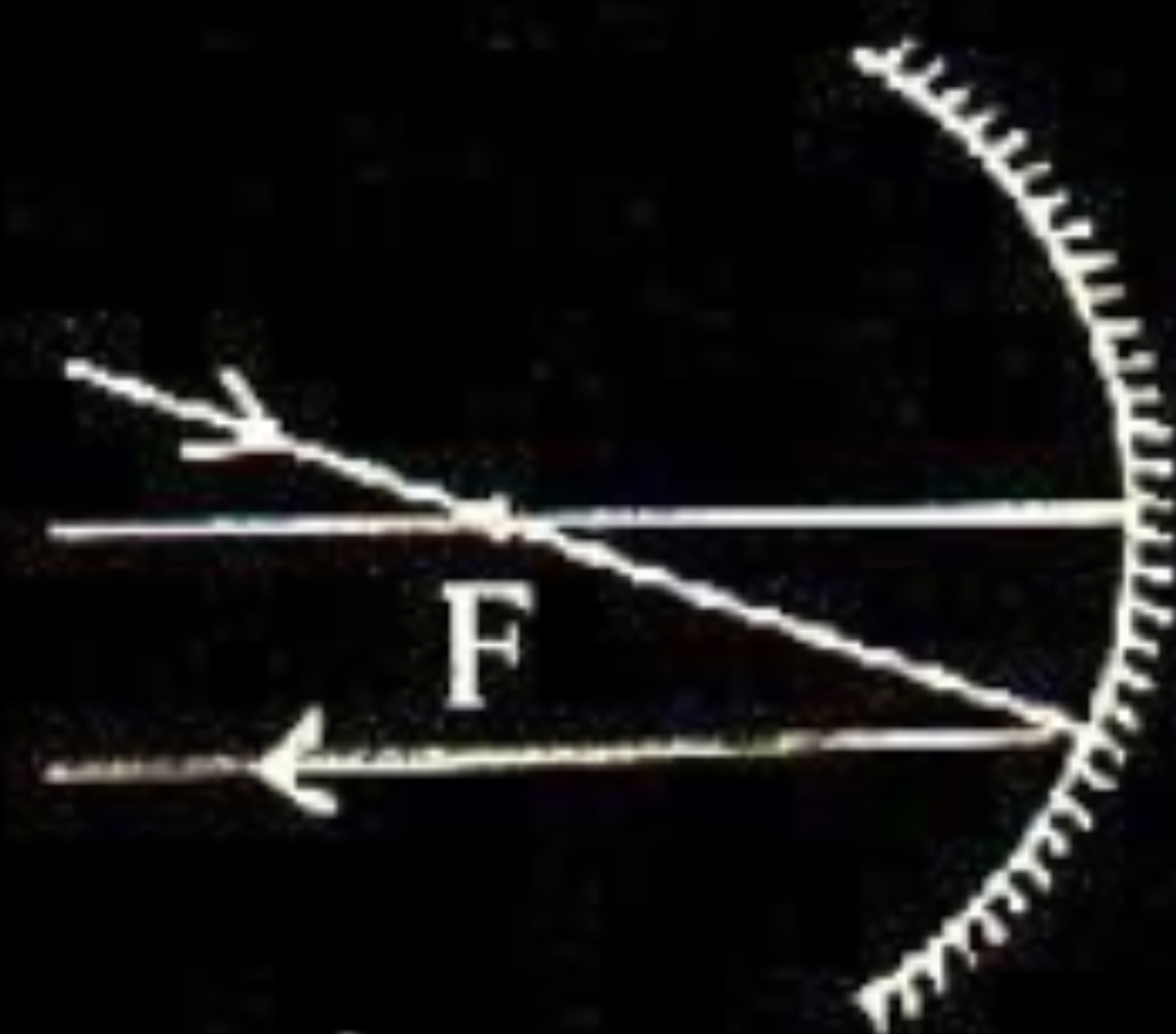
Draw the following diagram, in which a ray of light is incident on a concave/convex mirror, on your answer sheet. Show the path of this ray, after reflection, in each case.



Draw the following diagram, in which a ray of light is incident on a concave/convex mirror, on your answer sheet. Show the path of this ray, after reflection, in each case.



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3x1

State two positions in which a concave mirror produces a magnified image of a given object. List two differences between the two images.

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- | | |
|--|-----|
| • When the object is placed in front of the mirror – | |
| (i) between its pole and focus | 1/2 |
| (ii) between the focus and centre of curvature | 1/2 |
| • In case (i) the image is virtual and erect | 1/2 |
| In case (ii) the image is real and inverted | 1/2 |